SCALE-COVARIANT ATTENTION: PER-HEAD RECEPTIVE FIELDS MEASURED AT THE TRAINING RESOLUTION

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ABSTRACT

Diffusion transformers trained near one megapixel fail at 4096^2: attention meets token counts and position offsets it never saw, at 256 times the cost. Training-free fixes set how far each head looks by hand or per target size, because at the target full attention fails and cannot serve as a reference. At the training resolution, where full attention is still valid, reach is a property of the head. After the first steps, the head explains 97.5% of the variance in how much attention stays near the query, and the noise level 2.5%. We introduce Scale-Covariant Attention (SCA), which gives each head its own mix of a native-size window and a native-size grid over the whole image. Reach emulation shrinks both views at 1024^2 to the target's image-to-window ratio, so that the model's own full attention fits each mix in closed form, once for every size. Inside HRDiT's three-stage schedule, SCA on FLUX.1-dev at 4096^2 beats AReA on all ten per-image metrics for each of three seeds and on HRGen-1K at $4.1\times$ its speed, and matches HRDiT's preference scores in 49.2 against 107.0 s per image. A single-pass mode keeps more fine detail and runs in 77.1 s, $2.63\times$ faster than AReA. Code: github.com/shreshthsaini/scale-covariant-attention.

Figure 1: **Progressive SCA at 4096^2.** FLUX.1-dev with HRDiT's three-stage schedule and SCA attention, Aesthetic-4K evaluation prompts at seed 0; insets show native pixels. Figure S.4 shows more samples beside AReA, HRDiT, and single-pass SCA.

1 INTRODUCTION

Diffusion transformers (Peebles & Xie, 2022) such as FLUX.1 (Black Forest Labs, 2024) and Stable Diffusion 3 (Esser et al., 2024) are trained near one megapixel. At 4096×4096 the image has 16 times as many tokens, full attention costs about 256 times as much, and every layer meets token counts and position offsets it never saw in training. Full attention then returns stripes or a flat field (Figure S.1). Fixes that restrict attention must decide how far each head looks, and at the target nothing can check that decision, because full attention there is the failure. We ask whether the decision can be measured where full attention is still correct, at the training resolution.

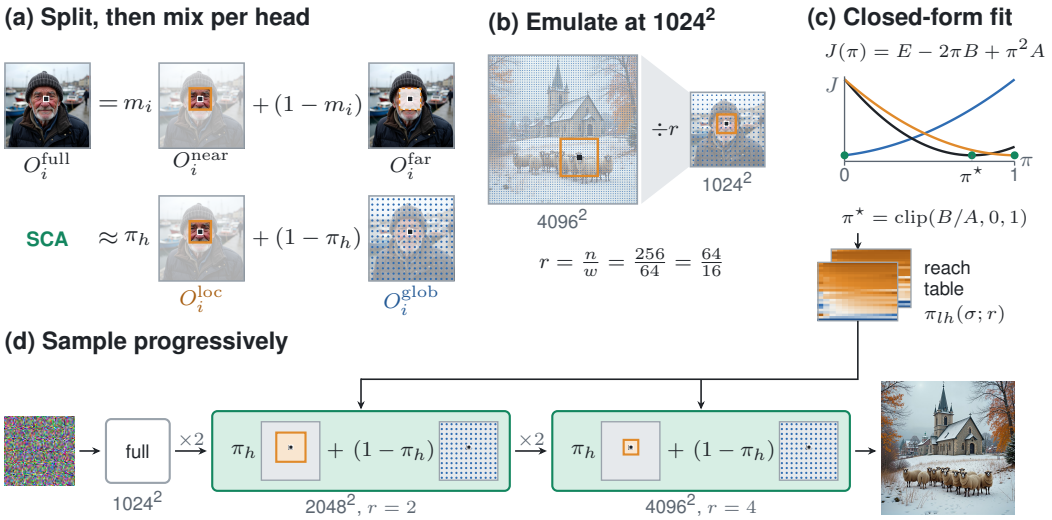

Figure 2: **Each head's mix, fitted once at 1024^2.** (a) Full attention splits exactly into outputs over near and far keys, mixed by each query's near mass m_i. SCA uses a local window and a grid of block centres over the whole image, mixed by one weight π_h per head. (b) Reach emulation shrinks both views by the target's image-to-window ratio r onto 1024^2 images of its own prompts. (c) The squared error $J(\pi)$ against full attention is quadratic, so $\pi^\star = \text{clip}(B/A, 0, 1)$ fills the reach table (three heads, rescaled). (d) Progressive SCA reads the table at $r = 2$ and $r = 4$ in HRDiT's upper two stages; Figure D.1 draws the operators at component level.

Some fixes edit the positional encoding (Zhuo et al., 2024; Issachar et al., 2025; Zhao et al., 2025; Rajabi et al., 2026b); others restrict the keys each query sees (Zhang et al., 2023; Ma et al., 2025; Xue et al., 2026; Rajabi et al., 2026a). AReA (Rajabi et al., 2026a) finds that attention narrows during denoising, averaged over heads, and gives every head one schedule: pooled keys for 8 of 28 steps, then windows that shrink to native size. HRDiT searches per-head scopes for each target size, and UltraImage types heads at run time (Xue et al., 2026; Zhao et al., 2025). Measured head by head at 1024^2, reach varies far more between heads than between steps (Figure 3c). After the first 8 steps, the head accounts for 97.5% of the variance in the share of attention a query keeps within 16 tokens, and the noise level for 2.5%.

We introduce Scale-Covariant Attention (SCA), which gives each head its own reach, measured once at the training resolution (Figure 2). Each head mixes two views that keep the training geometry: an exact native-size window around the query, and a native-size grid of block-centre keys over the whole image. We fit the mix by *reach emulation*. At 1024^2, shrinking the window and the grid to the target's image-to-window ratio lets the model's own full attention score both views, and the best weight has a closed form. One emulation pass fills the *reach table*, one weight per block, head, noise level, and ratio, which serves every target size. We call this scale covariance: the weight depends on the head and the image-to-window ratio, not on the absolute size.

SCA replaces only the attention, so it runs inside any sampler. Our main configuration keeps HRDiT's released three-stage schedule (Xue et al., 2026), which sets the layout at 1024^2 and refines it at 2048^2 and 4096^2, and swaps HRDiT's four attention changes for SCA. We call it progressive SCA; the schedule is HRDiT's (Section 2.5). A single-pass mode samples at the target in 28 steps: it opens with the global view for 8 steps, as AReA does, then shrinks a window to native size over four steps. The opening, the window sizes, and 1,024 bridge tokens that show each window the whole image are set by hand; only the weights are fitted (Table F.1).

The fitted weights are stable across disjoint prompts (correlation 0.99), track each head's attention mass within 16 tokens ($\rho = 0.77$), and transfer across ratios and resolutions (Section 3). At 4096^2 on FLUX.1-dev, progressive SCA beats AReA on all ten per-image metrics for each of three seeds and on HRGen-1K, at $4.1\times$ its speed; at seed 0 ImageReward rises from 0.92 to 1.32. On set-level scores it is level with AReA on FID, and its patch FID is higher at two of three seeds. It matches HRDiT on ImageReward, HPSv2.1, and PickScore in 46% of HRDiT's time, and trails it on KID and patch FID. Against single-pass SCA it gains preference and loses fine detail, with MUSIQ on 1024^2 crops -6.0 and patch FID $+9.2$.

- We show that the reach of attention heads, measured at the training resolution, varies far more between heads than between denoising steps, on FLUX.1-dev, FLUX.1-Krea-dev, and FLUX.2-dev (Section 3).
- We fit each head's mix of a native window and a whole-image view once at 1024^2, in closed form, by emulating the target's image-to-window ratio against the model's own full attention (Section 2.4).
- We show that at 4096^2 progressive SCA beats AReA on all ten per-image metrics for three seeds at $4.1\times$ its speed and matches HRDiT in less than half its time (Section 4.1).

2 Scale-Covariant Attention

Each head gets two views of the image that keep the training geometry, and one weight for how much of its output comes from each. Only the weight is measured: it is fitted at 1024^2 (Section 2.4) and read at the target's image-to-window ratio (Section 2.3).

2.1 Setting and notation

A multimodal diffusion transformer such as FLUX.1 concatenates T text tokens $\mathcal{T}$ with $N = HW$ image tokens on the grid $\mathcal{G} = [H] \times [W]$, where $[n] = \{0, \dots, n-1\}$. Each block applies joint attention (57 blocks of 24 heads in FLUX.1). It is trained at about 1024^2 pixels, a native grid of $n_0 \times n_0$ tokens with $n_0 = 64$. Rotary embeddings (Su et al., 2021) give token j a position $p_j \in \mathbb{Z}^2$ (FLUX.1 puts all text tokens at 0) and a rotation R_p with $R_p^\top R_{p'} = R_{p'-p}$. For one head, a pre-rotation query q placed at u, and a set $\mathcal{K}$ of (key, value, key position) triples,

$$\mathcal{A}(q, u; \mathcal{K}) \;=\; \textstyle\sum_{(k,v,u') \in \mathcal{K}} \operatorname{softmax}_{\mathcal{K}}\big(q^\top R_{u'-u}\, k/\sqrt{d}\big)\; v, \tag{1}$$

where d is the head dimension. Full attention is $O_i^{\text{full}} = \mathcal{A}\big(q_i, p_i; \{(k_j, v_j, p_j)\}_{j \in \mathcal{T} \cup \mathcal{G}}\big)$. The target grid is $H = s_H n_0$, $W = s_W n_0$ with integer factors (s when equal), so full attention meets offsets s times the trained range. The latent at noise level σ is $x_\sigma = (1-\sigma)x_0 + \sigma\epsilon$. Splitting the keys into near and far sets gives exactly $O_i^{\text{full}} = m_i O_i^{\text{near}} + (1-m_i) O_i^{\text{far}}$ (Lemma 1), where each O attends to its own set and m_i is the near set's softmax mass. Every head thus mixes a near and a far field per query, which SCA approximates with two separately normalized operators and one weight per head.

2.2 Two operators

Global operator. Let $\mathcal{G}_0 = [H/s_H] \times [W/s_W]$ be the $n_0 \times n_0$ grid of blocks. Block-centre subsampling keeps for each block b the token at $\kappa(b) = (s_H b_1 + \lfloor s_H/2 \rfloor,\, s_W b_2 + \lfloor s_W/2 \rfloor)$. The coarse map $c(p) = (\lfloor p_1/s_H \rfloor, \lfloor p_2/s_W \rfloor)$ sends a position to its block, so $c(\kappa(b)) = b$, and

$$\mathcal{K}^{\text{glob}} = \{(k_j, v_j, 0)\}_{j \in \mathcal{T}} \cup \{(k_{\kappa(b)}, v_{\kappa(b)}, b)\}_{b \in \mathcal{G}_0}, \qquad O_i^{\text{glob}} = \mathcal{A}\big(q_i,\, c(p_i);\, \mathcal{K}^{\text{glob}}\big), \tag{2}$$

with $c(p_i) = 0$ for text queries. This is AReA's first phase at pooling factor s, the image as one native canvas. It models content spanning a fixed fraction of the image: layout, object count, identity.

Local operator. Image queries form tiles of $t \times t$ tokens ($t = 8$); query i lies in tile $\tau = \lfloor p_i/t \rfloor$. The window $\mathcal{W}_w(i)$ of tile τ has side w, at most the grid side. It starts at $a(\tau) = \operatorname{clip}\big(t\tau + \lfloor t/2 \rfloor - \lfloor w/2 \rfloor, 0, H - w\big)$ per axis, centred on the tile and clamped to the grid. Its keys are the text tokens, a bridge set $\mathcal{B}$ of image tokens, and the window, all at their original positions:

$$\mathcal{K}_i^{\text{loc}}(w) = \{(k_j, v_j, 0)\}_{j \in \mathcal{T}} \cup \{(k_j, v_j, p_j)\}_{j \in \mathcal{B} \cup \mathcal{W}_w(i)}, \qquad O_i^{\text{loc}}(w) = \mathcal{A}\big(q_i,\, p_i;\, \mathcal{K}_i^{\text{loc}}(w)\big). \tag{3}$$

The bridge takes, in each cell c of an 8×8 partition of the grid, the 16 tokens whose pre-rotation keys lie farthest from the mean image key $\bar{k}^h$, summed over heads h:

$$\mathcal{B} = \textstyle\bigcup_c \operatorname{top}_{16}\big\{ \sum_h \|k_j^h - \bar{k}^h\|^2 : j \in c \big\}. \tag{4}$$

It is recomputed at every block and step with AReA's selection rule (Rajabi et al., 2026a), so each window also sees tokens from the whole image. Keys form a multiset; a bridge token inside the window enters twice. With $w = n_0$ the window is an exact native-size sliding window, which models content spanning a fixed number of tokens: edges, texture, fine structure. Appendix A proves the two propositions below.

Algorithm 1: **Reach emulation**, once per model

```
# Input: prompts; full-attention sampler at 1024^2
# Output: pi*[s_c, l, h, beta]; full in App. E
def emulate(q, k, v, l, sigma):     # every block
  O_full = attn(q, k, v)          # Eq. 1, all keys
  beta = min(int(14 * sigma), 13)     # noise bin
  for s_c in (2, 4):              # r = n / w = s_c
    f, w = s_c, n // s_c          # subsample, window
    t_c = round(t * w / n0)       # tile t n/(s_c n0)
    O_glob = global_branch(q, k, v, f)    # Eq. 2
    O_loc = local_branch(q, k, v, w, t_c) # Eq. 3
    dl, df = O_loc - O_glob, O_full - O_glob
    A[s_c, l, :, beta] += sum(dl * dl)    # Eq. 7
    B[s_c, l, :, beta] += sum(df * dl)  # over rows
    E[s_c, l, :, beta] += sum(df * df)
  return O_full                   # sampler unchanged
pi = clip(B / A, 0, 1); pi[A == 0] = 1  # Prop. 3
pi = fill_empty_bins(pi)  # read at r, Eq. 6
```

Algorithm 2: **Sampling**, one SCA attention call

```
# Input: q, k, v (T text, then H x W image), l, σ
# Output: O for all T + HW queries; App. E
fh, fw = H // n0, W // n0       # subsampling s
kb = k_img[fh//2::fh, fw//2::fw]  # κ(b); vb alike
Q = cat(rope(q_txt, 0), rope(q_img, c(p)))
K = cat(rope(k_txt, 0), rope(kb, b))
O_glob = sdpa(Q, K, cat(v_txt, vb))     # Eq. 2
if sigma > sigma_g:                  # 0.89, Eq. 5
  return O_glob       # opening phase: no local
w = window(sigma)  # 128,112,96,80,n0; Sec. 2.3
B = bridge(k_img, cells=8, k=16)  # Eq. 4, 1,024
Kl = cat(k_txt, k_img[B], k_img)  # Vl alike
mask = flex_layout(T + len(B), t=8, w=min(w, H))
O_loc = flex_attention(rope(q_img, p),  # Eq. 3
    rope(Kl, p_Kl), Vl, block_mask=mask)  # App. D
pi = interp(sigma, max(H,W)/w, table[l])  # Eq. 6
O_img = pi*O_loc + (1-pi)*O_glob[T:]    # Eq. 5
return cat(O_glob[:T], O_img)     # text: O_glob
```

Proposition 1 (Trained position range)**.** *Assume exact integer positions and* $t \le n_0 \le \min(H, W)$. *Every query-key offset of the global operator, and every offset between an image query and a key of its native window* $\mathcal{W}_{n_0}(i)$ *(bridge keys and wider windows excluded), is at most* $n_0 - 1$ *per axis. Each global-operator query sees the native* $T + n_0^2$ *keys.*

Proposition 2 (Cost)**.** *Per head and layer, full attention scores* $(N+T)^2$ *query-key pairs, the global operator* $(N + T)(T + N_0)$ *with* $N_0 = |\mathcal{G}_0| = n_0^2$*, and the local operator* $N(T + R + w^2)$ *with* $R = |\mathcal{B}|$*. For fixed* n_0*,* w*,* R*, and* T *the total is* $O\big(N(T + R + w^2 + N_0)\big)$*, linear in* N.

Design choices. A block centre is a real token, so the global branch attends to keys and values the model produced; mean pooling does worse (Section 4.3). Native windows keep offsets in the trained range, so SCA keeps FLUX.1's RoPE and softmax scale; text keys sit at position 0 in both branches (Remark 1). Bridge keys keep their true positions and windows wider than n_0 reach offsets up to $w-1$, so the local operator leaves the trained range for 1,024 keys and during the four wide-window steps. FLUX.1 also rounds positions in bf16 beyond 256 tokens per axis (Appendix D).

Worked example. At 4096^2 the latent grid is 256×256 tokens ($s = 4$, $N = 65{,}536$, $T = 512$). The global branch keeps one centre per 4×4 block, 4,096 keys on a 64×64 grid, so with the text a query sees 4,608 keys. In the native phase each 8×8 query tile attends to its 64×64 window, the 1,024 bridge tokens, and the text: 5,632 keys, against 66,048 keys at offsets up to 255 in full attention. Over 28 steps SCA scores 6.9 times fewer query-key pairs than full attention (Table D.1).

2.3 Single-pass sampling: from the global view to the native window

The reach table (Section 2.4; Algorithm 1) gives each head h of block l a weight $\pi_{lh}(\sigma; r)$ for every noise level and emulated image-to-window ratio $r \in \{2, 4\}$. At the target (Algorithm 2), a window schedule $w(\sigma)$ sets the ratio $r(\sigma) = \max(H, W)/w(\sigma)$, and an image query takes

$$O_i = \begin{cases} O_i^{\text{glob}} & \sigma > \sigma_g, \\ \pi\, O_i^{\text{loc}}\big(w(\sigma)\big) + (1 - \pi)\, O_i^{\text{glob}} & \text{otherwise,} \end{cases} \qquad \pi = \pi_{lh}\big(\sigma; r(\sigma)\big), \tag{5}$$

where σ_g is the gate and text queries take O_i^{glob}. Between ratios 2 and 4, π is linear in $\log r$:

$$\pi_{lh}(\sigma; r) = (1 - \lambda)\, \pi_{lh}(\sigma; 2) + \lambda\, \pi_{lh}(\sigma; 4), \qquad \lambda = \text{clip}(\log_2 r - 1,\, 0,\, 1). \tag{6}$$

We use $\sigma_g = 0.89$, the first 8 of 28 steps at maximum shift as in AReA. The windows are $w = 128$, 112, 96, and 80 tokens for σ in $(0.88, 0.89]$, $(0.86, 0.88]$, $(0.84, 0.86]$, and $(0.82, 0.84]$, then $w = n_0$. At 4096^2 these are the four steps after the opening, at ratios 2, 2.29, 2.67, and 3.2, and the last 16 steps run at ratio 4. Setting $w = n_0$ after the gate and dropping the bridge gives a one-step switch, our first operator, which we call SCA without the schedule. The four sizes and their σ bins were set by hand, and each bin holds one step only for the 28-step sampler; only the weights are fitted (Table F.1). The 42- and 56-step runs of Appendix N therefore use SCA without the schedule.

2.4 REACH EMULATION

On an in-distribution $n \times n$ grid ($n = 64$ at 1024^2), emulated ratio r uses subsampling factor r, window side n/r, and tile side $\mathrm{round}(tn/(rn_0))$. The grid-to-window ratio is then r, as at a target of scale r. The sampler runs unmodified full attention (Algorithm 1), and at every block and step both emulated operators see the same queries, keys, and values. For block l, head h, noise bin β, and ratio r, sums over prompts, steps with $\sigma \in \beta$, image queries, and channels give

$$A = \sum \|O^{\mathrm{loc}} - O^{\mathrm{glob}}\|^2, \quad B = \sum \langle O^{\mathrm{full}} - O^{\mathrm{glob}},\, O^{\mathrm{loc}} - O^{\mathrm{glob}} \rangle, \quad E = \sum \|O^{\mathrm{full}} - O^{\mathrm{glob}}\|^2, \tag{7}$$

where the emulated $O^{\mathrm{loc}}, O^{\mathrm{glob}}$ omit the bridge. Weight π has squared error $J(\pi) = E - 2\pi B + \pi^2 A$.

Proposition 3 (Closed-form fit and transfer cost)**.** *In each cell (l, h, β, r), J is minimized on $[0, 1]$ by $\pi^\star = \mathrm{clip}(B/A, 0, 1)$, with $\pi^\star = 1$ if $A = 0$. Any other weight $\pi' \in [0, 1]$ constant in the cell has $J(\pi') - J(\pi^\star) \geq A(\pi' - \pi^\star)^2$, with equality when $0 < \pi^\star < 1$. This covers weights fitted at another ratio or resolution, per layer, or shared, and a hard switch.*

The same sums score every alternative rule without another forward pass, and one pass collects ratios 2 and 4. At inference, each weight is interpolated linearly in σ between the mean noise levels of occupied bins (14 of width $1/14$), constant beyond them, and in $\log r$ by Eq. 6. The native phase reads the ratio nearest $\max(s_H, s_W)$ in log space, here always 4. The local operator is one exact FlexAttention call with text and bridge keys as a shared prefix (Dong et al., 2024, Algorithm 2).

Scale covariance. Emulated ratio r keeps two proportions of a target r times larger, image side to window side and global keys to window keys, with r^2 times fewer keys per branch. We assume the weight depends on r, not on the absolute size, and test this in Section 3.

2.5 PROGRESSIVE SCA

Replacing only the attention, SCA fits inside a multi-stage sampler. HRDiT's released code (Xue et al., 2026) denoises at 1024^2 for 30 steps with the model's own attention. It then decodes, upsamples, sharpens, re-encodes, and re-noises the image. It runs the last 17 steps of its schedule again at 2048^2 and the last 10 at 4096^2, guided toward the low frequencies of the previous stage. Progressive SCA keeps all of this, including guidance and seeds, and replaces HRDiT's four attention changes (NTK-scaled RoPE, a proportional attention scale, per-head key pruning, and position alignment) with SCA (Algorithm 3). One reach table serves both stages: the 2048^2 stage reads ratio 2 and the 4096^2 stage ratio 4. The stages re-enter at $\sigma = 0.81$ and 0.61, below the gate and the last wide window. They run the native window, the bridge, and the fitted per-head mix, and the 1024^2 stage takes the place of the global opening.

Algorithm 3: **Progressive SCA**, HRDiT's schedule

```
# Input: prompt, seed, table pi; full: Alg. 6
sig = schedule(30)              # HRDiT's, all stages
x = denoise(noise(1024), sig, attn="full", g=3.5)
for res, k, g, r in ((2048, 13, 4.5, 2),
                     (4096, 20, 6.0, 4)):
  img = sharpen(bicubic(decode(x), res))
  x = renoise(encode(img), sig[k])   # 0.81, 0.61
  x = denoise(x, sig[k:], attn=sca(pi, r), g=g,
              guide="low-pass of last stage")
return decode(x)                # 4096^2 image
```

3 REACH IS SET BY THE HEAD

Reach emulation on FLUX.1-dev uses 32 purpose-written prompts at 1024^2 and 16 at 1280^2, none from the evaluation sets. Errors are relative, $\sum J / \sum \|O^{\mathrm{full}}\|^2$, pooled over blocks, heads, and steps.

Reach varies by head, not by step. For 256 random queries per block and step we record the share of each head's image attention within 16 tokens of the query. After the first 8 steps, the head accounts for 97.5% of the variance in this share, and the noise level for 2.5%. Most of the head's part lies between heads of the same block. The head-averaged curve accounts for 0.7%: it rises from 0.57 to 0.62 as denoising proceeds, the narrowing AReA reports. FLUX.1-Krea-dev and FLUX.2-dev give 97.6% and 99.1% for the head (Table B.1).

Heads differ, and each head's weight depends on the ratio. At scale 2, $\pi_h(\sigma)$ has a standard deviation of 0.24 to 0.26 over the 1368 heads at every noise level, with weights in $[0, 1]$ (Figures 3c

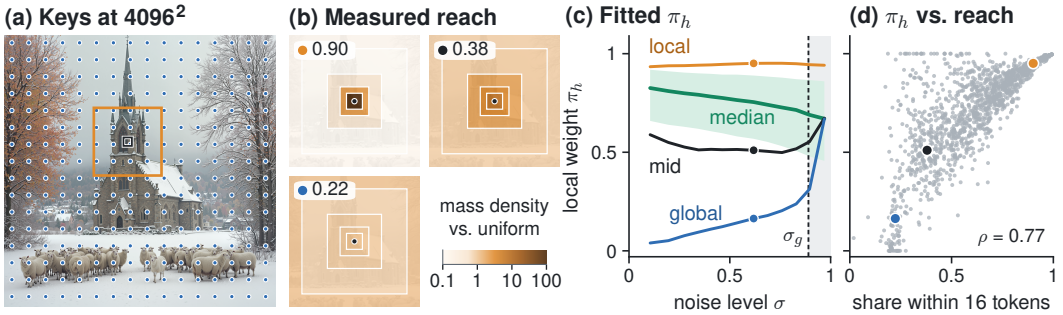

Figure 3: **Measured reach tracks the fitted π_h.** (a) Local window (amber) and every fourth block-centre key per axis (blue) on a 4096^2 output; (b) full-attention mass density of three heads at 1024^2 and $\sigma = 0.62$, with each head's share within 16 tokens. (c) Their π_h with the median and interquartile range of all 1368 heads; (d) π_h against that share (Pearson $\rho = 0.77$).

and B.2). At $\sigma = 0.62$ each head's scale-2 weight tracks its full-attention mass within 16 tokens ($\rho = 0.77$; Figure 3d). The mean weight is lower at ratio 2 than 4 (0.65 against 0.69 at $\sigma = 0.90$): wider windows take less local weight.

The table is stable and transfers. Tables fitted on disjoint prompt halves correlate at 0.989 over all cells (0.986 at scale 4). Applied at scale 4, the scale-2 table gives 0.205 against 0.200 for the scale-4 table, below every shared rule fitted at scale 4; the 1024^2 table scored at 1280^2 gives 0.0589 against 0.0585. The two resolutions emulate scale 2 with windows of 32 and 40 tokens, so the weights appear to depend mainly on the image-to-window ratio, as Eq. 6 assumes.

Per-head beats coarser rules; output error misses layout. Against the best hard switch, per-head mixing lowers the relative error from 0.092 to 0.057 at scale 2 and from 0.261 to 0.200 at scale 4. Per-layer tables reach 0.071 and 0.223, and one shared weight 0.077 and 0.230 (Figure B.3). Output error cannot see layout: fitted on output error alone, the best hard switch is local attention from the first step. The per-head mix without an opening loses 1.33 ImageReward at 4096^2 (Section 4.3). The global opening, or the 1024^2 stage of progressive SCA, therefore sets the layout.

4 Experiments

Prompts and seeds. We follow the field's protocol at 4096^2: the full 195-prompt Aesthetic-4K evaluation split with its 4K references (Zhang et al., 2025a), which AReA, DyPE, TIDE, and SigMa report (Rajabi et al., 2026a; Issachar et al., 2025; Liu et al., 2026; Zhao et al., 2026), and all 1000 HRGen-1K prompts, as AReA does. Seeds hash the prompt id and a seed label shared by all methods; labels 0, 1, and 2 each cover all 195 prompts. Four prompts (boats, Moai, woman and bear, underwater surfer) were used in development, which favours SCA slightly, and a fixed 46-prompt subset, chosen before any ablation ran, serves the ablations. Other sizes are 2048×4096 and 2048×6144 (height×width), plus 6144^2 and 8192^2 without the schedule.

Configurations and baselines. The main model is FLUX.1-dev (Black Forest Labs, 2024), and every baseline runs from its released code. Progressive SCA runs HRDiT's recipe of 30, 17, and 10 steps at 1024^2, 2048^2, and 4096^2 with guidance 3.5, 4.5, and 6.0, and SCA at the upper two stages (Section 2.5). Single-pass SCA, full attention, AReA (Appendix H), DyPE (Issachar et al., 2025), SEGA (Rajabi et al., 2026b), and SigMa (Zhao et al., 2026) share AReA's sampler: 28 Euler flow-matching steps at guidance 4.5, shift $\mu = 1.15$, bf16, and tiled VAE decoding. The multi-stage methods UltraImage (Zhao et al., 2025) (30 steps per stage) and HRDiT (Xue et al., 2026) use their own recipes (Appendix I). FLUX.1-Krea-dev (195 prompts, single-pass SCA against AReA) and FLUX.2-dev (50 prompts, against full attention, without the schedule) test other checkpoints, each with its own reach table from 16 prompts at 1024^2. One reach table per checkpoint serves every size and both configurations (Table F.2).

Table 1: **Main comparison at 4096^2.** FLUX.1-dev, 195 Aesthetic-4K prompts, seed label 0. IR ImageReward, HPS HPSv2.1 $\times 100$, Pick PickScore, M_c MUSIQ on 1024^2 crops, FID_p patch FID by our crop rule (†, Appendix G). **Best** and second within each group. Colours mark each row against progressive SCA: blue where the row is significantly better, orange where it is significantly worse (95% paired intervals, Table L.1). Time on one GH200: clean batch-1 median after a warm-up (AReA and full attention from sweep 1, single-pass SCA from sweep 2, HRDiT and progressive SCA from sweep 3 with AReA re-timed at 202.4 s) and campaign mean on shared nodes (camp., not ranked); n/m: not measured. At $n = 195$ FID is dominated by finite-sample bias; Table K.1 adds KID and all metrics. Key result: progressive SCA is significantly better than every single-pass baseline on every per-image column, matches HRDiT's preference scores in less than half its time, and trades crop-level detail for preference against single-pass SCA.

		Time (s)								
Method	Steps	clean↓	camp.	IR↑	HPS↑	Pick↑	CLIP↑	M_c↑	FID↓	$FID_p^\dagger$↓
Single pass, shared sampler (28 steps, guidance 4.5)										
Full attention	28	186.5	196.2	−2.18	7.60	17.94	15.37	33.1	312.4	252.2
DyPE	28	n/m	195.9	−0.10	20.55	20.51	24.92	50.3	205.5	171.9
SEGA	28	n/m	190.4	0.56	24.34	21.64	26.95	50.3	164.9	125.8
AReA	28	202.4	264.4	0.92	27.16	22.36	27.69	50.4	145.8	102.9
Single-pass SCA (ours)	28	**77.1**	78.4	**1.25**	**29.73**	**23.12**	**29.14**	**60.1**	**140.2**	**98.4**
Multi-stage, own recipe and budget										
UltraImage	30+30	n/m	308.5	1.28	30.77	23.33	28.84	41.7	142.0	109.5
HRDiT	30+17+10	107.0	110.4	**1.34**	30.92	**23.46**	29.12	53.5	**138.4**	**103.0**
Progressive SCA (ours)	30+17+10	**49.2**	50.7	1.32	**31.02**	23.41	**29.39**	**54.1**	141.9	107.7

Metrics and intervals. Preference and alignment scores are ImageReward, HPSv2, HPSv2.1, PickScore, CLIP score, and aesthetic score (Xu et al., 2023; Wu et al., 2023; Kirstain et al., 2023; Radford et al., 2021; Schuhmann, 2022). No-reference quality is MUSIQ on the full image and over four 1024^2 crops ($MUSIQ_c$), CLIP-IQA, and ARNIQA (Ke et al., 2021; Wang et al., 2022; Agnolucci et al., 2024). FID and KID compare against all 195 Aesthetic-4K references at 299^2, as AReA does, alongside the Inception score (Heusel et al., 2017; Salimans et al., 2016). Patch FID and KID use the same 1024^2 crops; patch FID is our own definition, never compared across papers. Per-image intervals are 95% paired percentile bootstraps over prompts with 2000 draws; FID, KID, and their patch versions use 1000 draws that also resample the 168 reference-image clusters (Appendix G). In every table, blue marks a value whose interval against its row's reference excludes zero in the row's favour and orange one that excludes zero against it; bold and underline rank rows within a group.

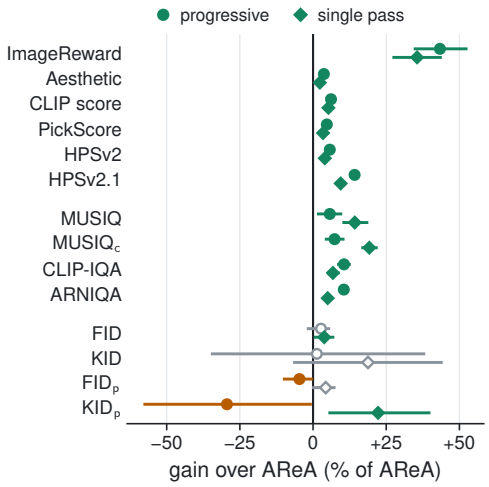

Figure 4: **Both modes beat AReA on every per-image metric.** Difference from AReA at 4096^2 ($n = 195$) as percent of AReA's mean, with 95% paired intervals; right is better (FID, KID, and their patch versions are negated). Orange: significantly worse; grey: the interval spans zero.

Timing. Clean latency is the batch-1 median of three prompts on one idle NVIDIA GH200 after an excluded warm-up image. At 4096^2 progressive SCA takes 49.2 s, HRDiT 107.0 s, and AReA 202.4 s, timed back to back on one GPU; earlier sweeps on the same hardware give single-pass SCA and the other sizes (Table D.2). Campaign means on shared nodes are reported separately and never ranked.

4.1 Main results

Progressive SCA beats AReA on all ten per-image metrics. ImageReward rises by $+0.40$ $[0.32, 0.49]$, PickScore by $+1.05$ $[0.93, 1.19]$, and HPSv2.1 by $+3.86$ $[3.41, 4.30]$, $\times 100$ (Table 1; Figures 4 and 5). Seeds 1 and 2 give the same result on all ten, with ImageReward $+0.37$ $[0.29, 0.46]$

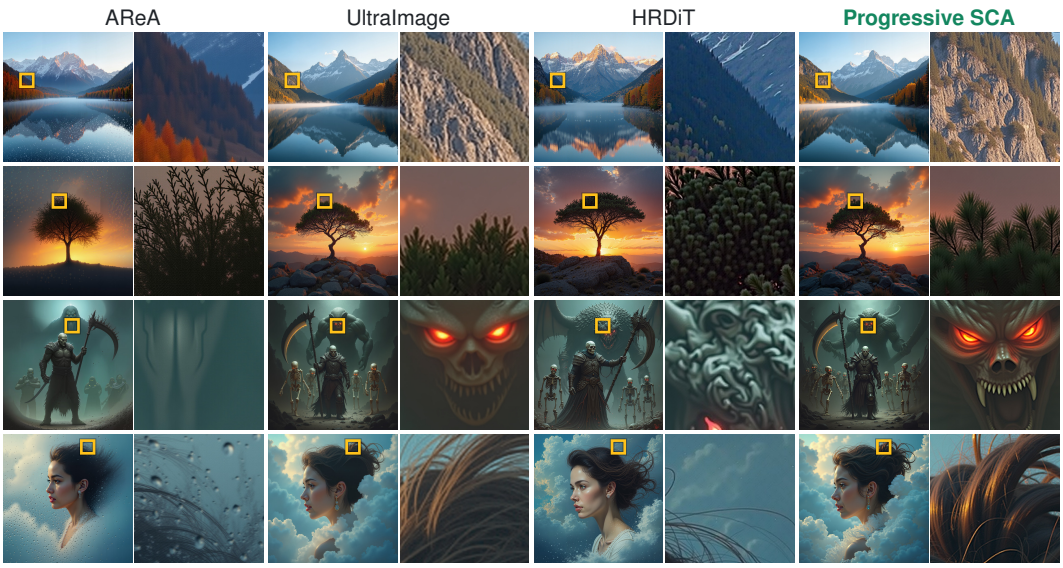

Figure 5: **Progressive SCA against AReA, UltraImage, and HRDiT at 4096^2.** Four Aesthetic-4K evaluation prompts at seed 0. Each image is followed by a 1:1 crop of the outlined window, and UltraImage's layouts match progressive SCA's. Figures S.2 and S.4 show more samples.

and $+0.35$ $[0.27, 0.45]$ and PickScore $+1.00$ and $+0.99$ (Table 3). On set-level scores at seed 0, FID (-3.9 $[-8.5, 3.1]$) and KID are unresolved, while patch FID ($+4.8$ $[0.1, 10.6]$) and patch KID are worse. The patch FID gap is significant at seeds 0 and 2 (seed 2: $+5.4$ $[0.8, 10.8]$) but not at seed 1 ($+1.5$ $[-2.7, 6.9]$). Progressive SCA also beats DyPE, SEGA, and full attention on all ten per-image metrics and UltraImage on nine, with the aesthetic score unresolved (Table L.1). Timed back to back on one GPU, it takes 49.2 s per image against AReA's 202.4 s, $4.1\times$ faster.

Progressive SCA matches HRDiT in less than half the time. Both runs follow HRDiT's schedule, guidance, and seeds; besides the attention, they differ in precision (HRDiT's released runner uses fp16, ours bf16) and in the runner. ImageReward differs by -0.02 $[-0.09, 0.03]$, HPSv2.1 by $+0.10$ $[-0.14, 0.35]$, and PickScore by -0.05 $[-0.14, 0.04]$; intervals that include zero do not establish equivalence. Progressive SCA is better on HPSv2, ARNIQA, and MUSIQ ($+1.3$ $[0.2, 2.4]$) and worse on CLIP-IQA (-0.9 $[-1.8, -0.1]$, $\times 100$). On set-level scores FID is unresolved, and HRDiT has lower KID, patch FID ($+4.6$ $[0.6, 7.7]$ for progressive SCA), and patch KID. In the same timing it takes 49.2 s per image against HRDiT's 107.0 s.

Against single-pass SCA, the progressive schedule trades fine detail for preference. Progressive SCA gains ImageReward ($+0.07$ $[0.01, 0.13]$), PickScore ($+0.29$ $[0.20, 0.38]$), and HPSv2.1 ($+1.29$, $\times 100$). It loses MUSIQ (-3.4 $[-5.1, -1.6]$) and MUSIQ_c (-6.0 $[-7.3, -4.6]$), and its patch FID is higher by $+9.2$ $[5.1, 13.2]$, with FID unresolved. Its 4096^2 stage runs 10 steps on an upsampled image, where single-pass SCA runs all 28 at the target (Figure L.2).

Single-pass SCA beats AReA on 12 of 15 metrics at $2.63\times$ the speed. The wins cover all ten per-image metrics and two of five set-level ones, among them ImageReward (0.92 to 1.25, $+0.33$ $[0.25, 0.41]$) and FID (145.8 to 140.2, -5.6 $[-10.6, -0.4]$). KID, patch FID, and IS are unresolved (Table K.2). It runs in 77.1 s against AReA's 202.4 s (clean batch-1 latency). DyPE and SEGA trail it on all ten per-image metrics (Table M.1), and SigMa on every per-image column of Table N.1. Our AReA run reproduces AReA's published FLUX.1-Krea-dev scores within 0.3 (Table H.1).

4.2 Other sizes and latency

Single-pass SCA is faster than every baseline we timed at every size; quality depends on the aspect ratio. At 2048×6144 it beats AReA on seven of 15 metrics and loses on the aesthetic score and patch FID; at 2048×4096 it wins three and loses three (Table K.3). Without the schedule

Table 2: **Ablations, single-pass SCA.** 4096^2, 46-prompt subset, columns as in Table 1. The shaded default gives absolute values; schedule rows give changes against it, base-operator rows against SCA without the schedule. Table K.5 adds all metrics. Key result: every base-operator change lowers HPSv2.1 and PickScore, while changes to the hand-set schedule move ImageReward by at most 0.05.

Variant	IR	HPS	Pick	M_c
Single-pass SCA	1.17	30.07	23.02	60.7
Schedule and bridge				
without the schedule	−0.05	−0.09	−0.06	+1.7
bridge only	+0.01	+0.07	−0.04	+0.5
first window 160	−0.01	−0.05	−0.01	−0.2
2,048 bridge tokens	−0.02	−0.09	−0.04	−0.1
five-step schedule	−0.01	−0.05	−0.02	−0.2
Base operator, against SCA without the schedule				
no opening (v0)	−1.33	−4.07	−2.18	+2.5
local only	−0.36	−2.86	−1.20	−0.3
global only	−0.40	−5.66	−1.44	−11.7
shared $\pi(\sigma)$	−0.09	−1.83	−0.38	−10.4
per-layer $\pi_\ell(\sigma)$	−0.05	−0.70	−0.16	−2.4
mean pooling	−1.52	−9.75	−2.60	−4.2

Table 3: **Generality.** Paired differences at 4096^2 over n prompts, 95% intervals below them; HPS is HPSv2.1 $\times 100$, Pick PickScore. The progressive sampler does not run FLUX.1-Krea-dev. Tables M.3 and M.4 add all metrics, Table M.2 the emulation-budget and gate checks. Key result: both configurations beat AReA at all three seeds and on HRGen-1K; on Krea, single-pass ImageReward is unresolved.

Setting	ΔIR	ΔHPS	ΔPick
Progressive SCA minus AReA			
Seed 0	+0.40	+3.86	+1.05
$n = 195$	[0.32, 0.49]	[3.41, 4.30]	[0.93, 1.19]
Seed 1	+0.37	+3.67	+1.00
$n = 195$	[0.29, 0.46]	[3.26, 4.06]	[0.88, 1.12]
Seed 2	+0.35	+3.61	+0.99
$n = 195$	[0.27, 0.45]	[3.20, 4.03]	[0.87, 1.12]
HRGen-1K	+0.90	+6.59	+1.40
$n = 1000$	[0.84, 0.96]	[6.31, 6.88]	[1.33, 1.46]
Single-pass SCA minus AReA			
HRGen-1K	+0.76	+4.60	+1.09
$n = 1000$	[0.71, 0.82]	[4.37, 4.83]	[1.04, 1.14]
Seed 1	+0.31	+2.49	+0.77
$n = 195$	[0.24, 0.38]	[2.16, 2.83]	[0.67, 0.88]
Seed 2	+0.26	+2.53	+0.74
$n = 195$	[0.20, 0.34]	[2.13, 2.90]	[0.62, 0.86]
FLUX.1-Krea	+0.02	+0.91	+0.17
$n = 195$	[−0.02, 0.06]	[0.70, 1.10]	[0.11, 0.24]
Without the schedule, minus full attention			
FLUX.2-dev	+0.46	+1.84	+0.56
$n = 50$	[0.32, 0.60]	[1.10, 2.55]	[0.29, 0.86]

it also wins 14 of 15 metrics at 6144^2 (Table M.3). At 8192^2, 12 exploratory prompts favour it on six of ten scores, with grid artifacts (Appendix O).

Single-pass time grows nearly linearly with pixel count. Fitting clean seconds $\propto N^b$ over the N image tokens gives $b = 0.99$ for single-pass SCA and 1.12 for AReA (Table D.2; Figure D.2). The schedule and the bridge add 8% at 4096^2 (77.1 against 71.1 s without them), timed back to back on one GPU. At 8192^2 single-pass SCA takes 309.7 s against AReA's 952.0 s, 3.07× faster.

4.3 Ablations

Table 2 changes one component of single-pass SCA at a time (examples in Figure S.11). **The opening phase is essential.** Without it (v0), ImageReward falls by 1.33 $[1.06, 1.60]$, while CLIP-IQA and ARNIQA rise (Table K.5). **The per-head mix carries global context after the opening.** Local-only attention in the last 20 steps costs 0.36 $[0.21, 0.52]$ ImageReward. **Per-head weights beat shared and per-layer weights, in the order output error predicts.** One shared weight per noise level lowers ImageReward by 0.09 $[0.01, 0.17]$ and $MUSIQ_c$ by 10.4, and per-layer weights still lose on HPSv2, HPSv2.1, PickScore, ARNIQA, and $MUSIQ_c$ (Section 3). **Neither branch works alone.** Global only loses everywhere; mean pooling lowers ImageReward by 1.52. **The hand-set schedule and bridge change per-image scores little.** Without them, $MUSIQ_c$ is 1.7 higher and no other column moves significantly. Of three variants of their settings, 2,048 bridge tokens lowers HPSv2.1 and PickScore and a five-step schedule lowers $MUSIQ_c$ by 0.2.

4.4 Generality and sensitivity

Progressive SCA on more seeds and prompts. With seeds 1 and 2 and on all 1000 HRGen-1K prompts, progressive SCA again beats AReA on all ten per-image metrics, on HRGen-1K with ImageReward +0.90 $[0.84, 0.96]$ (Table 3). **Single-pass SCA on FLUX.1-dev.** Gains over AReA hold

with seeds 1 and 2 and on HRGen-1K (Tables 3 and M.4). **FLUX.1-Krea-dev, AReA's checkpoint, single pass: quality on par in a third of the time.** Over 195 prompts single-pass SCA wins seven of ten per-image metrics and loses none, ImageReward is unresolved ($+0.020$ $[-0.020, 0.063]$), and patch FID is worse ($+3.4$ $[1.6, 7.1]$; Table M.3). **FLUX.2-dev, without the schedule.** SCA beats full attention on nine of ten per-image metrics in half the time. **Sensitivity, without the schedule.** Reach tables fitted on 4 to 16 prompts move ImageReward by ≤ 0.012 (Table M.2).

5 Related work

Training-free high-resolution sampling. Methods alter convolutions, spectra, or attention temperature (He et al., 2023; Zhang et al., 2023; Huang et al., 2024; Jin et al., 2023). Others fuse global and patch paths (Du et al., 2023; Yang et al., 2024), by frequency in FreeScale (Qiu et al., 2024). Others add native-resolution guidance (Du et al., 2024; Bu et al., 2025; Koh et al., 2025) or adapt RoPE (Zhuo et al., 2024; Issachar et al., 2025; Zhao et al., 2026; Rajabi et al., 2026b; Liu et al., 2026). Multi-stage samplers draw the image at native size first and refine it at larger sizes (Du et al., 2023; Xue et al., 2026); progressive SCA uses HRDiT's schedule as released and changes only its attention. Of the closest methods (Table J.1, Appendix J), AReA (Rajabi et al., 2026a) sets one schedule for all heads by hand and adds bridge tokens; SCA borrows its opening, block-centre keys, and bridge selection rule. In SCA's notation AReA's phases set every head's window weight to 0, then 1. ResDiT fuses position-scaled global and native-size patch attention between global-only and local-only phases (Ma et al., 2025).

Head- and layer-specific attention. HRDiT searches per-head scopes under a cost budget for each target size (Xue et al., 2026); UltraImage types heads by entropy at run time and noted stable local and global head roles before us (Zhao et al., 2025). Sparse attention varies windows, spans, or key scales per layer or head in diffusion transformers (Yuan et al., 2024; Xi et al., 2025; Zhang et al., 2025b), language models (Fu et al., 2024; Xiao et al., 2024), and vision transformers (Pan et al., 2022). These settings are often fitted against dense attention at the size where it runs, which at 4096^2 is the failure; SCA fits once, at the training resolution. Our local branch is neighbourhood attention (Hassani et al., 2022). Head roles, two branches, a global opening, and shrinking windows have precedents; reach emulation and the measured comparison of heads and steps are ours.

6 Limitations and conclusion

Limitations. Fidelity. Progressive SCA trades fine detail for preference against single-pass SCA (MUSIQ_c -6.0, patch FID $+9.2$). Its patch FID and patch KID are worse than HRDiT's, and its patch FID is higher than AReA's at two of three seeds (Section 4.1). In single-pass mode, small additional figures occasionally appear in scenes with people or animals; the progressive configuration reduces this to the level of the multi-stage baselines. **Hand-set settings.** The opening, window sizes, and bridge are set by hand because output error cannot see layout: fitted on output error alone, the best hard switch is local attention from the first step. We chose the window sizes on the 46-prompt subset, part of the evaluation set. **Other limits.** Progressive SCA depends on HRDiT's schedule, and its sampler does not support FLUX.1-Krea-dev. The 6144^2, 8192^2, FLUX.2-dev, and sensitivity results use SCA without the schedule. Reach emulation omits the bridge and sees 16 times fewer keys per branch than the target; the deployed operator's error is untested. Metrics are automatic; intervals ignore multiple comparisons.

Conclusion. A head's reach can be measured once at the training resolution, where full attention is still correct, by shrinking both views to the target's image-to-window ratio. Measured this way, reach varies far more between heads than between steps, so SCA gives each head its own fitted mix where AReA gives all heads one schedule. Inside HRDiT's schedule, SCA beats AReA on all ten per-image metrics at 4096^2 for three seeds and on HRGen-1K at $4.1\times$ its speed and matches HRDiT's preference scores in less than half its time. In one pass it runs $2.63\times$ faster than AReA and keeps more fine detail. Code: github.com/shreshthsaini/scale-covariant-attention.

REPRODUCIBILITY STATEMENT

Section 4 states the protocol, including timing (Section 4.2), and Appendix G defines every metric and interval. Appendix B gives the reach-emulation geometry, statistics, and noise bins; Algorithms 1, 2, and 3 give the procedures, and Appendix E gives them in full. Appendix L gives the progressive protocol and every paired comparison of progressive SCA. Appendix F lists every hyperparameter, including the window schedule, and Appendix D the kernel layout, numerical checks, and position rounding. Appendices H and I give each baseline's code revision, recipe, and seeds, and Appendix Q the hardware and logged compute. Code, reach tables, per-image metrics, and the paired-difference files behind every table will be released in the code repository (link in the abstract).

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APPENDIX CONTENTS

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A PROOFS

This section proves Lemma 1 and Propositions 1 to 3 of Section 2 and treats text-image offsets in Remark 1.

Lemma 1 (Softmax split)**.** *Let the keys of a query be partitioned into a near set $\mathcal{K}^{\text{near}}$ and a far set $\mathcal{K}^{\text{far}}$, with logits ℓ_j. Then $O^{\text{full}} = m\,O^{\text{near}} + (1-m)\,O^{\text{far}}$ with $m = \sum_{j\in\mathcal{K}^{\text{near}}} e^{\ell_j} / \sum_j e^{\ell_j}$ and $O^{\text{near}}, O^{\text{far}}$ the softmax outputs restricted to $\mathcal{K}^{\text{near}}$ and $\mathcal{K}^{\text{far}}$.*

Proof. Write $Z^{\text{near}} = \sum_{j\in\mathcal{K}^{\text{near}}} e^{\ell_j}$, so that $Z^{\text{near}}O^{\text{near}} = \sum_{j\in\mathcal{K}^{\text{near}}} e^{\ell_j} v_j$, and likewise for the far set. Then $O^{\text{full}} = (Z^{\text{near}}O^{\text{near}} + Z^{\text{far}}O^{\text{far}})/(Z^{\text{near}} + Z^{\text{far}})$, and $Z^{\text{near}}/(Z^{\text{near}} + Z^{\text{far}}) = m$. □

The lemma motivates the two-branch form; SCA's key sets are not a partition. Block centres can fall inside the window, text keys enter both branches, and a bridge token inside the window appears twice in the local softmax, which doubles its weight there.

Proof of Proposition 1. Global operator. Image queries sit at $c(p_i) \in \mathcal{G}_0 = [n_0]^2$ and text queries at $0 \in \mathcal{G}_0$; subsampled keys sit at $b \in \mathcal{G}_0$ and text keys at 0. Any two points of $[n_0]^2$ differ by at most $n_0 - 1$ per axis. The key count is $T + |\mathcal{G}_0| = T + n_0^2$.

Local operator. Fix one axis of length $H \geq w$ and a tile $[t\tau,\, t\tau + t)$ with $t \leq w$. The unclamped start $a' = t\tau + \lfloor t/2 \rfloor - \lfloor w/2 \rfloor$ satisfies $a' \leq t\tau$ because $\lfloor t/2 \rfloor \leq \lfloor w/2 \rfloor$, and $a' + w \geq t\tau + t$ because $w - \lfloor w/2 \rfloor = \lceil w/2 \rceil \geq \lceil t/2 \rceil = t - \lfloor t/2 \rfloor$. If $a' < 0$ the window is $[0, w)$, and $w \geq a' + w \geq t\tau + t$. If $a' > H - w$ the window is $[H - w, H)$, with $H - w < a' \leq t\tau$ and $H \geq t\tau + t$. In every case the window contains the tile, so it contains each query of the tile, and two positions in a window of side $w \leq n_0$ differ by at most $n_0 - 1$. The proposition makes no claim about the bridge keys, which keep their true positions. □

Remark 1 (Text-image offsets). FLUX.1 places all text tokens at position 0, so the offset between an image query at p_i and a text key is p_i. In the global operator the query sits at $c(p_i) \in [n_0]^2$, inside the trained range. In the local operator it keeps p_i, up to $sn_0 - 1$ per axis, exactly as in full attention and AReA's windows at the target. Mapping text offsets through $c(\cdot)$ in the local operator is a possible refinement that we have not tested.

Proof of Proposition 2. Full attention scores every query against every key, $(N + T)^2$ pairs. The global operator scores $N + T$ queries against $T + |\mathcal{G}_0|$ keys with $|\mathcal{G}_0| = N/(s_H s_W) = n_0^2$. The local operator scores N image queries against T text keys, R bridge keys, and w^2 window keys. The sum is at most $(2N + T)(T + R + w^2 + N_0)$, which is $O(N(T + R + w^2 + N_0))$ for fixed n_0, w, R, and T. Value aggregation has the same pair counts. In the opening phase the local term is absent. The FlexAttention kernel visits whole blocks and masks the extra pairs. A query block of 8×16 tokens spans two tiles whose windows are 8 columns apart, so it visits a 64×72 key region, 12.5% more than one window, plus partial blocks at clamped borders. This changes the constant, not the order. □

Proof of Proposition 3. $J(\pi) = E - 2\pi B + \pi^2 A$ with $A \geq 0$. If $A = 0$ then $O^{\mathrm{loc}} = O^{\mathrm{glob}}$ on every summand, so $B = 0$ and J is constant; any π, in particular $\pi^\star = 1$, is optimal and the bound holds trivially. If $A > 0$, J is a strictly convex quadratic with unconstrained minimizer B/A, and its minimizer on $[0, 1]$ is the projection $\mathrm{clip}(B/A, 0, 1)$. For any π',

$$J(\pi') - J(\pi^\star) = A(\pi' - \pi^\star)^2 + 2(\pi' - \pi^\star)(A\pi^\star - B).$$

If $0 < \pi^\star < 1$ then $A\pi^\star = B$ and the second term vanishes. If $\pi^\star = 0$ then $B \leq 0$, so $A\pi^\star - B \geq 0$ and $\pi' - \pi^\star \geq 0$. If $\pi^\star = 1$ then $B \geq A$, so $A\pi^\star - B \leq 0$ and $\pi' - \pi^\star \leq 0$. In both clipped cases the second term is nonnegative.

Any rule that assigns one weight per cell is a feasible π' in every cell. Examples are a per-layer table, a weight shared per noise level, a hard switch, or a table fitted elsewhere. Its total error is therefore at least the per-head table's plus $\sum_{\text{cells}} A(\pi' - \pi^\star)^2$. A table interpolated in σ between the mean sampled noise levels of occupied bins is not constant within a bin; the statement applies to the bin-constant table used to score rules in Section 3. □

B REACH EMULATION: DETAILS AND STATISTICS

This section expands Section 3 and Figure 3. Table B.1 backs the section's first finding, that reach varies by head and not by step. Figure B.1 shows the reach tables, and Figure B.2 every head's $\pi_h(\sigma)$ with the split-half agreement. Figures B.3 and B.4 give the output errors per rule and per noise level, and Figure B.5 every head's weight per block.

Table B.1: **Reach is set by the head.** Variance of the share of each head's image attention within 8 or 16 tokens of the query (Chebyshev distance), over all (block, head, noise bin) cells. It is split into the head (between blocks plus between heads of one block) and the noise level within a head; the two sum to 100%. Avg. is the variance of the head-averaged curve over noise levels, the part one schedule shared by all heads could follow. 256 random queries per block and step, full attention at each row's resolution. AReA's statistic is the distance to dominant keys; ours is the mass share, which sets the output by Lemma 1. Key result: after the gate the head carries at least 97.0% of the variance in every model and size.

		within 8 tokens (%)			within 16 tokens (%)		
Model and size	Prompts	Head	Noise	Avg.	Head	Noise	Avg.
After the gate ($\sigma < 0.89$)							
FLUX.1-dev, 1024^2	32	97.0	3.0	1.02	97.5	2.5	0.65
FLUX.1-dev, 1280^2	16	97.4	2.6	0.75	97.8	2.2	0.46
FLUX.1-Krea-dev, 1024^2	16	97.2	2.8	1.03	97.6	2.4	0.79
FLUX.2-dev, 1024^2	16	99.0	1.0	0.10	99.1	0.9	0.04
All noise levels							
FLUX.1-dev, 1024^2	32	92.2	7.8	4.02	93.3	6.7	3.29
FLUX.1-dev, 1280^2	16	93.3	6.7	3.09	94.3	5.7	2.44
FLUX.1-Krea-dev, 1024^2	16	93.8	6.2	3.03	94.7	5.3	2.55
FLUX.2-dev, 1024^2	16	96.3	3.7	1.29	97.0	3.0	0.76

Statistics. For every emulated scale, block, head, and noise bin, the pass stores A, B, E, the error of the local branch alone $\sum \|O^{\mathrm{full}} - O^{\mathrm{loc}}\|^2$, $\sum \|O^{\mathrm{full}}\|^2$, and the query count. These are float64 sums over image queries and channels. The sampler's trajectory is that of unmodified full attention; the emulated operators only observe. Statistics are written per prompt and merged by summation, so prompts can be added later and split-half fits need no extra pass. Text-query outputs are not in the objective, because SCA sends text queries through the global branch only.

Geometry at emulated scale. At 1024^2 the grid is 64×64. Emulated scale 2 uses a 32×32 window, 2×2 subsampling onto a 32×32 grid, and 4×4 query tiles. Emulated scale 4 uses a 16×16 window, 4×4 subsampling onto a 16×16 grid, and 2×2 tiles. At 1280^2 the grid is 80×80, and emulated scale 2 uses a 40×40 window, 2×2 subsampling onto a 40×40 grid, and 5×5 tiles.

Noise bins. With 28 steps and $\mu = 1.15$, σ runs from 1.0 to 0.105, so the lowest of the 14 bins is empty and copies its neighbour; 13 bins are populated. Under this schedule 8 steps have $\sigma > 0.89$ (the ninth has $\sigma = 0.888$) and 13 have $\sigma > 0.80$.

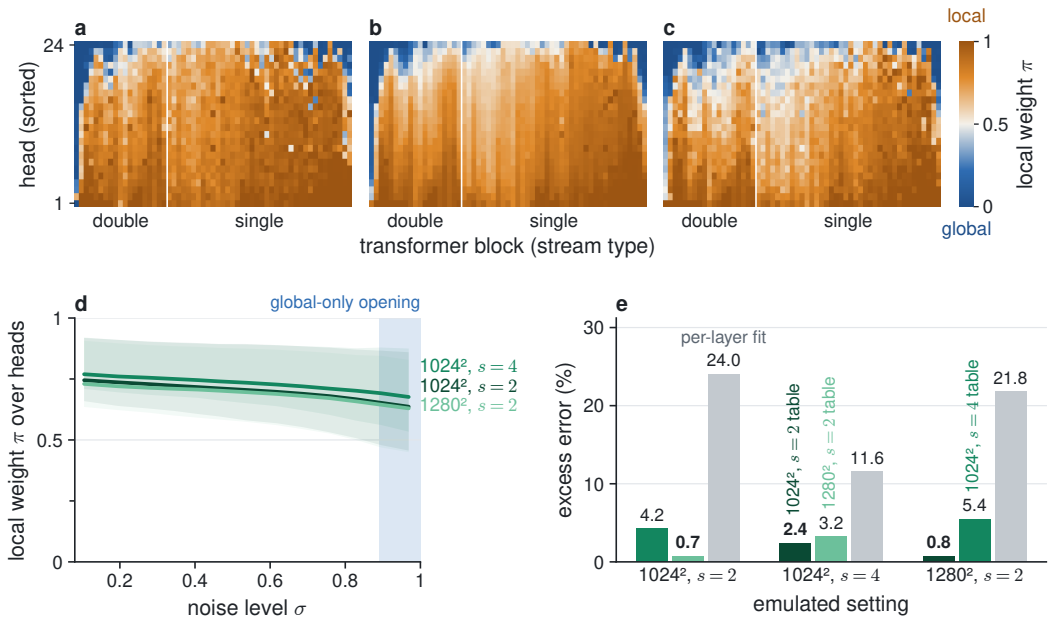

Figure B.1: **Heads differ widely, yet a reach table transfers across scale and resolution.** (a to c) $\pi_h(\sigma)$ of the scale-4 reach table at $\sigma = 0.10, 0.46, 0.83$ ($\sigma = 1$: pure noise); (d) mean and interquartile range across heads, shaded: global-only opening. (e) Excess error over each setting's own table: at most 5.4% transferred, 12 to 24% per-layer.

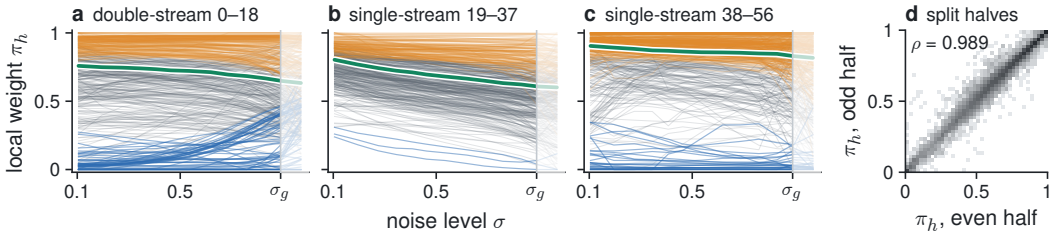

Figure B.2: **Heads use different mixes.** (a to c) $\pi_h(\sigma)$ of all 1368 heads at emulated scale 2; amber: mean above 0.8, blue: below 0.2, green: median; $\sigma > \sigma_g$ shaded. (d) Tables from disjoint prompt halves agree (Pearson $\rho = 0.989$).

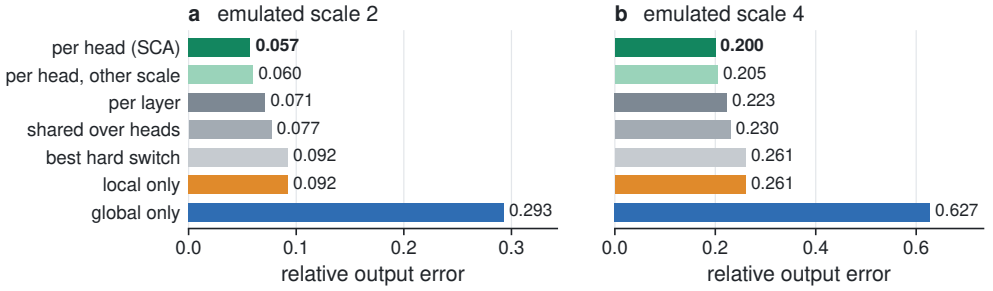

Figure B.3: **Per-head weights explain full attention best, and transfer across scales.** Relative output error per mixing rule at emulated scales 2 (a) and 4 (b), 32 prompts; "other scale" swaps the scale-2 and scale-4 tables; the best hard switch ($\sigma^\star = 1$) equals local only. In-sample and held-out errors agree within 0.0003.

No global phase in output energy. The best single shared weight falls from 0.88 at $\sigma = 0.10$ to 0.72 at $\sigma = 0.97$, and the mean per-head weight from 0.74 to 0.64 (Figure B.4). The median within-block range of the per-head weights is 0.66 to 0.84 at every noise level (Figure B.5). Among hard switches from global to local, the best threshold is $\sigma^\star = 1$, which equals local only. A switch at $\sigma^\star = 0.89$ between our two branches raises the error from 0.092 to 0.152 at scale 2 and from 0.261

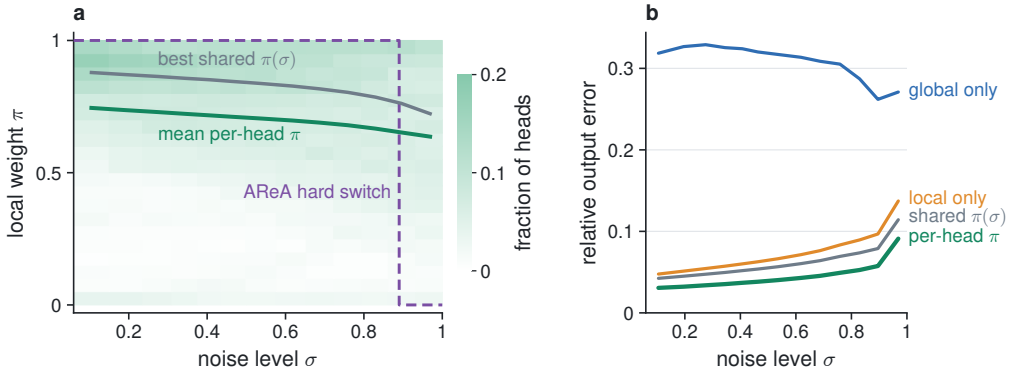

Figure B.4: **In output energy, no noise level prefers the global branch.** (a) Fraction of heads per π and noise bin, with mean $\pi_h(\sigma)$, best shared $\pi(\sigma)$ and AReA's switch; scale 2, 32 prompts, $\sigma = 1$: pure noise. (b) Error per noise bin: local beats global everywhere; per-head mixing beats both.

to 0.376 at scale 4. In the noisiest bin, the global branch alone has relative error 0.27 at scale 2, the local branch alone 0.14, and the per-head mix 0.09. Generation needs the opening (Section 4.3); output error cannot see layout.

Attention mass. For 256 random image queries per block and step, the pass also records the softmax mass on text keys and on image keys within Chebyshev distance 2, 4, 8, and 16. At 1024^2, averaged over heads and queries, text receives 15 to 19% of the mass. Image keys within distance 16 receive 52% at $\sigma = 0.10$ and 40% at $\sigma = 0.97$, and keys outside that 33×33 neighbourhood still receive 29% and 41%. A purely local receptive field cannot account for any step, even at native resolution. Figure 3b and d use the per-head shares in the noise bin at $\sigma = 0.62$; panel b colours mass density on a log scale clipped at 0.1 and 100 times the uniform density.

Emulation budget. We merge the stored sufficient statistics from the first 4, 8, or 16 emulation prompts and rerun the 46 ablation prompts with each resulting table. Table M.2 reports generation-level differences from the 32-prompt table, all for SCA without the schedule. The test reuses the emulation statistics but uses separate prompts for generation evaluation; it is not a held-out output-error benchmark.

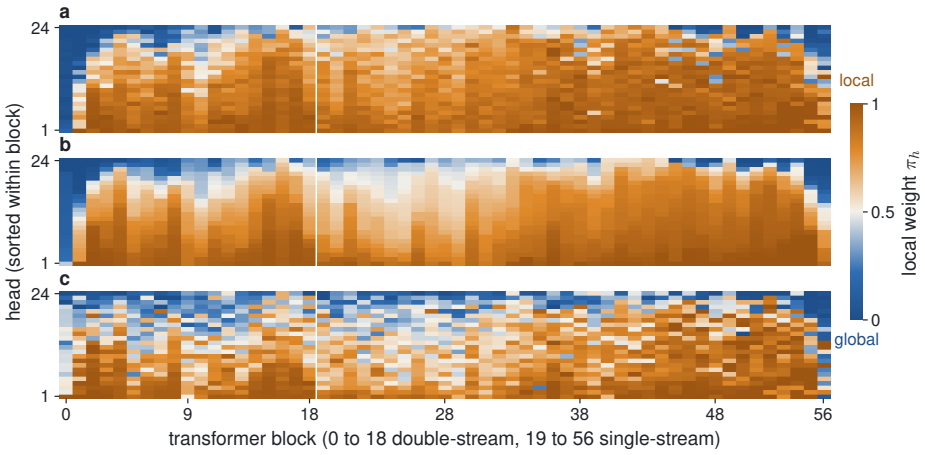

Figure B.5: **Heads in the same block differ widely at every noise level.** $\pi_h(\sigma)$ of the scale-2 table at $\sigma = 0.10$ (a), 0.54 (b), 0.97 (c), 57 blocks by 24 heads, 32 prompts, heads sorted per block by (b). Median within-block range 0.66 to 0.84; Figure B.1 shows the scale-4 reach table that sampling reads.

C EMULATION PROMPTS

This section lists the prompts behind the reach emulation of Section 2.4 and Algorithm 1. The 1024^2 table uses prompts 1 to 32 and the 1280^2, FLUX.1-Krea-dev, and FLUX.2-dev tables prompts 1 to 16; prompts 33 to 48 are spares that no reported table uses. None of them is an evaluation prompt.

1. A candid street portrait of an elderly fisherman with a weathered face and a knitted wool cap, harbor in the background, overcast light
2. Two children laughing while running through a lawn sprinkler on a summer afternoon, backlit by the sun
3. A professional headshot of a young woman with curly red hair and freckles, soft studio lighting, neutral grey backdrop
4. A jazz trio performing in a dim basement club, saxophone player in the foreground, warm stage lights and haze
5. A chef plating a dessert in a busy restaurant kitchen, steam rising, stainless steel counters
6. A ballet dancer mid-leap in an empty rehearsal studio with tall windows and wooden floors
7. A red fox standing in fresh snow at the edge of a birch forest, early morning light
8. A macro photograph of a honeybee covered in pollen on a purple lavender flower
9. A herd of elephants crossing a dusty savanna river at sunset, long shadows
10. A tabby cat sleeping curled up on a stack of old books next to a window
11. A school of tropical fish swimming around a coral reef, sun rays through clear blue water
12. A bald eagle perched on a dead pine branch above a misty lake
13. A neon sign reading "OPEN 24 HOURS" in the window of a rainy diner at night, reflections on the wet sidewalk
14. A hand-painted wooden sign that says "Fresh Eggs" leaning against a farm fence, chickens nearby
15. A vintage movie theater marquee with the words "NOW SHOWING" lit by hundreds of bulbs
16. A chalkboard menu in a cafe listing coffee drinks and prices in neat handwriting
17. A highway road sign reading "Exit 42 Pine Valley" against a clear blue sky
18. A bookstore storefront with gold lettering "Second Chapter Books" on the glass, autumn leaves on the ground
19. A wide alpine valley with a winding river, snow-capped peaks and wildflower meadows, midsummer
20. Sand dunes in the desert at golden hour with sharp ridges and wind ripples
21. A rugged coastline with sea stacks and crashing waves under a stormy sky
22. A terraced rice field on a hillside in morning fog, farmers in conical hats
23. The northern lights over a frozen lake with a small wooden cabin, stars visible
24. A dense tropical rainforest with a waterfall and hanging vines, light filtering through the canopy
25. A cozy Scandinavian living room with a fireplace, sheepskin rug, and large windows showing snow outside
26. A modern open-plan kitchen with a marble island, pendant lights, and potted herbs
27. A cluttered artist studio with canvases, paint tubes, brushes in jars, and afternoon sunlight
28. A grand library reading room with long oak tables, green lamps, and arched ceilings
29. A small Japanese tatami room with a low table, sliding paper doors, and a garden view
30. A product photograph of a matte black wristwatch on a slate surface, dramatic side light
31. A pair of white running shoes floating against a pastel gradient background, studio product shot
32. A glass bottle of olive oil with a sprig of rosemary on a rustic wooden table, soft daylight
33. A flat lay of a leather notebook, fountain pen, glasses, and a cup of espresso on a desk
34. A ceramic teapot with a blue glaze and matching cups arranged on a linen tablecloth
35. A crowded night market in Taipei with food stalls, lanterns, and hundreds of people
36. A packed football stadium during a night match, floodlights and cheering fans
37. Commuters crossing a huge intersection in Tokyo during rush hour, seen from above
38. A music festival crowd at sunset with raised hands and confetti in the air
39. A busy train station concourse with travelers, departure boards, and a glass roof
40. A gothic cathedral facade with intricate stone carvings and a rose window, low angle
41. A row of colorful Victorian townhouses on a hilly street with a cable car
42. A brutalist concrete apartment block with repeating balconies under a pale sky
43. A glass skyscraper reflecting clouds, shot looking straight up from the street
44. A traditional Chinese pagoda beside a lake with cherry blossoms in full bloom
45. An old stone bridge over a canal in Venice with gondolas and laundry lines
46. An astronaut riding a horse across a red desert planet with two moons, cinematic
47. A detailed watercolor illustration of a lighthouse on a rocky island with seagulls
48. A steampunk airship floating above a Victorian city at dusk, intricate brass details

D IMPLEMENTATION DETAILS AND COST

This section expands Section 2.2 and Figure 2. Figure D.1 draws the two operators, the gate, and the mix at component level, together with the reach table for all 57 blocks. Table D.1 gives attention cost, and Table D.2 and Figure D.2 end-to-end latency; Appendix E writes out the corresponding algorithms.

FlexAttention window. Image tokens are reordered tile-major, so that a sparse block is a small spatial tile. Query blocks cover 8×16 tokens (128 queries) and key blocks 8×8 (64 keys), shifted so that unclamped windows start on a block edge. The T text keys and the 1,024 bridge keys come first, as a prefix of $T + 1{,}024$ keys shared by every query; the bridge keys keep their true positions. The wider windows of the four steps after the opening use the same tiling with more key blocks per query block. The mask function states window membership per query-key pair, so the result does not depend on the block layout, only the amount of skipped work does. For the operator without the schedule on GH200, 8×16 query blocks with 8×8 key blocks were fastest at 2048^2, 4096^2, and

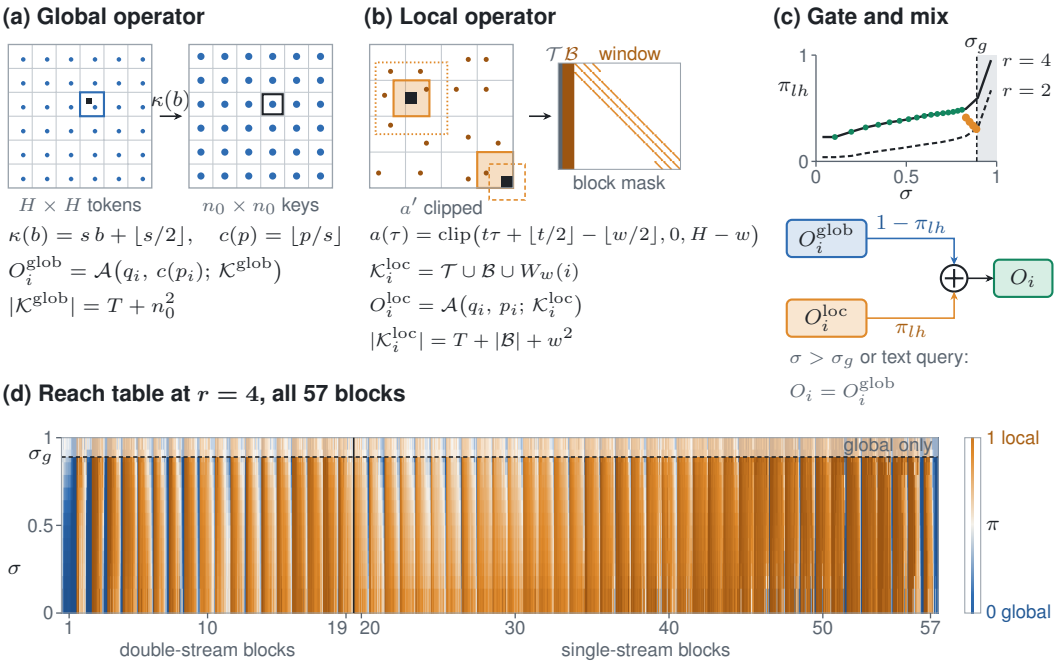

Figure D.1: **The operators at component level.** On a toy grid ($H = 24$, $s = 4$, $n_0 = 6$, $t = 2$), (a) the global operator keys each block by its centre token. (b) A query tile attends to the text keys, the bridge tokens (dots), and a clamped window whose side shrinks from $2n_0$ to n_0, shown with its block mask (Equations 2 to 4). (c) One head mixes the two by π_{lh} (solid: $r = 4$; dashed: $r = 2$; amber: wide-window steps; Equation 5). (d) The $r = 4$ table holds all 57×24 heads by 14 noise bins.

2048×6144. RoPE, subsampling, both branches, and the mix compile into one graph per layer; the mix is in fp32.

Numerical checks. The 18 CPU tests in `tests/test_sca.py` run in float32. They cover full-mode agreement with stock FLUX attention, both branches against brute-force Equations 2 and 3 without the bridge, and full-grid and factor-1 identities. Reach emulation when the local branch is exact, the gate, and FlexAttention against an explicit gather reference complete them. Further CPU tests cover the bridge selection (Equation 4), the local branch with bridge keys against brute force, the window schedule, and the reduction to the operator without the schedule when these options are off. A separate GPU check in `scripts/bench_attn.py` compares the gather and FlexAttention implementations in bf16. The recorded maximum absolute difference is 0.015625, and both have the same maximum error against the fp32 reference on the tested grids. This is an observed absolute-error check; no per-element ULP bound was measured.

Position ids in bf16. The FLUX.1 pipeline builds image position ids in bf16, which cannot represent every integer above 256. Our frozen runs retain these rounded ids for RoPE. Passing the true grid from the sampler fixes window membership and subsampling geometry, but does not restore exact RoPE coordinates: the coarse-query positions are also computed from the rounded ids. At a 384-token axis, index 383 rounds to 384 and its scale-6 coarse coordinate becomes 64; at a 512-token axis, index 511 rounds to 512 and its scale-8 coarse coordinate also becomes 64. A local 64-token window can likewise have a rounded endpoint difference of 64 instead of the ideal bound 63. Proposition 1 therefore applies exactly to the mathematical operator, not to these finite-precision positions at 2048×6144, 6144^2, or 8192^2. The 4096^2 grid uses indices 0–255, which bf16 represents exactly. FLUX.2 uses integer position ids. Rebuilding FLUX.1 positions in higher precision would require a separate controlled experiment; the reported runs are unchanged.

Non-divisible grids. When H/n_0 is not an integer, the subsampling factor per axis is the divisor of H nearest to H/n_0, and the tile side the divisor nearest to t. Neither case arises at the sizes we evaluate.

Table D.1: **Cost.** Top: analytic attention FLOPs per image (57 blocks, 24 heads, head dimension 128, 28 steps, $T = 512$), with full attention over each row in parentheses. Single-pass SCA runs the global branch alone for the 8 opening steps, local windows of side 128, 112, 96, and 80 for the next four, and the native window $n_0 = 64$ for the last 16. Every local window adds $T + 1{,}024$ text and bridge keys. Without the schedule the window is always n_0 and there is no bridge; "gated" skips the local branch in the first 8 steps. Bottom: measured time of one attention layer for SCA without the schedule (both branches and the mix) in bf16 on GH200, minimum over interleaved runs, with the speedup over full attention in parentheses. Key result: at 4096^2 single-pass SCA needs 6.9 times fewer attention FLOPs than full attention.

FLOPs per image grid, (s_H, s_W)	2048×4096 128×256, (2, 4)	4096×4096 256×256, (4, 4)	2048×6144 128×384, (2, 6)
Full attention	2.17×10^{16}	8.56×10^{16}	4.84×10^{16}
SCA without the schedule, no gate	5.97×10^{15} (3.6)	1.19×10^{16} (7.2)	8.93×10^{15} (5.4)
SCA without the schedule, gated	5.12×10^{15} (4.2)	1.02×10^{16} (8.4)	7.66×10^{15} (6.3)
Single-pass SCA	6.24×10^{15} (3.5)	1.24×10^{16} (6.9)	9.34×10^{15} (5.2)
One layer, bf16 (ms) token grid	2048^2 128×128	4096^2 256×256	2048×6144 128×384
Full attention (SDPA)	7.1	87.3	49.6
SCA without the schedule	5.2 (1.4)	18.7 (4.7)	14.0 (3.5)

Table D.2: **Seconds per image** of single-pass SCA, AReA, and full attention: batch 1 on an idle GH200, 28 steps with text encoding and VAE decoding. Median of three timed images after an excluded warm-up; full attention not run at 8192^2. Key result: single-pass SCA is the fastest method at every size, 2.6 times faster than AReA at 4096^2.

Size (N tokens)	Full attention	AReA	Single-pass SCA
2048^2 (16,384)	22.9	41.0	**19.8**
3072^2 (36,864)	73.9	114.3	**44.3**
4096^2 (65,536)	186.5	202.4	**77.1**
6144^2 (147,456)	762.7	482.3	**174.7**
8192^2 (262,144)	not run	952.0	**309.7**

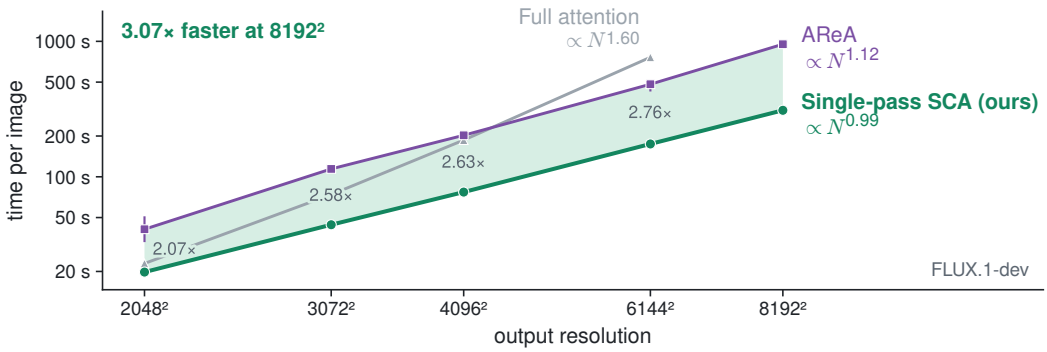

Figure D.2: **Latency at every size.** Single-pass SCA against AReA and full attention: batch 1 on one GH200, median of three timed images, whiskers min to max; full attention was not run at 8192^2. Grey ratios: speedup over AReA per size; exponents fit seconds $\propto N^b$ (N image tokens).

End-to-end latency. Table D.2 and Figure D.2 follow one matched protocol on idle GH200s. At each size, each method generates one excluded warm-up image and then three timed images in the same process, at batch 1 and 28 steps; we report the median. Full attention and AReA come from a first sweep (`results/latency_curve_batch1.csv`, which also lists the excluded warm-up images; `results/latency_curve_summary.csv` summarizes all its cells but full attention at 6144^2). Single-pass SCA comes from a second sweep that timed it and SCA without the schedule back to back on one GPU (`results/latency_curve_v2/summary.csv`). There SCA without the schedule reproduces its first-sweep medians within 0.8% at every size (71.1 s at 4096^2 in both). The schedule and the bridge add 8.4% at 4096^2 and 8 to 11% across sizes. At 4096^2 the medians are 186.5 s for full attention (186.0, 186.6, 186.5), 202.4 s for AReA (192.7, 202.4, 204.3), and 77.1 s for single-pass SCA (77.3, 77.1, 77.1; Figure D.2). At 2048^2 and 3072^2 they are 22.9 and 73.9 s for full attention, 41.0 and 114.3 s for AReA, and 19.8 and 44.3 s for single-pass SCA. At 6144^2 and 8192^2, AReA takes 482.3 and 952.0 s and single-pass SCA 174.7 and 309.7 s, a reduction of 63.8% and 67.5%. At 6144^2 full attention takes 762.7 s (763.5, 762.7, 762.5), $4.4\times$ the time of single-pass SCA. An earlier 4096^2 benchmark, kept as a separate run (`results/latency_4096_batch1.csv`), gave 184.6 s for full attention (184.1, 184.7, 184.6) and 212.6 s for AReA (207.3, 225.8, 212.6). It gave 71.0 s for SCA without the schedule (71.0, 83.1, 70.9). At the non-square sizes we have only campaign means over 195 prompts on shared nodes. Single-pass SCA takes 41.1 s (38.1 s without the schedule) and AReA 112.4 s at 2048×4096, and 60.5 s (55.8 s) and 150.5 s at 2048×6144. A third sweep timed progressive SCA, HRDiT, and AReA back to back on one GH200 at 4096^2, each with one excluded warm-up image and three timed images (`results/prog_latency/latency_summary.csv`). Its medians are 49.2, 107.0, and 202.4 s, and Tables 1 and K.1 use it for HRDiT and progressive SCA.

E Algorithms in full

Algorithms 4 and 5 expand Algorithms 1 and 2 to every step the code performs, and Algorithm 6 does so for Algorithm 3, progressive SCA (Section 2.5). Each step follows the statistics and attention modules in `sca/` or `scripts/sample_progressive.py` in the code repository (link in the abstract), and each comment names the equation or setting that a line implements.

Algorithm 6: **Progressive SCA**, HRDiT's schedule with SCA attention (Section 2.5)

```
# Input: prompt, seed; FLUX.1-dev in bf16; reach table at scales 2 and 4
# Output: 4096^2 image; scripts/sample_progressive.py
sig = cat(shift(linspace(1, 1/30, 30), mu=1.15), [0])  # one 30-step schedule for all stages, as in HRDiT
x = noise(1024^2, seed)                                # stage 1: stock dense attention, native RoPE
for i in range(30):                                    # Euler; ref keeps each step's x0 estimate
  v = flux(x, sig[i], g=3.5); ref[i] = x - sig[i] * v; x = x + (sig[i+1] - sig[i]) * v
for res, k, g, s_c, a in ((2048, 13, 4.5, 2, 1.0), (4096, 20, 6.0, 4, 0.25)):
  img = sharpen(bicubic(decode(x), res))               # 2 img - blur(img), blur width ~ U(0.1, 2.0)
  x = sig[k] * noise + (1 - sig[k]) * encode(img)      # re-enter at sigma 0.81, then 0.61
  use_sca(table[s_c], n0=64, t=8, sigma_g=0.89, bridge=(8, 16))  # replaces NTK RoPE, attention scale,
                                                       # pruning, alignment; w = n0, as sig[k] < 0.82 (Sec. 2.3)
  N = 30 - k                                           # 17, then 10 steps; sig[k] < sigma_g: no opening phase
  for n, i in enumerate(range(k, 30)):
    v = flux(x, sig[i], g); x0 = x - sig[i] * v; r, ref[i] = ref[i], x0    # SCA in every block
    x0 = x0 + a * (N - n) / N * (lowpass(up(r)) - lowpass(x0))  # Butterworth, order 4, cut-off 0.2
    u = momentum((x - x0) / sig[i], 0.5 * (N - n) / N)          # HRDiT's velocity momentum
    x = x + (sig[i+1] - sig[i]) * u
return decode(x)
```

F Hyperparameters

Table F.1 separates the hand-set settings of single-pass SCA from its fitted weight, and Table F.2 lists every setting; both expand the Configurations and baselines paragraph of Section 4.

G Evaluation protocol and metrics

This section expands the Prompts and seeds and the Metrics and intervals paragraphs of Section 4: Table G.1 summarizes the protocol, and the paragraphs below define each metric and interval.

Algorithm 4: **Reach emulation**, in full

```
# Input: 32 prompts; FLUX.1-dev at 1024^2 (n = 64),
#   28 steps, guidance 4.5, full attention
# Output: pi[s_c, l, h, beta], centres[beta]
def bin(sigma): return min(int(14 * sigma), 13)
def on_step(sigma):           # once per step
  S[bin(sigma)] += sigma; C[bin(sigma)] += 1
def emulate(q, k, v, l, sigma):    # every block
  O_full = attn(q, k, v)      # Eq. 1, all keys
  beta = bin(sigma)           # 14 bins, width 1/14
  for s_c in (2, 4):          # emulation, App. B
    f, w = s_c, n // s_c      # subsample, window
    t_c = round(t * w / n0)   # tile sides 4 and 2
    O_glob = global_branch(q, k, v, f)    # Eq. 2
    O_loc = local_branch(q, k, v, w, t_c) # Eq. 3
    dl, df = O_loc - O_glob, O_full - O_glob
    # Eq. 7: per head, over image rows, channels
    A[s_c, l, :, beta] += sum(dl * dl)
    B[s_c, l, :, beta] += sum(df * dl)
    E[s_c, l, :, beta] += sum(df * df)
    F[s_c, l, :, beta] += sum(O_full * O_full)
  return O_full               # sampler unchanged
sample(prompts, on_step, emulate)
pop = C > 0                   # 13 of 14 bins
centres = where(pop, S / C, (arange(14)+0.5)/14)
pi = clip(B / A, 0, 1); pi[A == 0] = 1  # Prop. 3
pi = fill_empty_bins(pi, pop)  # nearest bin
J = E - 2 * pi * B + pi**2 * A # error of the mix
rel_err = J.sum() / F.sum()    # Section 3
```

Algorithm 5: **Sampling**, in full

```
# Input: q, k, v before RoPE (T text, then H x W
#   image), block l, noise level sigma
# Output: O for all T + HW queries; sca/attn.py
# Per image: s_H, s_W = H // n0, W // n0; pi_s is
#   the table entry nearest max(s_H, s_W) in log
# Per step, all 57 x 24 heads, ends held:
w = n0                        # Sec. 2.3
for sig_k, w_k in ((.88, 128), (.86, 112),
                   (.84, 96), (.82, 80)):
  if sigma > sig_k: w = w_k; break
if w == n0: pi = interp(sigma, centres, pi_s)
else:                  # Eq. 6, log-linear
  r = max(H, W) / w; u = clip(log2(r / 2), 0, 1)
  pi = (1-u) * interp(sigma, centres, table[2]) \
     + u * interp(sigma, centres, table[4])
kb = k_img[s_H//2::s_H, s_W//2::s_W]  # κ(b)
vb = v_img[s_H//2::s_H, s_W//2::s_W]
cp = p // (s_H, s_W)          # c(p_i); text at 0
Q = cat(rope(q_txt, 0), rope(q_img, cp))
K = cat(rope(k_txt, 0), rope(kb, b))
O_glob = sdpa(Q, K, cat(v_txt, vb))   # Eq. 2
if sigma > sigma_g:                   # 0.89, Eq. 5
  return O_glob               # opening phase
# bridge (Eq. 4), per block and step:
d = ((k_img - k_img.mean(0))**2).sum(heads, ch)
B = top_k(d, 16) per cell, 8 x 8 cells  # 1,024
kr = rope(k_img, p)           # true positions
kl = cat(rope(k_txt, 0), kr[B], kr)
vl = cat(v_txt, v_img[B], v_img)  # a bridge key
     in the window enters twice
P = T + len(B)                # prefix, seen by all
w = (min(w, H), min(w, W))    # window sides
a = clip(t*(p//t) + t//2 - w//2, 0, (H, W) - w)
keep(i, j) = j < P or a_i <= p_j < a_i + w  # Eq. 3
perm = tile_major(H, W)       # 8 x 16 query blocks
mask = block_mask(keep, perm) # 8 x 8 key blocks
O_loc = flex_attention(rope(q_img, p)[perm],
    kl, vl, mask)[inverse(perm)]
O_img = pi[l] * O_loc + (1 - pi[l]) * O_glob[T:]
return cat(O_glob[:T], O_img) # fp32 mix, Eq. 5
```

Table F.1: **Hand-set and fitted settings.** Only the mixing weight is fitted; every other setting of single-pass SCA was chosen by hand. The four window bins hold one Euler step each only for the 28-step sampler at $\mu = 1.15$.

Setting	Value	Source
gate σ_g	0.89 (8 of 28 steps)	hand-set, AReA's boundary
window sizes	128, 112, 96, 80, then 64	hand-set on the 46-prompt subset
window σ bins	$(0.88, 0.89]$, $(0.86, 0.88]$, $(0.84, 0.86]$, $(0.82, 0.84]$	hand-set, tied to 28 steps
tile t	8	hand-set
bridge	8×8 cells, 16 tokens each (1,024)	hand-set, AReA's selection rule
emulated scales	2 and 4	hand-set
noise bins	14 of width $1/14$	hand-set
interpolation	linear in $\log_2 r$ and in σ	hand-set
weight $\pi_{lh}(\sigma; r)$	per block, head, bin, and ratio	fitted (Proposition 3)

Table F.2: **All hyperparameters** of single-pass SCA (code commit `7bd64c0`), produced by `scripts/sample.py --method sca --gate-sigma 0.89 --bridge-cells 8 --bridge-k 16 --win-sched 0.88:128,0.86:112,0.84:96,0.82:80 --win-pi-match --win-pi-interp` with the FLUX.1-dev reach table `k1_1024.npz` as `--pi-table`. Progressive SCA uses the same attention settings inside HRDiT's schedule (Appendix L); Appendix I gives the settings that differ for UltraImage, HRDiT, SigMa, and FLUX.2-dev.

Group	Setting	Value
Model	checkpoint, precision	FLUX.1-dev, bf16 weights and activations
	blocks, heads, head dim	19 double-stream + 38 single-stream, 24, 128
	native grid n_0, text tokens T	64 (1024^2 pixels, 16 pixels per token), 512
Sampler	steps, solver	28, flow-matching Euler
	guidance, shift	4.5 (distilled guidance), $\mu = 1.15$ at every size
	seed, decoding	seed label 0 (one prompt-derived seed per prompt, shared across methods); tiled VAE decoding
Local branch	tile t	8
	window $w(\sigma)$	128, 112, 96, 80 for σ in $(0.88, 0.89]$, $(0.86, 0.88]$, $(0.84, 0.86]$, $(0.82, 0.84]$, then 64; clamped at borders; original positions
	bridge	16 tokens per cell of an 8×8 grid (1,024), farthest from the mean key, recomputed per block and step, true positions
	kernel	FlexAttention; query blocks 8×16, key blocks 8×8, tile-major order
Global branch	subsampling	block centre, factor (s_H, s_W) = target grid / n_0
	positions	queries at $c(p)$, keys at block index, text at 0
Mixing	table	per (block, head, σ bin, ratio), $57 \times 24 \times 14 \times 2$, ratios 2 and 4
	ratio	$r = \max(H, W)/w(\sigma)$, weight linear in $\log_2 r$ between ratios 2 and 4, nearest outside
	interpolation	linear in σ between the mean sampled noise levels of occupied bins, constant outside
	gate	$\sigma_g = 0.89$ (8 of 28 steps global only, then 4 wide-window steps)
Emulation	prompts, resolution	32 at 1024^2; prompts 1 to 16 also at 1280^2
	emulated scales	2 and 4, one pass; 14 bins of width $1/14$
	hardware	one GH200; complete emulation runtime was not retained

Table G.1: **Evaluation protocol.** The FLUX.1-dev attention comparisons share sampler settings and prompt seeds. UltraImage and HRDiT use their own multi-stage protocols, and progressive SCA uses HRDiT's (Appendix L); FLUX.2-dev uses its own native-resolution schedule (Appendix I). Prompt counts include the four development prompts. Table F.2 lists the hyperparameters; the paragraphs below define the metrics.

Models	FLUX.1-dev (Black Forest Labs, 2024) (main results); FLUX.1-Krea-dev (Black Forest Labs & Krea AI, 2025) (195 prompts); FLUX.2-dev (Black Forest Labs, 2025) (50 prompts); bf16.
Sampler	28 Euler flow-matching steps, guidance 4.5, shift $\mu = 1.15$ at every size, seed label 0 per prompt, tiled VAE decoding; progressive SCA follows HRDiT's three stages (Appendix L), and the UltraImage, HRDiT, and FLUX.2-dev exceptions are in Appendix I.
Sizes	4096^2, 2048×4096, 2048×6144 (195 prompts each); 6144^2 (100-prompt subset); 8192^2 (12 exploratory paired prompts, metrics and qualitative examples).
Prompts	Aesthetic-4K evaluation set with 4K references (Zhang et al., 2025a), 195 prompts, fixed 46-prompt ablation subset; HRGen-1K (Rajabi et al., 2026a), all 1000 prompts.
Baselines	Full attention; AReA, released code (Rajabi et al., 2026a); DyPE (Issachar et al., 2025), SEGA (Rajabi et al., 2026b), UltraImage (two-stage defaults; Zhao et al., 2025), HRDiT (released progressive protocol; Xue et al., 2026); SigMa (Zhao et al., 2026) on 195 prompts.
Metrics	ImageReward, HPSv2/2.1, PickScore, CLIP, aesthetic; FID, KID, IS, patch FID/KID; MUSIQ (full, 1024^2 crops), CLIP-IQA, ARNIQA; 95% paired bootstrap intervals over prompts.
Hardware	One NVIDIA GH200 per process; verified timing logs identify the GH200 120 GB model. Clean latency uses batch 1; campaign timings are marked separately.

Prompts and references. The Aesthetic-4K evaluation split (Zhang et al., 2025a) has 195 prompts with 4K reference images. Four of them (boats, Moai, woman and bear, underwater surfer) were used to choose the opening phase and native window, and the 46-prompt subset to set the four wide-window sizes. They stay in the evaluation set for every method, which favours both SCA configurations slightly. The 195 reference rows hold 168 distinct images (27 rows reuse an image under their own caption); distribution metrics keep the official rows and weights, as Diffusion-4K's statistics do. The 46-prompt ablation subset was fixed before any ablation was run. HRGen-1K (Rajabi et al., 2026a) has 1000 prompts and no references; we use all of them. The 48 emulation prompts were written for this purpose and share no prompt with Aesthetic-4K (the largest word-set Jaccard overlap with any evaluation prompt is 0.26).

Implementations. torch-fidelity 0.3.0, pyiqa 0.1.16 (MUSIQ) and 0.1.15 (CLIP-IQA, ARNIQA), hpsv2 1.2.0, and image-reward 1.5.

Preference and alignment metrics. ImageReward v1.0 (Xu et al., 2023) (bicubic resize to 224, centre crop). HPSv2 and HPSv2.1 (Wu et al., 2023): cosine of normalized image and text features with the official validation transform (longest side 224, padded). PickScore (Kirstain et al., 2023): $\exp(\text{logit scale})$ times the cosine. CLIP score (Radford et al., 2021): $100 \max(\cos, 0)$ with ViT-L/14. Aesthetic: the LAION improved aesthetic predictor (sac+logos+ava1, linear MSE) on the L2-normalized CLIP ViT-L/14 image embedding (Schuhmann, 2022).

No-reference quality. MUSIQ (Ke et al., 2021) (pyiqa defaults) on the full image at native resolution, and MUSIQ_c, its mean over four fixed random 1024^2 crops per image. Crop positions depend only on the prompt id, so they are identical across methods. CLIP-IQA (Wang et al., 2022) and ARNIQA (Agnolucci et al., 2024) (pyiqa defaults) on the image downscaled with antialiased bicubic resampling to a long side of 1024 pixels, the scale both were trained on.

Distribution metrics and reference correction. FID (Heusel et al., 2017) and KID compare the generated images with all 195 official reference rows, using Inception-v3 features (torch-fidelity) after bilinear resizing to 299^2. The generated sets hold 195 images on the main set of Table 1 and 195 on the panel of Table N.1. The reference rows contain 168 distinct image contents; rows that share an image keep their official weights. Patch FID and KID are our own definition: the same Inception features on four fixed random 1024^2 crops per generated image and per reference row (780 generated and 780 reference crops on the main set). AReA and SEGA report a patch FID without defining its crops, so patch FID is never compared across papers. An audit found that the original feature-bank assembly keyed reference crops by resolved image path, so rows that share an image reused the last row's crop features. We corrected the lookup to retain each row's intended cached crops. Whole-image FID/KID are unchanged; all patch point estimates reported here use the corrected assembly. Inception score uses full images at 299^2, with 10 splits.

Intervals. Per-image intervals are 95% paired percentile bootstraps over prompt ids: original summaries use 1000 draws; separately recomputed comparisons use 2000 draws with seed 0. For the main FID/KID and patch-FID/KID comparison of single-pass SCA with AReA, 1000 draws resample generated prompt ids jointly across methods and resample the 168 distinct reference-image identities. Each draw keeps all rows and crops of a sampled identity together. This retains the 195-row weighted point estimate while accounting for rows that share a reference image. For single-pass SCA the resulting FID difference is -5.6 $[-10.6, -0.4]$, patch FID -4.4 $[-7.9, 0.4]$, and patch KID ($\times 10^3$) -2.53 $[-4.57, -0.60]$ (Table K.2). The whole-image and patch FID/KID comparisons at both non-square sizes, against every added baseline, and for the sampling-budget and progressive runs also use reference-image clusters. Other whole-image FID/KID intervals are labelled historical reference-row bootstraps; they have not been recomputed with image-identity clusters. Historical patch intervals are not attached to corrected estimates and are omitted where recomputation is incomplete. FID's finite-sample bias can place point estimates outside percentile intervals. Intervals are unadjusted for multiple comparisons and measure prompt/reference sampling uncertainty, not variability over independent model training runs.

H REPRODUCING AREA

This section expands the AReA comparison of Section 4.1: our protocol for AReA and our runs beside its published numbers at 4096^2 (Table H.1) and on HRGen-1K (Table H.2). We run AReA's released code with its FLUX.1 defaults: block-centre pooling, a first phase for 8 of 28 steps, halo width 8, and bridge ratio 0.5. We use the FLUX.1-dev checkpoint, guidance 4.5, 28 steps, μ at its maximum, and the seeds shared with the other methods. AReA's published numbers match FLUX.1-Krea-dev, the default of its released code. Rerunning that code on Krea-dev reproduces its published ImageReward, aesthetic score, and PickScore within 0.1 and its CLIP, HPS, and MUSIQ within 0.28 (Table H.1). On FLUX.1-dev the same code gives ImageReward 0.92 against the published 1.32. Our direct comparison of single-pass SCA with AReA holds the FLUX.1-dev settings fixed. UltraImage and FLUX.2-dev use separate protocols (Appendix I). AReA's paper reports 151 s for AReA and 314 s for full attention at 4096^2 on an H100; the matched sweep measures 202.4 s and 186.5 s on one GH200 (medians of three images). We use commit `9e38ec2` of the released repository, which includes a cost cap on halo and bridge tokens and a jagged halo edge. Neither appears in AReA's paper, which does not say whether its Table 1 used them. We recomputed the first-phase pooling factors from the code with 512 text tokens. They are 4, 4, 4, 4, 2, 2, 2, 2 at 4096×4096 and 2048×4096, and 8, 8, 8, 4, 4, 4, 2, 2 at 2048×6144. The second-phase windows shrink over 20 steps from 180, 151, and 192 tokens respectively to 64. At 2048×6144 no window is cheaper than full attention once the text tokens are counted. The first window therefore falls back to 192 tokens, and the cap drops the bridge and halo on the first three second-phase steps. Other differences from the release defaults are the FLUX.1-dev checkpoint instead of FLUX.1-Krea-dev, and one seed per prompt shared across methods instead of one seed for all prompts. On FLUX.1-Krea-dev, AReA's default checkpoint, we rerun AReA and fit a new reach table for SCA (16 prompts at 1024^2, scale-4 table, other settings unchanged) with the same 28-step, guidance-4.5 sampler on all 195 prompts. DyPE and SEGA run the Krea transformer through the frozen baseline harness on the 100 prompts of rows 4 to 103. There single-pass SCA wins eight of the 15 metrics and loses only patch FID, at a third of AReA's time (Table M.3). Its FID is 1.0 higher, an interval that spans zero ($[-3.6, 6.0]$). The progressive sampler does not support FLUX.1-Krea-dev. Table H.2 expands the HRGen-1K block of Table H.1 to every row of AReA's Table 2.

I OTHER BASELINES AND FLUX.2-DEV

This section gives the settings of the other baselines in Tables 1 and N.1 and of the FLUX.2-dev check (Section 4.4); Appendix L gives those of progressive SCA.

DyPE and SEGA. We use the existing baseline harness frozen at commit `c05ebdc`, with FLUX.1-dev, 28 steps, guidance 4.5, maximum shift, tiled VAE decoding, and the same prompt-derived seeds as SCA. These settings isolate the attention/position treatment on the checkpoint used in our main comparison. They do not reproduce the methods' reported FLUX.1-Krea-dev results; on Krea weights both trail AReA on ImageReward (Table M.3). Their low FLUX.1-dev scores should therefore not be read as a ranking under each method's preferred checkpoint.

UltraImage. We run a guided two-stage configuration from the FLUX branch of the released implementation through `scripts/run_ultraimage.py`. It uses a 1024^2 first stage and a 4096^2 second stage, 30 steps per stage, guidance 3.5 then 7, the `super_imax` guidance schedule, shifted-window patchification, proportional attention, RIFLEx positions, and attention decay. Each prompt also needs a 2+2-step block-average preparation pass. The reported campaign time includes preparation when not cached and the two-stage call; main-stage and preparation times are retained separately in sidecars. Each call reseeds both the generator and the global RNG, including the VAE sampling path, from the prompt id and seed label. Thus seed labels are paired, but the two-stage method does not share an identical latent trajectory with the single-pass methods. The wrapper preserves, without tuning, an upstream scheduler behaviour: assigning `scheduler.config.shift` does not update the scheduler's internal shift in the installed diffusers version, so both stages retain shift 3.0. We report this system-level comparison separately from the common-sampler attention comparisons.

Table H.1: **Published numbers at 4096^2 next to our runs.** Aesthetic-4K, 195 prompts: AReA's Table 1 (HPS and MUSIQ from its Table 6; "FLUX.1"), our runs with FLUX.1-Krea-dev, the checkpoint AReA's released code uses (28 steps, guidance 4.5), and our FLUX.1-dev protocol. Last block: HRGen-1K (all rows in Table H.2). HPSv2 $\times 100$; MUSIQ on the full image; FID at 299^2 against the 195 reference rows; patch FID is omitted because AReA does not state its crops. †Fewer than all prompts; not ranked. **Bold** / underline: best / second best within each block (bold only in two-row blocks); ours shaded. Key result: our run of AReA's code on FLUX.1-Krea-dev reproduces its published ImageReward, Aesthetic, and PickScore within 0.1 and its CLIP, HPS, and MUSIQ within 0.28. Against that run single-pass SCA is higher on CLIP, PickScore, HPS, and MUSIQ, within 0.1 on ImageReward and Aesthetic, and worse on FID. On HRGen-1K, where we did not rerun AReA on FLUX.1-Krea-dev, single-pass SCA on that checkpoint exceeds every published AReA score.

Method	n	Image-Reward↑	Aes-thetic↑	CLIP↑	Pick-Score↑	HPSv2↑	MUSIQ↑	FID↓
Published by AReA, single pass (FLUX.1)								
Full attention	195	−0.72	5.80	25.72	20.34	–	–	179.8
UltraImage	195	−0.14	6.11	26.77	20.85	–	–	169.4
DyPE	195	1.11	6.32	28.98	22.79	–	–	155.2
SigMa	195	1.01	6.16	28.36	22.47	–	–	156.8
SEGA	195	1.27	6.31	29.16	23.16	–	–	150.6
AReA	195	**1.32**	**6.37**	**29.71**	**23.31**	**29.69**	**45.2**	**148.5**
Our runs, FLUX.1-Krea-dev, same prompts and metrics								
AReA	195	1.28	6.32	29.49	23.21	29.41	45.0	**142.6**
DyPE	100†	0.96	6.41	28.73	22.50	28.39	42.7	171.6
SEGA	100†	1.13	6.31	29.19	23.01	29.15	46.0	164.2
Single-pass SCA (ours)	195	**1.30**	**6.37**	**29.64**	**23.38**	**29.81**	**46.9**	143.6
Our runs, FLUX.1-dev, shared sampler								
AReA	195	0.92	6.27	27.69	22.36	28.62	39.4	145.8
Single-pass SCA (ours)	195	**1.25**	**6.42**	**29.14**	**23.12**	**29.77**	**45.0**	**140.2**
HRGen-1K, 1000 prompts without references: AReA's Table 2 and our FLUX.1-Krea-dev run								
AReA, published (FLUX.1)	1000	0.60	5.87	29.89	21.44	27.50	–	–
Single-pass SCA (ours), FLUX.1-Krea-dev	1000	**0.84**	**5.96**	**30.02**	**21.68**	**28.06**	44.5	–

Table H.2: **HRGen-1K: AReA's published Table 2 next to our runs.** All 1000 prompts at 4096^2, no reference images. Top: AReA's Table 2 ("FLUX.1", which its released code runs as FLUX.1-Krea-dev). Bottom: single-pass SCA on FLUX.1-Krea-dev with its own reach table (28 steps, guidance 4.5), and our paired FLUX.1-dev runs of AReA and single-pass SCA. HPSv2 $\times 100$; IS: Inception score; MUSIQ on the full image. **Bold** / underline: best / second best among the published rows and our FLUX.1-Krea-dev row. Table H.1 holds the Aesthetic-4K comparison. Key result: single-pass SCA on FLUX.1-Krea-dev exceeds AReA's published ImageReward, Aesthetic, CLIP, PickScore, HPS, and IS.

Method	n	Image-Reward↑	Aes-thetic↑	CLIP↑	Pick-Score↑	HPSv2↑	IS↑	MUSIQ↑
Published by AReA (FLUX.1)								
Full attention	1000	−1.46	5.12	23.36	19.04	23.20	7.2	–
UltraImage	1000	−0.03	5.77	28.26	20.46	26.90	10.2	–
DyPE	1000	0.33	5.83	29.35	20.94	26.60	10.9	–
SigMa	1000	0.06	5.63	27.91	20.54	25.80	10.7	–
SEGA	1000	0.44	5.82	29.63	21.18	27.00	12.0	–
AReA	1000	0.60	5.87	29.89	21.44	27.50	13.9	–
Our runs								
Single-pass SCA (ours), FLUX.1-Krea-dev	1000	**0.84**	**5.96**	**30.02**	**21.68**	**28.06**	**16.4**	44.5
AReA, our run, FLUX.1-dev	1000	−0.03	5.92	26.08	20.43	26.16	12.7	37.4
Single-pass SCA (ours), FLUX.1-dev	1000	0.73	6.19	28.43	21.52	27.96	15.8	45.1

HRDiT. We run the released progressive protocol of the official repository (commit `f3be935`) on FLUX.1-dev in fp16, with its shipped per-head scope plan and attention settings. It takes 30 steps at 1024^2 with guidance 3.5, then 17 steps at 2048^2 with guidance 4.5 and 10 steps at 4096^2 with guidance 6, 57 steps in all, with pixel-space upsampling between stages. Its sampling budget therefore differs from the shared protocol, and we report it with the multi-stage recipes. HRDiT's global Torch random state is reset per prompt because its Gaussian blur draws a random width. Its clean time, 107.0 s, is the median of three timed images after an excluded warm-up in the third latency sweep (Appendix D). The first sweep gave 108.6 s over three batch-1 runs, the

first of which, 144.4 s, includes warm-up. The 195-prompt HRDiT evaluation used the original crop assembly, under which patch FID is 109.9 for HRDiT and 105.9 for SCA without the schedule. Recomputed from the cached crop features with the corrected assembly of Table 1 (`repro/evaluate_corrected.py`), HRDiT's patch FID is 103.0. The same assembly gives 107.7 for progressive SCA, 98.4 for single-pass SCA, 99.1 for SCA without the schedule, and 102.9 for AReA.

Progressive SCA. It runs HRDiT's schedule with SCA attention; Appendix L gives its protocol.

SigMa. The dev-matched profile runs the released code (commit `fbe3a86`) on FLUX.1-dev under the shared sampler: 28 steps, guidance 4.5, bf16, YaRN positions, and the shift capped at its maximum from the target sequence length. SigMa's paper states FLUX.1-dev with guidance 3.5, while its released command line hardcodes FLUX.1-Krea-dev with guidance 4.5; we ran only the profile that matches our protocol. The outputs are finite, fully decoded 4096^2 images, but the images we inspected show a regular lattice of dots, as DyPE's do at 4096^2 on FLUX.1-dev. This may reflect our port or profile rather than the method, so the SigMa row should not be read as the method's best result. We did not run ResDiT (no verified official code) or UltraImage's direct-extrapolation variant.

Adapter for HRDiT and SigMa. SigMa and HRDiT both run on all 195 prompts through one adapter with campaign per-prompt CPU generators, resident models, and tiled VAE decoding. These differ from the upstream command-line seed, offload, and decode choices, so the runs are controlled adaptations, not reproductions of the authors' tables. Campaign time includes text encoding, decoding, and finite-value checks after every step, which add synchronization overhead; it excludes model loading and is not a clean latency benchmark.

FLUX.2-dev. The model has 56 attention blocks and 48 heads per block. We reuse the two branches and the reach-emulation procedure, with a native grid of 64^2 tokens, local window 64, and query tile 8. The four-axis positions preserve the time and text axes while coarsening the two spatial axes. Reach emulation uses 16 prompts at 1024^2, 14 noise bins, and emulated scales 2 and 4; sampling uses the scale-4 table. Both full attention and SCA without the schedule use 28 steps, guidance 4.0, seed label 0, and a shift fixed to the native sequence length, $\mu = 2.1514$. The pipeline's default extrapolated shift at 4096^2 would be about 11.55; we therefore do not interpret this comparison as a default-pipeline benchmark. The gate is 0.959 so that the first 8 steps use only global attention. The 50 evaluation prompts are Aesthetic-4K rows 4–53, a 50-prompt subset. Prompt embeddings are cached for both methods; reported times include denoising and tiled VAE decoding, and exclude text encoding and model loading. At emulated scale 4, fitted per-head relative output error is 0.1790, versus 0.1889 for per-layer weights and 0.1924 for local-only attention; split-half correlation is 0.950 and cross-half error 0.1794. These are reach-emulation diagnostics, not generation ablations of the individual components on FLUX.2.

J MECHANISMS OF RELATED METHODS

This section expands Section 5: Table J.1 compares how the closest training-free methods change attention.

Table J.1: **How training-free methods change attention.** Entries follow each paper's text or, where marked *, its mechanism. Offsets: whether image query-key offsets stay in the trained range (text keys aside, Remark 1); SCA stays in range in the native window but not for its bridge keys or four wide-window steps (Section 2.2). Cost: order in the image tokens N at a fixed native size. AReA's entry follows its code: its pooled steps at factor 2 attend to $N/4$ keys, and its first window is the largest whose cost stays below full attention. Fitted latency exponents are 1.12 for AReA and 0.99 for single-pass SCA (Section 4.2). †Also inside HRDiT's cascade, as progressive SCA. Key result: only SCA fits per-head weights once at the training resolution, and it runs at linear cost in a single pass or inside HRDiT's cascade.

Method	Local view	Global view	Per-head reach	Offsets	Cost	Passes
FreeScale	overlapping crops	full attention	×	n/a	$O(N^2)^*$	cascade
DyPE	×	full, scaled RoPE	×	×*	$O(N^2)^*$	single
SEGA	×	full, NTK, logit scale	×	×	$O(N^2)$	single
UltraImage	×	full, edited RoPE	entropy at run time	×*	$O(N^2)^*$	one or two
HRDiT	per-head scopes	×	budget, fitted per size	✓*	$O(N^2)^*$	cascade
ResDiT	native patches	full, interpolated	×	✓	$O(N^2)^*$	single
AReA	shrinking windows	pooled keys	×	×	$O(N^2)^*$	single
SCA (ours)	shrinking windows	block-centre keys	fitted at 1024^2	native	$O(N)$	single†

K Extended results with intervals

This section expands Tables 1 and 2 to all metrics and sizes. Table K.1 adds every per-image metric to Table 1 and covers the non-square sizes and 6144^2. Table K.2 pairs single-pass SCA with AReA at 4096^2, Table K.3 does so at the non-square sizes, and Table K.4 checks the reach table there. Table K.5 expands Table 2. Appendix L gives the full comparison of progressive SCA.

Table K.1: **Both SCA configurations beat AReA on all ten per-image metrics at 4096^2; the non-square sizes are mixed.** FLUX.1-dev, 195 prompts (100 at 6144^2, where the row is SCA without the schedule); HPS, CLIP-IQA, and ARNIQA ×100; lower is better for latency and FID. Bold: best per size. Blue / orange: significantly better / worse than AReA. Full attention's FID and FID_p are unmarked; DyPE, SEGA, UltraImage, and HRDiT are paired with single-pass SCA in Table M.1 instead. Progressive SCA is paired over 2000 prompt draws, and for FID and FID_p over 1000 draws that also resample reference-image clusters. Latency: batch-1 median of three on an idle GH200 (HRDiT and progressive SCA: third sweep, Appendix D); ‡campaign mean on shared nodes; $^\circ$UltraImage's two-stage defaults. Key result: at 4096^2 progressive SCA is the fastest method (49.2 s) and best on CLIP, both HPS scores, and ARNIQA; single-pass SCA is best on patch FID and both MUSIQ scores.

Method	Lat. (s)↓	Image Reward	Image Aesth.	Pick CLIP	Pick Score	HPSv2	HPSv2.1	FID↓	FID_p↓	MUSIQ	MUSIQ_c	CLIP-IQA	ARNIQA
4096 × 4096, scale factors (4, 4)													
Full attention	186.5	−2.18	4.50	15.37	17.94	20.75	7.60	312.4	252.2	22.9	33.1	49.0	54.4
DyPE	195.9‡	−0.10	5.72	24.92	20.51	25.53	20.55	205.5	171.9	36.3	50.3	59.6	60.2
SEGA	190.4‡	0.56	6.02	26.95	21.64	27.13	24.34	164.9	125.8	37.3	50.3	66.7	64.8
UltraImage$^\circ$	308.5‡	1.28	6.50	28.84	23.33	29.96	30.77	142.0	109.5	33.6	41.7	72.8	72.3
HRDiT	107.0	**1.34**	**6.53**	29.12	**23.46**	29.88	30.92	**138.4**	103.0	40.3	53.5	**75.8**	71.6
AReA (released code)	202.4	0.92	6.27	27.69	22.36	28.62	27.16	145.8	102.9	39.4	50.4	67.7	67.4
Single-pass SCA	77.1	1.25	6.42	29.14	23.12	29.77	29.73	140.2	**98.4**	**45.0**	**60.1**	72.3	70.7
Progressive SCA	**49.2**	1.32	6.51	**29.39**	23.41	**30.26**	**31.02**	141.9	107.7	41.6	54.1	74.9	**74.5**
2048 × 4096, scale factors (2, 4)													
AReA (released code)	112.4‡	**1.13**	**6.41**	**28.59**	**22.66**	28.36	**27.64**	144.6	**111.7**	51.9	61.6	**74.9**	**73.0**
Single-pass SCA	**41.1**‡	1.11	6.17	28.09	22.61	**28.57**	27.49	**142.5**	114.9	**56.2**	**65.3**	72.5	72.6
2048 × 6144, scale factors (2, 6)													
AReA (released code)	150.5‡	0.39	**6.14**	26.52	21.51	26.41	23.36	163.0	**105.7**	42.8	53.0	67.5	69.3
Single-pass SCA	**60.5**‡	**0.64**	5.88	**26.78**	**21.66**	**26.59**	**23.60**	**142.8**	109.7	**47.7**	**59.4**	**67.8**	**69.5**
6144 × 6144, scale factors (6, 6), 100 prompts													
AReA (released code)	507.9‡	−0.42	5.80	23.84	20.30	24.95	18.78	216.7	188.6	28.3	44.5	53.8	47.8
SCA without the schedule	**164.7**‡	**0.84**	**6.24**	**27.12**	**21.89**	**27.48**	**25.16**	**169.9**	**122.4**	**37.5**	**48.3**	**60.1**	**62.6**

Appendix M pairs the added baselines with single-pass SCA. All these tables are populated from archived evaluation CSV files. Per-image means were checked against the per-image records; distribution-level estimates use cached features and the corrected crop assembly (Appendix G). Set-level metrics (FID, KID, their patch versions, Inception score) use the whole set. FID/KID and patch-FID/KID intervals on the 195-prompt sets resample reference-image clusters. Other FID/KID

intervals retain the historical reference-row bootstrap unless a corrected image-cluster interval exists, and Inception-score intervals resample only generated images.

Table K.2: **All metrics at 4096^2.** 195 prompts; Δ is method minus AReA with its 95% paired interval, coloured as in Table K.1. Intervals use 1000 prompt draws for single-pass SCA (FID, KID, and patch versions also resample reference-image clusters) and 2000 for full attention's per-image metrics. Full attention's set-level differences and campaign time have none. ‡Campaign mean (clean latency: Table D.2). Table L.1 pairs progressive SCA with every baseline. Key result: single-pass SCA beats AReA on all ten per-image metrics, FID, and patch KID; KID, patch FID, and Inception score are unresolved.

		Single-pass			
Metric	AReA	SCA	Δ [95% CI]	Full attention	Δ [95% CI]
ImageReward	0.921	1.248	+0.327 [+0.247, +0.405]	−2.180	−3.101 [−3.218, −2.984]
Aesthetic	6.27	6.42	+0.15 [+0.10, +0.20]	4.50	−1.77 [−1.87, −1.67]
CLIP score	27.69	29.14	+1.45 [+1.13, +1.82]	15.37	−12.32 [−13.12, −11.54]
PickScore	22.36	23.12	+0.76 [+0.64, +0.88]	17.94	−4.42 [−4.63, −4.22]
HPSv2 (×100)	28.62	29.77	+1.15 [+0.96, +1.34]	20.75	−7.87 [−8.24, −7.50]
HPSv2.1 (×100)	27.16	29.73	+2.56 [+2.20, +2.97]	7.60	−19.56 [−20.41, −18.72]
MUSIQ	39.4	45.0	+5.6 [+3.7, +7.4]	22.9	−16.5 [−18.0, −14.9]
MUSIQ_c	50.4	60.1	+9.7 [+8.2, +11.1]	33.1	−17.3 [−19.1, −15.5]
CLIP-IQA (×100)	67.7	72.3	+4.6 [+3.1, +6.1]	49.0	−18.7 [−20.5, −16.9]
ARNIQA (×100)	67.4	70.7	+3.4 [+2.5, +4.4]	54.4	−12.9 [−14.5, −11.3]
Inception score	6.66	6.44	−0.22 [−0.65, +0.39]	3.82	−2.84
FID↓	145.8	140.2	−5.6 [−10.6, −0.4]	312.4	+166.6
KID↓ ($\times 10^3$)	7.17	5.82	−1.35 [−3.17, +0.49]	164.15	+156.98
FID_p ↓	102.9	98.4	−4.4 [−7.9, +0.4]	252.2	+149.3
KID_p ↓ ($\times 10^3$)	11.40	8.87	−2.53 [−4.57, −0.60]	171.01	+159.61
Seconds per image‡ ↓	264.4	78.4	−186.0 [−199.6, −173.4]	196.2	−68.1

Table K.3 gives this pairing at the two non-square sizes, Table K.4 checks the scale-2 table there (Appendix P), and Table K.5 gives the intervals behind the base-operator rows of Table 2.

Table K.3: **All metrics at the non-square sizes.** 195 prompts each, single-pass SCA paired against AReA (released code). Conventions as in Table K.2: evaluator intervals over 1000 prompt draws, and for FID, KID, and their patch versions 1000 draws that also resample reference-image clusters. Key result: at 2048×6144 single-pass SCA wins seven of 15 metrics and loses two, at 2048×4096 it wins three and loses three, and it is faster at both.

	2048×4096, factors (2, 4)			2048×6144, factors (2, 6)		
		Single-pass			Single-pass	
Metric	AReA	SCA	Δ [95% CI]	AReA	SCA	Δ [95% CI]
ImageReward	1.132	1.106	−0.026 [−0.107, +0.047]	0.391	0.638	+0.247 [+0.125, +0.376]
Aesthetic	6.41	6.17	−0.23 [−0.29, −0.18]	6.14	5.88	−0.26 [−0.37, −0.15]
CLIP score	28.59	28.09	−0.49 [−0.89, −0.15]	26.52	26.78	+0.26 [−0.30, +0.79]
PickScore	22.66	22.61	−0.06 [−0.17, +0.06]	21.51	21.66	+0.15 [−0.03, +0.32]
HPSv2 (×100)	28.36	28.57	+0.21 [+0.09, +0.32]	26.41	26.59	+0.17 [+0.06, +0.31]
HPSv2.1 (×100)	27.64	27.49	−0.16 [−0.45, +0.12]	23.36	23.60	+0.23 [−0.14, +0.63]
MUSIQ	51.9	56.2	+4.3 [+2.9, +5.8]	42.8	47.7	+4.9 [+3.1, +6.8]
MUSIQ_c	61.6	65.3	+3.7 [+2.5, +4.8]	53.0	59.4	+6.4 [+4.9, +8.0]
CLIP-IQA (×100)	74.9	72.5	−2.4 [−3.4, −1.5]	67.5	67.8	+0.3 [−1.1, +1.8]
ARNIQA (×100)	73.0	72.6	−0.4 [−1.1, +0.1]	69.3	69.5	+0.2 [−0.6, +1.0]
Inception score	6.27	6.82	+0.55 [−0.12, +1.04]	5.48	6.41	+0.93 [+0.28, +1.39]
FID↓	144.6	142.5	−2.1 [−6.9, +3.3]	163.0	142.8	−20.2 [−24.7, −11.9]
KID↓ ($\times 10^3$)	7.81	7.02	−0.79 [−3.25, +1.95]	17.92	7.01	−10.91 [−14.53, −7.26]
FID_p ↓	111.7	114.9	+3.2 [−0.5, +7.1]	105.7	109.7	+4.0 [+1.0, +10.8]
KID_p ↓ ($\times 10^3$)	14.26	15.20	+0.95 [−0.46, +2.59]	13.56	15.09	+1.54 [−2.01, +5.42]
Seconds per image‡ ↓	112.4	41.1	−71.3 [−73.3, −69.2]	150.5	60.5	−90.0 [−92.2, −87.8]

L PROGRESSIVE SCA: PROTOCOL AND FULL RESULTS

This section expands Section 2.5 and the progressive rows of Table 1. It gives the protocol, every difference against the baselines with intervals (Table L.1), the comparisons with HRDiT and single-pass SCA (Table L.2), and two randomly drawn prompts (Figure L.1). Algorithm 6 writes out the sampler, and Figure S.4 shows six more randomly drawn prompts.

Protocol. Progressive SCA is HRDiT's released three-stage schedule (Xue et al., 2026) with SCA attention: it keeps HRDiT's schedule, transitions, guidance, sharpening, and momentum, and

Table K.4: **Reach table at 2048 × 4096.** 46 ablation prompts, SCA without the schedule throughout, paired against the same operator with the scale-4 table; blue / orange: significantly better / worse than this reference. Sources: `results/ps2_*`. Key result: the scale-2 table changes three metrics significantly, by at most 0.17.

Metric	SCA w/o sched., scale-2 table	AReA (released code)
ImageReward	+0.012 [−0.025, +0.050]	+0.027 [−0.141, +0.192]
Aesthetic	−0.00 [−0.03, +0.02]	+0.29 [+0.16, +0.43]
CLIP score	−0.02 [−0.22, +0.17]	+0.19 [−0.53, +0.91]
PickScore	+0.05 [+0.002, +0.09]	+0.14 [−0.09, +0.39]
HPSv2 (×100)	+0.09 [+0.04, +0.13]	−0.19 [−0.44, +0.06]
HPSv2.1 (×100)	+0.17 [+0.08, +0.26]	+0.39 [−0.21, +1.01]
MUSIQ	+0.0 [−0.5, +0.5]	−3.1 [−7.5, +1.2]
MUSIQ_c	−0.0 [−0.4, +0.2]	−4.1 [−6.7, −1.3]
CLIP-IQA (×100)	−0.6 [−1.2, +0.0]	+1.3 [−0.8, +3.4]
ARNIQA (×100)	−0.2 [−0.6, +0.1]	+0.4 [−1.5, +2.8]

Table K.5: **Ablation intervals for the base-operator rows of Table 2.** Fixed 46-prompt ablation subset at 4096^2, paired against SCA without the schedule (2000 bootstrap draws), whose column gives absolute values; every variant also runs without the schedule. Blue / orange: significantly better / worse. Panels split the variants by component. Each panel's last two rows give patch FID (set-level, no paired interval) and the relative output error of the weight rule at emulated scale 4 (Section 3). The v0 column keeps the per-head table, and reach emulation scores block-centre keys only, so mean pooling has no output-error entry. Sources: `results/ablnb_*` and the statistics file `k1_1024.json`. Key result: every variant is significantly worse than SCA without the schedule on five to ten per-image metrics.

Weight rule after the opening phase

Metric	SCA w/o sched.	local only	shared $\pi(\sigma)$	per-layer
ImageReward	1.124	−0.357 [−0.524, −0.207]	−0.090 [−0.173, −0.012]	−0.054 [−0.159, +0.030]
Aesthetic	6.45	−0.21 [−0.35, −0.07]	−0.09 [−0.18, −0.02]	+0.01 [−0.06, +0.07]
CLIP score	29.41	−1.93 [−2.68, −1.20]	−0.38 [−0.78, +0.08]	−0.13 [−0.54, +0.28]
PickScore	22.96	−1.20 [−1.44, −0.99]	−0.38 [−0.51, −0.26]	−0.16 [−0.28, −0.05]
HPSv2 (×100)	29.71	−1.78 [−2.07, −1.50]	−0.96 [−1.19, −0.73]	−0.50 [−0.63, −0.37]
HPSv2.1 (×100)	29.98	−2.86 [−3.35, −2.36]	−1.83 [−2.24, −1.43]	−0.70 [−0.98, −0.41]
MUSIQ	46.1	−3.5 [−6.5, −0.7]	−4.9 [−7.9, −1.9]	+0.7 [−1.6, +3.0]
MUSIQ_c	62.5	−0.3 [−2.5, +1.7]	−10.4 [−12.8, −8.3]	−2.4 [−3.6, −1.1]
CLIP-IQA (×100)	71.3	−4.6 [−6.3, −3.0]	−7.7 [−9.4, −6.0]	−0.9 [−2.0, +0.1]
ARNIQA (×100)	70.5	−1.0 [−2.0, +0.0]	−6.5 [−8.5, −4.8]	−1.5 [−2.3, −0.9]
FID_p ↓	168.0	192.2	166.7	167.7
Output error↓	0.200	0.261	0.230	0.223

Global branch and opening phase

Metric	SCA w/o sched.	global only	mean pooling	no opening (v0)
ImageReward	1.124	−0.395 [−0.507, −0.295]	−1.518 [−1.847, −1.193]	−1.328 [−1.595, −1.064]
Aesthetic	6.45	−0.76 [−0.87, −0.64]	−0.61 [−0.77, −0.44]	−0.08 [−0.17, +0.01]
CLIP score	29.41	−3.48 [−4.43, −2.59]	−3.59 [−4.76, −2.49]	−1.65 [−2.58, −0.84]
PickScore	22.96	−1.44 [−1.66, −1.22]	−2.60 [−3.00, −2.17]	−2.18 [−2.51, −1.85]
HPSv2 (×100)	29.71	−3.32 [−3.72, −2.94]	−4.04 [−4.66, −3.43]	−1.89 [−2.34, −1.45]
HPSv2.1 (×100)	29.98	−5.66 [−6.35, −4.98]	−9.75 [−11.30, −8.17]	−4.07 [−5.11, −3.04]
MUSIQ	46.1	−3.5 [−6.7, −0.2]	−1.8 [−5.3, +1.9]	−5.6 [−9.0, −2.1]
MUSIQ_c	62.5	−11.7 [−13.6, −9.8]	−4.2 [−8.4, −0.5]	+2.5 [−0.4, +5.5]
CLIP-IQA (×100)	71.3	−27.0 [−30.2, −23.8]	−10.1 [−13.0, −7.1]	+2.9 [+1.0, +4.9]
ARNIQA (×100)	70.5	−45.1 [−47.6, −42.5]	−4.8 [−6.7, −2.6]	+4.3 [+2.7, +6.1]
FID_p ↓	168.0	237.0	184.6	202.6
Output error↓	0.200	0.627	–	0.200

Figure L.1: **Progressive SCA keeps a large subject, as HRDiT does; single-pass SCA draws it smaller.** Two prompts drawn at random from the fixed 46-prompt ablation subset, seed 0; same 1:1 512×512 crop per row; times in Table 1.

changes the attention. `scripts/sample_progressive.py` reimplements the schedule with the stock diffusers FLUX components. One 30-step schedule with dynamic shift $\mu = 1.15$ (from the 1024^2 sequence length) serves all stages and reproduces the recorded HRDiT noise levels to 10^{-6}. The first stage runs all 30 steps at 1024^2 with FLUX.1-dev's stock attention and native RoPE, at guidance 3.5. Each transition decodes the latent, upsamples it bicubically, applies HRDiT's unsharp mask with a blur width drawn from the per-prompt global seed, re-encodes it, and re-noises it with the per-prompt generator. The 2048^2 stage re-enters at $\sigma = 0.81$ and runs the last 17 steps at guidance 4.5; the 4096^2 stage re-enters at $\sigma = 0.61$ and runs the last 10 at guidance 6.0. Against HRDiT's own runs, three things change together: the attention, the precision (HRDiT's released runner uses fp16, ours bf16), and the runner, so the comparison does not isolate the attention.

In both high-resolution stages SCA replaces HRDiT's four attention changes: NTK-scaled RoPE, a proportional attention scale, per-head key pruning, and position alignment. It uses $n_0 = 64$, $t = 8$, the 1,024 bridge tokens, and the FLUX.1-dev table read at ratio 2 at 2048^2 and at ratio 4 at 4096^2. Both re-entry levels lie below the gate $\sigma_g = 0.89$ and below the wide-window steps ($0.82 < \sigma \leq 0.89$), so no step runs the opening phase or a wide window. Every high-resolution step therefore uses the native window, the bridge, and the fitted per-head mix. HRDiT's low-band structure guidance and velocity momentum are kept, and both methods use the same per-prompt seeds. Deliberate differences from HRDiT's runner are bf16 instead of fp16 and a finite-value check at the end of each stage instead of after every step. With bf16 and stock components the 1024^2 images, and hence the layouts, differ from HRDiT's for the same seed (Figure L.1). Reported time is CUDA-synchronized and covers text encoding, all stages, transitions, and the final decode, excluding model loading. The gate and every other setting were fixed before generation.

Results. Table L.1 gives progressive SCA minus every baseline with intervals, and Table K.1 its absolute values. Against AReA it is significantly better on all ten per-image metrics. Its FID and KID differences against AReA span zero, while its patch FID ($+4.8$ $[0.1, 10.6]$) and patch KID are significantly higher. Against HRDiT it is significantly better on HPSv2, MUSIQ, and ARNIQA, worse only on CLIP-IQA, and within noise on the other six per-image metrics (Table L.2). HRDiT has lower KID, patch FID (103.0 against 107.7; progressive minus HRDiT $+4.6$ $[0.6, 7.7]$), and patch KID, while the FID difference (141.9 against 138.4) spans zero. Progressive SCA takes 49.2 s per image and HRDiT 107.0 s in the clean sweep of Appendix D, and 50.7 and 110.4 s as campaign means on shared nodes. Against single-pass SCA, progressive SCA gains seven per-image metrics, ties on CLIP ($+0.25$ $[-0.02, 0.52]$), and loses both MUSIQ scores (Table L.2). Its patch FID ($+9.2$ $[5.1, 13.2]$) and patch KID are significantly higher, and its FID and KID differences span zero. It takes 27.7 s less per image in campaign means (50.7 against 78.4 s) and 27.9 s less in the clean sweeps (49.2 against 77.1 s). Seeds 1 and 2 and the 1000 HRGen-1K prompts give the same per-image result against AReA (Table 3); the patch FID gap to AReA is significant at seeds 0 and 2 but not at seed 1. The set-level intervals use 1000 draws over prompts and reference-image clusters (`results/prog195_vs_baselines/cfid`). Figure L.2 compares progressive and single-pass SCA crop by crop.

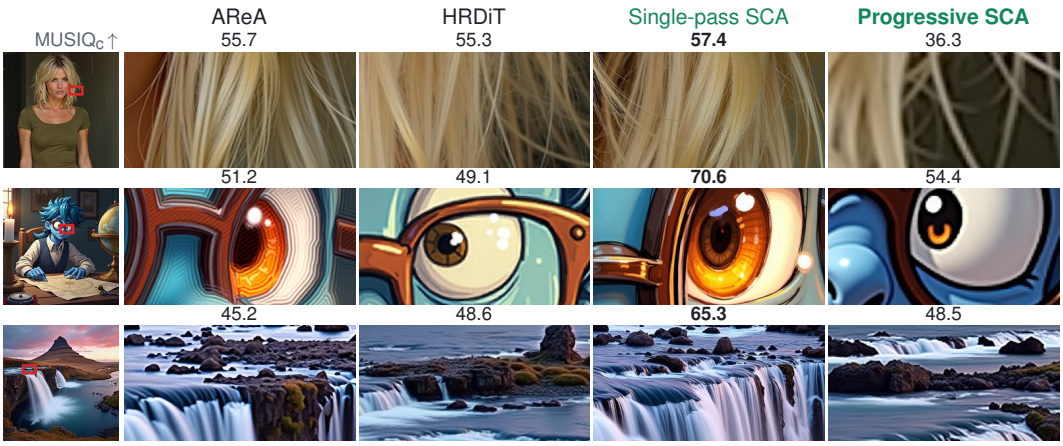

Figure L.2: **Single-pass SCA keeps the most native detail.** Three Aesthetic-4K evaluation prompts at seed 0: progressive SCA's image with a red window, then that window at 1:1 from each method under its image's MUSIQ_c (best bold). Progressive SCA is softer than single-pass SCA in every crop, the fidelity trade-off of Section 4.1.

Table L.1: **Progressive SCA against every 4096^2 baseline.** FLUX.1-dev, 195 Aesthetic-4K prompts, seed label 0. Each cell is progressive SCA minus the column's method, with its 95% paired interval below it. Per-image metrics use 2000 prompt draws; FID, KID and their patch versions use 1000 draws that also resample the 168 reference-image clusters. Single-pass: single-pass SCA. Blue / orange: the interval excludes zero in favour of / against progressive SCA; lower is better for FID and KID, so their signs read the other way. Inception score has no paired interval (Table K.1). The point estimate lies outside its percentile interval for Full attn. patch FID, as it can for a biased statistic. Key result: progressive SCA is significantly better than AReA, DyPE, SEGA, and full attention on every per-image metric and than UltraImage on nine. Its differences from HRDiT on ImageReward, HPSv2.1, and PickScore span zero, and its patch FID is higher than AReA's, HRDiT's, and single-pass SCA's.

Metric	AReA	HRDiT	UltraImage	DyPE	SEGA	Full attn.	Single-pass
ImageReward	+0.399	−0.022	+0.037	+1.417	+0.760	+3.500	+0.072
	[0.317, 0.486]	[−0.088, 0.034]	[0.008, 0.067]	[1.266, 1.561]	[0.632, 0.884]	[3.402, 3.590]	[0.010, 0.134]
Aesthetic	+0.23	−0.03	+0.00	+0.78	+0.49	+2.00	+0.09
	[0.17, 0.30]	[−0.07, 0.02]	[−0.02, 0.03]	[0.69, 0.87]	[0.41, 0.56]	[1.91, 2.09]	[0.04, 0.13]
CLIP score	+1.70	+0.27	+0.56	+4.47	+2.44	+14.02	+0.25
	[1.33, 2.07]	[−0.01, 0.54]	[0.38, 0.73]	[3.85, 5.10]	[1.98, 2.88]	[13.21, 14.79]	[−0.02, 0.52]
PickScore	+1.05	−0.05	+0.08	+2.90	+1.76	+5.47	+0.29
	[0.93, 1.19]	[−0.14, 0.04]	[0.03, 0.12]	[2.71, 3.10]	[1.61, 1.92]	[5.27, 5.69]	[0.20, 0.38]
HPSv2 (×100)	+1.64	+0.38	+0.30	+4.73	+3.12	+9.51	+0.48
	[1.42, 1.86]	[0.24, 0.51]	[0.21, 0.38]	[4.43, 5.05]	[2.87, 3.37]	[9.18, 9.84]	[0.32, 0.64]
HPSv2.1 (×100)	+3.86	+0.10	+0.25	+10.47	+6.68	+23.42	+1.29
	[3.41, 4.30]	[−0.14, 0.35]	[0.14, 0.37]	[9.70, 11.26]	[6.07, 7.24]	[22.64, 24.21]	[0.99, 1.59]
MUSIQ	+2.3	+1.3	+8.1	+5.4	+4.3	+18.7	−3.4
	[0.5, 3.9]	[0.2, 2.4]	[7.0, 9.1]	[3.6, 7.2]	[2.7, 5.9]	[17.4, 20.0]	[−5.1, −1.6]
MUSIQ, crops	+3.7	+0.6	+12.4	+3.8	+3.9	+21.0	−6.0
	[2.0, 5.4]	[−0.7, 1.8]	[11.5, 13.2]	[2.0, 5.8]	[2.2, 5.6]	[19.2, 22.8]	[−7.3, −4.6]
CLIP-IQA (×100)	+7.2	−0.9	+2.1	+15.3	+8.2	+25.9	+2.6
	[5.6, 8.7]	[−1.8, −0.1]	[1.4, 2.7]	[13.4, 17.3]	[6.5, 10.0]	[24.4, 27.4]	[1.6, 3.5]
ARNIQA (×100)	+7.1	+2.9	+2.1	+14.2	+9.7	+20.0	+3.7
	[6.1, 8.2]	[2.3, 3.4]	[1.7, 2.6]	[12.9, 15.5]	[8.5, 10.9]	[18.7, 21.3]	[3.2, 4.3]
FID	−3.9	+3.4	−0.2	−63.6	−23.1	−170.6	+1.6
	[−8.5, 3.1]	[−1.8, 7.2]	[−3.8, 2.5]	[−74.6, −49.8]	[−34.4, −15.8]	[−175.4, −144.2]	[−2.3, 7.3]
KID (×1000)	−0.09	+2.51	+0.70	−39.51	−3.81	−157.07	+1.25
	[−2.75, 2.50]	[0.84, 4.34]	[−0.50, 1.80]	[−52.24, −29.83]	[−9.34, 0.46]	[−175.94, −138.99]	[−0.48, 3.10]
Patch FID	+4.8	+4.6	−1.9	−64.2	−18.2	−144.5	+9.2
	[0.1, 10.6]	[0.6, 7.7]	[−3.7, 0.5]	[−73.8, −50.2]	[−27.1, −9.8]	[−141.9, −113.8]	[5.1, 13.2]
Patch KID (×1000)	+3.35	+2.82	−1.66	−43.69	−8.33	−156.26	+5.88
	[0.05, 6.61]	[1.00, 4.63]	[−2.85, −0.37]	[−54.62, −33.89]	[−14.21, −3.30]	[−173.40, −140.53]	[3.55, 8.25]

Table L.2: **Per image, progressive SCA is significantly worse than HRDiT only on CLIP-IQA.** 4096^2, 195 prompts, seed label 0. Per-image differences (first-named setting minus second) with 95% paired intervals over 2000 prompt draws; blue / orange: the interval excludes zero in favour of / against the first-named setting. Progressive: HRDiT's schedule with SCA attention (Section 2.5). SCA: 28 steps, single-pass SCA in the second column and SCA without the schedule in the third. 42 steps: SCA without the schedule at 42 steps, sized to HRDiT's campaign time. Intervals that include zero do not establish equivalence.

			SCA without the schedule	
Metric	Progressive vs. HRDiT	Progressive vs. single-pass	42 vs. 28 steps	HRDiT vs. 42 steps
ImageReward	−0.022 [−0.088, +0.034]	+0.072 [+0.010, +0.134]	+0.030 [−0.001, +0.059]	+0.083 [+0.023, +0.150]
Aesthetic	−0.027 [−0.070, +0.017]	+0.087 [+0.038, +0.130]	+0.020 [−0.007, +0.047]	+0.071 [+0.022, +0.119]
CLIP score	+0.27 [−0.01, +0.54]	+0.25 [−0.02, +0.52]	−0.01 [−0.19, +0.20]	−0.02 [−0.31, +0.27]
PickScore	−0.05 [−0.14, +0.04]	+0.29 [+0.20, +0.38]	+0.08 [+0.03, +0.13]	+0.29 [+0.19, +0.39]
HPSv2 (×100)	+0.38 [+0.24, +0.51]	+0.48 [+0.32, +0.64]	+0.23 [+0.15, +0.31]	−0.10 [−0.26, +0.07]
HPSv2.1 (×100)	+0.10 [−0.14, +0.35]	+1.29 [+0.99, +1.59]	+0.60 [+0.44, +0.75]	+0.65 [+0.38, +0.93]
MUSIQ	+1.3 [+0.2, +2.4]	−3.4 [−5.1, −1.6]	−0.4 [−1.5, +0.7]	−4.7 [−6.3, −2.9]
MUSIQ_c	+0.6 [−0.7, +1.8]	−6.0 [−7.3, −4.6]	+1.9 [+1.3, +2.5]	−9.6 [−10.9, −8.4]
CLIP-IQA (×100)	−0.9 [−1.8, −0.1]	+2.6 [+1.6, +3.5]	+0.4 [−0.1, +0.8]	+2.4 [+1.5, +3.3]
ARNIQA (×100)	+2.9 [+2.3, +3.4]	+3.7 [+3.2, +4.3]	+1.1 [+0.7, +1.4]	−0.7 [−1.4, −0.1]

M ADDITIONAL PAIRED INTERVALS

This section expands Table 3 and the baseline rows of Tables 1 and N.1 with per-metric intervals. Every comparison here uses single-pass SCA, or SCA without the schedule where labelled; Appendix L pairs progressive SCA. Table M.1 pairs every baseline with single-pass SCA, and Figure M.1 plots four of its metrics for full attention, AReA, DyPE, SEGA, and UltraImage. Table M.2 covers the emulation-budget and gate checks, Table M.3 the 6144^2 and FLUX.1-Krea-dev runs, and Table M.4 the prompt-set, seed, and model checks. Every difference in the tables is method minus the named reference. Positive values favour the method for the ten per-image metrics, and blue / orange mark intervals that exclude zero in favour of / against the method. These comparisons resample matched prompt ids, with 2000 draws and seed 0 (Table M.3: 1000); they do not treat multiple crops as independent images. The HRGen-1K sample has 1000 prompts, the two extra seeds 195, the sensitivity experiments 46, FLUX.2-dev 50, and the main baseline comparison 195. Tables use the separately recomputed per-image CSVs; these intervals differ slightly from the original 1000-draw evaluator summaries. No multiplicity correction is applied.

Every baseline against single-pass SCA. Table M.1 pairs every baseline of Tables 1 and N.1 with single-pass SCA on the same 195 prompts and seeds. Per-image intervals use 2000 prompt draws with seed 0. FID, KID, and their patch versions are paired with SCA without the schedule, over 1000 draws that resample generated prompts and reference-image clusters. Table N.2 gives the matching intervals for the 56-step and guidance-7 runs of SCA without the schedule.

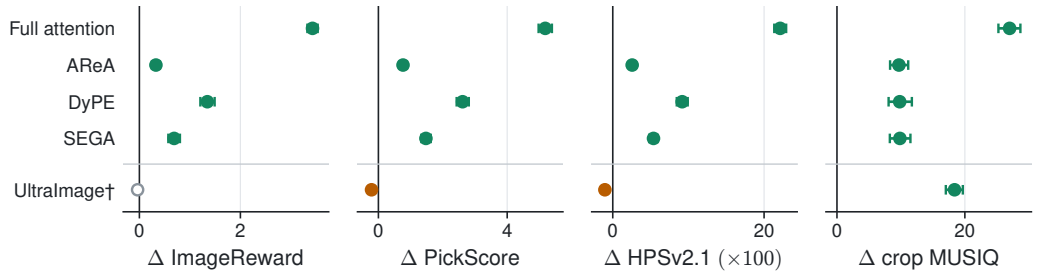

Figure M.1: **Single-pass SCA beats every shared-protocol baseline on all four metrics; against UltraImage the result is mixed.** Single-pass SCA minus baseline, mean over the same 195 prompts at 4096^2, whiskers 95% prompt-bootstrap intervals (2000 draws, one seed); open markers: interval contains zero; orange: significant loss. †Own two-stage recipe, not an equal-cost comparison.

6144^2 and FLUX.1-Krea-dev. Table M.3 gives every metric with its paired interval for the 100-prompt 6144^2 comparison, which uses SCA without the schedule, and for FLUX.1-Krea-dev. On FLUX.1-Krea-dev, DyPE and SEGA trail AReA on ImageReward and single-pass SCA does not. The Krea reach table follows the FLUX.1-dev procedure with 16 prompts at 1024^2 (Appendix H).

Table M.1: **Every shared-sampler baseline trails single-pass SCA on all ten per-image metrics; HRDiT and UltraImage lead on several preference metrics.** Baseline minus single-pass SCA (28 steps) at 4096^2 with 95% paired intervals; the set-level and time rows pair against SCA without the schedule. HPSv2, HPSv2.1, CLIP-IQA, and ARNIQA are $\times 100$, KID and KID_p $\times 10^3$. For FID, KID, their patch versions, and campaign time (seconds per image), lower is better, so negative favours the baseline. Blue / orange: baseline significantly better / worse than the reference. SigMa uses its dev-matched profile; UltraImage and HRDiT use their own recipes (Appendix I). Finite-sample bias can put a FID estimate outside its percentile interval (full attention's FID_p).

	Single pass, shared sampler				
Metric	Full attn.	DyPE	SigMa	SEGA	AReA
ImageReward	−3.428 [−3.530, −3.327]	−1.345 [−1.490, −1.207]	−1.771 [−1.920, −1.624]	−0.688 [−0.803, −0.573]	−0.327 [−0.405, −0.250]
Aesthetic	−1.92 [−2.00, −1.83]	−0.69 [−0.79, −0.60]	−0.82 [−0.92, −0.72]	−0.40 [−0.48, −0.33]	−0.15 [−0.20, −0.09]
CLIP score	−13.77 [−14.54, −13.01]	−4.22 [−4.83, −3.63]	−5.64 [−6.28, −4.98]	−2.19 [−2.62, −1.79]	−1.45 [−1.80, −1.12]
PickScore	−5.18 [−5.39, −4.98]	−2.61 [−2.81, −2.43]	−3.11 [−3.32, −2.90]	−1.48 [−1.63, −1.33]	−0.76 [−0.88, −0.66]
HPSv2	−9.02 [−9.34, −8.69]	−4.25 [−4.55, −3.97]	−4.99 [−5.31, −4.71]	−2.64 [−2.88, −2.40]	−1.15 [−1.34, −0.96]
HPSv2.1	−22.13 [−22.92, −21.35]	−9.18 [−9.91, −8.46]	−11.08 [−11.83, −10.35]	−5.39 [−5.91, −4.82]	−2.56 [−2.94, −2.18]
MUSIQ	−22.1 [−23.7, −20.5]	−8.7 [−10.8, −6.6]	−9.5 [−11.5, −7.4]	−7.7 [−9.5, −5.9]	−5.6 [−7.4, −3.9]
MUSIQ_c	−27.0 [−28.7, −25.2]	−9.8 [−11.7, −8.1]	−10.2 [−12.1, −8.5]	−9.8 [−11.5, −8.3]	−9.7 [−11.1, −8.3]
CLIP-IQA	−23.3 [−24.8, −21.9]	−12.8 [−14.6, −11.0]	−14.3 [−16.3, −12.4]	−5.7 [−7.3, −4.2]	−4.6 [−6.1, −3.2]
ARNIQA	−16.3 [−17.7, −14.9]	−10.5 [−11.9, −9.1]	−10.9 [−12.3, −9.5]	−6.0 [−7.2, −4.8]	−3.4 [−4.3, −2.5]
Baseline minus SCA without the schedule					
FID↓	+174.4 [+150.2, +179.4]	+67.4 [+56.2, +80.3]	+78.9 [+66.7, +91.0]	+26.9 [+21.1, +39.1]	+7.7 [+2.6, +13.1]
KID↓	+158.0 [+140.3, +176.0]	+40.4 [+30.6, +52.9]	+50.4 [+39.1, +64.0]	+4.7 [+0.1, +10.5]	+1.0 [−1.5, +3.2]
FID_p ↓	+153.1 [+122.3, +149.6]	+72.8 [+58.7, +81.6]	+85.5 [+70.2, +93.7]	+26.7 [+18.8, +35.3]	+3.8 [−1.3, +7.5]
KID_p ↓	+161.9 [+146.6, +178.2]	+49.3 [+40.1, +60.0]	+62.4 [+52.0, +74.7]	+14.0 [+9.3, +20.0]	+2.3 [−0.1, +4.6]
Time (s)↓	+124.7 [+120.8, +129.4]	+124.4 [+120.5, +128.6]	+121.0 [+119.0, +123.0]	+118.9 [+118.0, +119.7]	+192.9 [+180.3, +206.4]

	Multi-stage, own recipe	
Metric	UltraImage	HRDiT
ImageReward	+0.035 [−0.031, +0.102]	+0.094 [+0.032, +0.157]
Aesthetic	+0.08 [+0.04, +0.12]	+0.11 [+0.07, +0.16]
CLIP score	−0.31 [−0.60, −0.01]	−0.02 [−0.29, +0.27]
PickScore	+0.21 [+0.12, +0.31]	+0.34 [+0.25, +0.43]
HPSv2	+0.19 [+0.03, +0.35]	+0.11 [−0.05, +0.27]
HPSv2.1	+1.04 [+0.75, +1.33]	+1.19 [+0.92, +1.47]
MUSIQ	−11.4 [−13.3, −9.5]	−4.6 [−6.2, −2.9]
MUSIQ_c	−18.4 [−19.7, −17.0]	−6.6 [−7.9, −5.3]
CLIP-IQA	+0.5 [−0.6, +1.6]	+3.5 [+2.6, +4.4]
ARNIQA	+1.6 [+0.9, +2.3]	+0.9 [+0.2, +1.5]
Baseline minus SCA without the schedule		
FID↓	+4.0 [+1.1, +10.2]	+0.4 [−2.3, +6.4]
KID↓	+0.2 [−1.8, +2.2]	−1.6 [−3.6, +0.3]
FID_p ↓	+10.4 [+6.2, +13.9]	+3.9 [+0.7, +8.0]
KID_p ↓	+7.3 [+5.1, +9.5]	+2.8 [+0.9, +4.8]
Time (s)↓	+237.0 [+233.5, +241.7]	+38.8 [+37.2, +40.5]

Table M.2: **Reach tables fitted on 4 to 16 prompts change one of 30 differences significantly (MUSIQ +0.1); a gate at $\sigma_g = 0.85$ lowers seven metrics.** Variant minus default SCA, both without the schedule, 46 ablation prompts; blue / orange: variant significantly better / worse than default SCA. Budgets use the first 4, 8, or 16 of the 32 emulation prompts; the default gate is $\sigma_g = 0.89$.

	SCA without the schedule: variant minus default				
Metric	4 prompts	8 prompts	16 prompts	$\sigma_g = 0.85$	$\sigma_g = 0.93$
ImageReward	+0.012 [−0.007, +0.029]	+0.010 [−0.004, +0.024]	−0.007 [−0.020, +0.005]	−0.015 [−0.098, +0.064]	−0.004 [−0.097, +0.069]
Aesthetic	+0.01 [−0.00, +0.02]	+0.01 [−0.00, +0.02]	+0.00 [−0.01, +0.01]	−0.19 [−0.26, −0.12]	+0.00 [−0.05, +0.06]
CLIP score	+0.08 [−0.03, +0.19]	+0.05 [−0.04, +0.15]	+0.04 [−0.04, +0.13]	−0.27 [−0.82, +0.26]	−0.64 [−1.03, −0.29]
PickScore	−0.01 [−0.03, +0.01]	+0.00 [−0.01, +0.02]	+0.01 [−0.01, +0.03]	−0.12 [−0.27, +0.02]	−0.22 [−0.38, −0.07]
HPSv2 (×100)	+0.02 [−0.02, +0.05]	+0.02 [−0.00, +0.05]	+0.00 [−0.02, +0.03]	−0.48 [−0.69, −0.28]	+0.02 [−0.16, +0.20]
HPSv2.1 (×100)	−0.02 [−0.07, +0.04]	+0.01 [−0.03, +0.05]	−0.01 [−0.05, +0.04]	−0.85 [−1.22, −0.48]	−0.15 [−0.56, +0.24]
MUSIQ	−0.1 [−0.4, +0.3]	−0.1 [−0.5, +0.2]	+0.1 [+0.0, +0.3]	−1.9 [−3.5, −0.2]	+0.4 [−1.0, +2.0]
MUSIQ_c	+0.1 [−0.1, +0.3]	−0.0 [−0.2, +0.1]	−0.1 [−0.2, +0.0]	−2.0 [−3.0, −1.1]	+0.6 [−0.4, +1.9]
CLIP-IQA (×100)	+0.2 [−0.1, +0.4]	+0.2 [−0.1, +0.4]	−0.1 [−0.4, +0.1]	−3.2 [−4.8, −1.7]	+0.0 [−1.2, +1.1]
ARNIQA (×100)	+0.1 [−0.1, +0.3]	+0.0 [−0.1, +0.2]	−0.0 [−0.1, +0.1]	−4.0 [−5.9, −2.3]	+1.8 [+0.7, +3.1]

The progressive sampler does not support FLUX.1-Krea-dev, so every Krea row is single-pass SCA.

Table M.3: **At 6144^2 SCA without the schedule beats AReA on 14 of 15 metrics; on FLUX.1-Krea-dev single-pass SCA wins eight metrics and loses patch FID, in a third of the time.** Method minus AReA with 95% paired intervals over 100 prompts at 6144^2 and for DyPE and SEGA, which ran only those, and all 195 for single-pass SCA on FLUX.1-Krea-dev. The first column is FLUX.1-dev at 6144^2, the others FLUX.1-Krea-dev at 4096^2 with its own reach table. Per-image intervals use 1000 prompt draws; FID, KID, and their patch versions resample prompts and reference-image clusters. Lower is better for FID, KID, their patch versions, and time. Blue / orange: significantly better / worse than AReA. Time is the campaign mean on shared GH200 nodes.

	6144^2	FLUX.1-Krea-dev, 4096^2		
Metric	SCA w/o sched.	Single-pass SCA	DyPE	SEGA
ImageReward	+1.262 [+1.051, +1.450]	+0.020 [−0.021, +0.060]	−0.282 [−0.424, −0.173]	−0.110 [−0.189, −0.030]
Aesthetic	+0.44 [+0.31, +0.55]	+0.04 [+0.01, +0.07]	+0.05 [−0.02, +0.12]	−0.05 [−0.11, +0.02]
CLIP score	+3.28 [+2.32, +4.18]	+0.15 [−0.06, +0.36]	−0.95 [−1.47, −0.49]	−0.49 [−0.91, −0.07]
PickScore	+1.58 [+1.31, +1.88]	+0.17 [+0.11, +0.24]	−0.73 [−0.93, −0.55]	−0.23 [−0.36, −0.10]
HPSv2 (×100)	+2.53 [+2.12, +2.94]	+0.39 [+0.30, +0.48]	−1.02 [−1.24, −0.80]	−0.25 [−0.43, −0.09]
HPSv2.1 (×100)	+6.37 [+5.39, +7.38]	+0.91 [+0.70, +1.12]	−1.24 [−1.76, −0.76]	−0.25 [−0.61, +0.09]
MUSIQ	+9.2 [+6.9, +11.7]	+1.9 [+1.0, +2.8]	−2.6 [−4.6, −0.4]	+0.7 [−1.2, +2.8]
MUSIQ_c	+3.8 [+1.4, +6.2]	+2.7 [+2.1, +3.4]	+3.2 [+1.8, +4.6]	+1.3 [−0.1, +2.6]
CLIP-IQA (×100)	+6.3 [+4.3, +8.4]	−0.6 [−1.4, +0.3]	+0.9 [−0.7, +2.3]	+1.8 [+0.4, +3.2]
ARNIQA (×100)	+14.9 [+13.0, +17.0]	+0.8 [+0.3, +1.2]	−1.6 [−2.6, −0.7]	−0.4 [−1.3, +0.4]
FID↓	−46.8 [−58.6, −35.3]	+1.0 [−3.6, +6.0]	+5.0 [−2.8, +16.3]	−2.3 [−10.0, +7.4]
KID↓ (×10^3)	−23.20 [−31.77, −16.84]	−0.27 [−1.95, +1.16]	−0.81 [−4.63, +3.26]	−0.85 [−4.00, +1.53]
FID_p ↓	−66.2 [−70.2, −48.9]	+3.4 [+1.6, +7.1]	+7.7 [+5.0, +18.5]	+0.1 [−3.1, +6.7]
KID_p ↓ (×10^3)	−56.88 [−68.46, −47.50]	+0.66 [−0.57, +2.05]	+1.32 [−1.44, +4.94]	−0.81 [−2.48, +0.90]
Inception score	+0.45 [−0.06, +0.84]	+0.67 [+0.14, +1.08]	+0.25 [−0.22, +0.69]	+0.09 [−0.43, +0.56]
Time (s)↓	−343.1 [−349.3, −337.4]	−160.9 [−173.3, −149.4]	−24.3 [−28.9, −19.6]	−20.1 [−25.0, −15.5]

Table M.4: **The gains hold on HRGen-1K, two more seeds, and FLUX.2-dev.** Single-pass SCA minus AReA for HRGen-1K and seeds 1 and 2; SCA without the schedule minus full attention for FLUX.2-dev. All at 4096^2; blue: significantly better than the reference.

Metric	HRGen-1K	Seed 1	Seed 2	FLUX.2-dev w/o sched.
ImageReward	+0.764 [+0.714, +0.816]	+0.306 [+0.242, +0.378]	+0.264 [+0.197, +0.338]	+0.456 [+0.320, +0.598]
Aesthetic	+0.27 [+0.24, +0.31]	+0.11 [+0.06, +0.16]	+0.09 [+0.04, +0.14]	+0.30 [+0.18, +0.42]
CLIP score	+2.35 [+2.12, +2.59]	+1.22 [+0.89, +1.54]	+1.37 [+1.08, +1.66]	+0.38 [−0.38, +1.12]
PickScore	+1.09 [+1.04, +1.14]	+0.77 [+0.67, +0.88]	+0.74 [+0.62, +0.86]	+0.56 [+0.29, +0.86]
HPSv2 (×100)	+1.81 [+1.71, +1.90]	+1.17 [+1.01, +1.33]	+1.18 [+0.99, +1.38]	+0.85 [+0.50, +1.21]
HPSv2.1 (×100)	+4.60 [+4.37, +4.83]	+2.49 [+2.16, +2.83]	+2.53 [+2.13, +2.90]	+1.84 [+1.10, +2.55]
MUSIQ	+7.7 [+6.9, +8.5]	+5.5 [+3.8, +7.2]	+6.3 [+4.4, +8.1]	+16.0 [+11.9, +19.9]
MUSIQ$_c$	+12.6 [+11.8, +13.3]	+10.3 [+8.9, +11.6]	+10.9 [+9.3, +12.6]	+21.4 [+18.2, +24.8]
CLIP-IQA (×100)	+9.0 [+8.2, +9.7]	+4.6 [+3.2, +6.1]	+4.2 [+3.0, +5.4]	+6.6 [+4.7, +8.6]
ARNIQA (×100)	+6.5 [+6.0, +7.0]	+3.4 [+2.5, +4.3]	+3.5 [+2.6, +4.4]	+7.1 [+5.4, +8.8]

N SigMa, sampling budgets, and guidance

This section expands the single-pass comparison of Section 4.1 with SigMa and three sampling budgets of SCA without the schedule (Table N.1). Table N.2 gives the budget and guidance checks with intervals, and Figure N.1 plots ImageReward against time.

Table N.1: **SigMa and sampling budgets at 4096^2.** 195 prompts, columns as in Table 1, bold best within each group. Colours mark each row against single-pass SCA at 28 steps; the budget rows use SCA without the schedule and are coloured against its 28-step run, which this table does not show (Table N.2). The FID columns above them have no interval. Table 1 holds the multi-stage rows. Key result: SigMa trails single-pass SCA on every metric shown, and 56 steps raise the ImageReward of SCA without the schedule by 0.03 at 139.9 against 71.5 s.

		Time (s)								
Method	Steps	clean↓	camp.	IR↑	HPS↑	Pick↑	CLIP↑	M_c↑	FID↓	$FID_p^\dagger$↓
Single pass, shared sampler (28 steps, guidance 4.5; other rows in Table 1)										
SigMa	28	n/m	192.5	−0.52	18.65	20.01	23.51	49.9	216.9	184.6
Single-pass SCA	28	**77.1**	**78.4**	**1.25**	**29.73**	**23.12**	**29.14**	**60.1**	**140.2**	**98.4**
SCA without the schedule, other budgets										
56 steps	56	n/m	139.9	**1.26**	**30.55**	**23.22**	**29.14**	**64.1**	**139.8**	102.9
guidance 7	28	n/m	**71.9**	1.16	29.07	22.82	28.94	61.4	140.4	**98.4**
42 steps	42	n/m	106.8	**1.26**	30.27	23.17	**29.14**	63.1	140.5	100.6

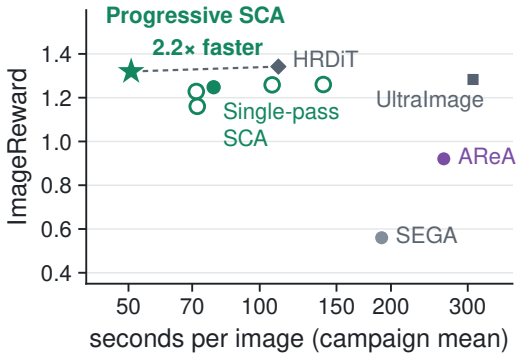

Figure N.1: **At 46% of HRDiT's time, progressive SCA trails its ImageReward by 0.02 $[-0.09, 0.03]$.** 195 prompts at 4096^2, campaign means. Star: progressive SCA; filled circle: single-pass SCA. Open circles: SCA without the schedule at 28, 42, and 56 steps and at guidance 7. Diamond and square: HRDiT and UltraImage. Full attention, DyPE, and SigMa fall below the range.

SigMa. With its dev-matched profile (Appendix I), SigMa trails single-pass SCA on all ten per-image metrics (Table M.1) and takes 192.5 s per image against 78.4 s (campaign means).

Sampling budgets. On the same 195 evaluation prompts, we compare the default 28-step run of SCA without the schedule with 56 steps at guidance 4.5 and with 28 steps at guidance 7 (Table N.2).

The 56-step run keeps the noise threshold at 0.89; it therefore changes the discretization of both the opening and mixed phases. It improves ImageReward by 0.032 $[0.0002, 0.062]$, HPSv2.1 by 0.88, PickScore by 0.13, and MUSIQ$_c$ by 2.9. It also increases corrected patch FID by 3.85 $[1.51, 6.33]$ with reference-image cluster resampling. Increasing guidance to 7 alone lowers seven per-image metrics, ImageReward by 0.068 $[0.025, 0.115]$ among them, and raises only MUSIQ. The default settings therefore do not maximize every metric: 56 steps gain eight per-image metrics but raise patch FID by 3.85.

Against UltraImage on these 195 prompts, 56-step SCA without the schedule has ImageReward difference -0.023 $[-0.089, 0.040]$, HPSv2.1 difference -0.22 $[-0.51, 0.05]$, and PickScore difference -0.11 $[-0.21, -0.01]$. The first two intervals span zero and do not establish equivalence; PickScore favours UltraImage. The MUSIQ advantage of 56-step SCA without the schedule is 11.6 $[9.5, 13.6]$ and its MUSIQ$_c$ advantage 22.4 $[21.2, 23.6]$. Campaign means are 71.5 s for 28 steps, 139.9 s for 56 steps, and 308.5 s for UltraImage on these 195 prompts. This check matches neither FLOPs nor wall-clock time: UltraImage takes 308.5 against 139.9 s, runs two resolutions with different guidance, and needs per-prompt preparation. Against HRDiT on the same prompts (Table 1), the 56-step run stays lower on ImageReward (1.26 against 1.34), HPSv2.1 (30.55 against 30.92), and PickScore (23.22 against 23.46). It also takes longer (139.9 against 110.4 s); these are point estimates without paired intervals.

Table L.2 adds a 42-step run of SCA without the schedule, sized to HRDiT's campaign time (106.8 against 110.4 s). The extra steps raise PickScore, HPSv2, HPSv2.1, ARNIQA, and MUSIQ$_c$ over 28 steps but leave ImageReward unresolved. HRDiT stays significantly ahead of the 42-step run on ImageReward, PickScore, HPSv2.1, aesthetic score, and CLIP-IQA, while trailing it on both MUSIQ scores and ARNIQA. Appendix L runs SCA inside HRDiT's schedule instead.

Table N.2: **Doubling the steps raises eight metrics; guidance 7 lowers seven and raises MUSIQ.** On 195 prompts, all with SCA without the schedule, the first two columns compare with its default 28-step run; the last compares its 56-step run with UltraImage. Per-image differences (first-named setting minus second) and 95% paired intervals, 2000 draws; blue / orange: the interval excludes zero in favour of / against the first-named setting.

	SCA without the schedule		
Metric	56 vs. 28 steps	Guidance 7 vs. 4.5	56 steps vs. UltraImage
ImageReward	$+0.032$ [$+0.0002$, $+0.062$]	-0.068 [-0.115, -0.025]	-0.023 [-0.089, $+0.040$]
Aesthetic	$+0.03$ [$+0.01$, $+0.06$]	-0.10 [-0.14, -0.07]	-0.03 [-0.07, $+0.02$]
CLIP score	-0.01 [-0.23, $+0.22$]	-0.21 [-0.45, $+0.03$]	$+0.30$ [-0.01, $+0.62$]
PickScore	$+0.13$ [$+0.07$, $+0.18$]	-0.27 [-0.34, -0.20]	-0.11 [-0.21, -0.01]
HPSv2 ($\times 100$)	$+0.32$ [$+0.23$, $+0.40$]	-0.25 [-0.35, -0.16]	$+0.11$ [-0.03, $+0.26$]
HPSv2.1 ($\times 100$)	$+0.88$ [$+0.71$, $+1.04$]	-0.60 [-0.80, -0.41]	-0.22 [-0.51, $+0.05$]
MUSIQ	-0.2 [-1.5, $+1.0$]	$+1.4$ [$+0.3$, $+2.6$]	$+11.6$ [$+9.5$, $+13.6$]
MUSIQ$_c$	$+2.9$ [$+2.2$, $+3.6$]	$+0.2$ [-0.4, $+0.8$]	$+22.4$ [$+21.2$, $+23.6$]
CLIP-IQA ($\times 100$)	$+0.9$ [$+0.4$, $+1.4$]	-1.0 [-1.7, -0.4]	$+1.1$ [$+0.0$, $+2.1$]
ARNIQA ($\times 100$)	$+1.5$ [$+1.0$, $+1.9$]	-1.2 [-1.5, -0.8]	$+0.4$ [-0.4, $+1.2$]

Distribution metrics of the 42-step run. With 195 generated images, 168 reference identities, and 1000 bootstrap draws (`results/a4k195_panel/corrected_fidp_ci/paired_deltas.csv`), 42-step SCA without the schedule has significantly higher FID than its 28-step run ($+2.4$ $[0.1, 5.7]$). KID, patch FID ($+1.5$ $[-0.5, 3.6]$), and patch KID span zero.

O Exploratory 8192^2 comparison

This section expands the 8192^2 sentence of Section 4.2. Table O.1 reports all 12 completed prompt pairs at 8192^2 under the shared FLUX.1-dev protocol, for SCA without the schedule against AReA. SCA without the schedule has higher ImageReward, CLIP, PickScore, both HPS variants, and ARNIQA, with paired intervals excluding zero. Aesthetic score and the three remaining quality metrics have intervals spanning zero. In particular, MUSIQ, MUSIQ$_c$, and CLIP-IQA have point estimates favouring AReA. These mixed quality scores and the grid and stripe failures in Figure S.15 limit the evidence for reliable 8K generation. With only 12 prompts, this comparison is exploratory; we do not report distributional metrics for this set.

All 8K scores were computed on CPU (Table O.1 notes the HPS precision). For native-resolution MUSIQ, attention queries are processed in chunks of 256 while retaining every key and native image patch. This bounds intermediate memory without changing the attention context; pretrained equivalence checks at three smaller image sizes reproduced the unchunked scores exactly.

Table O.1: **Exploratory 8192^2 comparison: SCA without the schedule is significantly better than AReA on six of ten metrics.** FLUX.1-dev, 12 matched prompts, seed 0, both methods scored on CPU with the same settings. HPS runs in float32 here and in mixed precision in the main evaluation, so its values are not comparable across resolutions. All metrics are higher-is-better. Intervals are paired prompt-bootstrap 95% percentile intervals (2000 draws, bootstrap seed 0); blue: significantly better than AReA.

Metric	AReA	SCA w/o sched.	Difference [95% CI]
ImageReward	−0.376	0.675	+1.051 [+0.386, +1.754]
Aesthetic	5.64	5.72	+0.08 [−0.26, +0.44]
CLIP score	22.38	25.52	+3.14 [+0.63, +6.11]
PickScore	20.35	21.27	+0.92 [+0.24, +1.75]
HPSv2 (×100)	24.60	26.13	+1.53 [+0.56, +2.66]
HPSv2.1 (×100)	16.59	20.51	+3.92 [+1.31, +6.85]
MUSIQ	28.6	25.7	−2.9 [−6.8, +0.8]
MUSIQ_c	44.0	40.1	−3.8 [−10.4, +1.6]
CLIP-IQA (×100)	57.1	52.0	−5.1 [−10.5, +0.6]
ARNIQA (×100)	37.3	54.6	+17.3 [+11.9, +22.6]

P REACH TABLE AT NON-SQUARE SIZES

This section checks the ratio rule of Section 2.4 at a non-square size. At $(s_H, s_W) = (2, 4)$ the two axes have different factors, and the native phase reads the table of the emulated scale nearest the larger factor, scale 4. On the 46 ablation prompts at 2048×4096, for SCA without the schedule, the scale-2 table makes three significant changes, at most 0.17 (Table K.4). They are HPSv2 +0.09, HPSv2.1 +0.17 (both ×100), and PickScore +0.05; ImageReward ($+0.01$ $[-0.03, 0.05]$) and the other metrics are within noise. Changes this small cannot resolve the mixed results of single-pass SCA against AReA at this size, so we keep the default rule.

Q COMPUTE AND REPRODUCIBILITY

This section expands the compute sentence of Section 4 and lists what the code release contains.

Compute. Generation and reach emulation used single NVIDIA GH200 GPUs on a shared cluster; the verified timing logs identify the GH200 120 GB model. Some metric evaluations ran on CPUs. The surviving logs do not establish the complete emulation runtime. At 4096^2 on shared nodes, campaign means are about 51 s per image for progressive SCA and 78 s for single-pass SCA (Table 1). They are 72 s for SCA without the schedule (Appendix N), 264 s for AReA, and 196 s for full attention. The main comparisons, ablations, and non-square sizes comprise more than 2000 generated images. Surviving worker logs establish a lower bound of 123.3 GH200 worker-hours up to 2026-10-17 06:59 UTC, including model loading, reach emulation, evaluation, and failed attempts. Later runs, among them single-pass and progressive SCA on all prompts, are not in this bound. The bound is the union of 581 closed task intervals per physical node, with timestamp precision treated conservatively; it measures occupied worker time, not GPU kernel activity. Missing historical logs, unmatched starts, and ongoing work prevent a complete campaign total. Overlapping scheduler allocation time and image-sidecar durations are not added to this bound.

Reproducibility. We will release the code: the attention processor, reach emulation, single-pass and progressive sampling (`scripts/sample_progressive.py`), evaluation wrappers, the AReA wrapper, and tests. We will also release the reach table `k1_1024.npz`, the per-image metric files, and the per-comparison summary and paired-delta CSV files behind every table. Main-text metric values and paired intervals are derived from those CSV files. Each SCA generation task ran from a frozen copy of the code named in the task. Per-image sidecar files record the sampling

settings and, where available, a code fingerprint; the inherited Krea baseline records do not establish one source revision for every image.

R BROADER IMPACT

SCA shortens generation at 4096^2 relative to full attention under our measured protocol, which helps applications such as print, design, and scientific visualization. Energy use remains unmeasured. The same reduction lowers the cost of producing high-resolution synthetic images that could be used to mislead. SCA adds no capability beyond the base model's and does not change its safety filters or watermarking; the risks and mitigations of the base model apply unchanged.

S QUALITATIVE GALLERIES

This section expands Figure 5 with more samples. Figure S.1 compares six methods on one prompt. Progressive SCA follows: sixteen evaluation prompts beside AReA and HRDiT (Figures S.2 and S.3), then six prompts drawn at random (Figure S.4). Figure S.5 shows single-pass SCA at three aspect ratios. The prompts of Figures S.6 to S.15, except Figure S.11, are drawn at random with a fixed seed from pools fixed without looking at images, which exclude the development prompts. Figures S.6 to S.9 also exclude 16 prompts that other figures showed when the pool was fixed. The selection seed, source hashes and crop coordinates of every gallery are archived in its `provenance.json`. Unless a caption says otherwise, images are FLUX.1-dev outputs for Aesthetic-4K evaluation prompts at seed 0 with 28 steps and guidance 4.5. In the randomly drawn galleries and in Figures L.1 and L.2, every crop shows native pixels of the outlined window, which is placed automatically at the highest edge energy averaged over the methods shown.

Figures S.2 to S.4 show progressive SCA; Figures S.1, S.4, S.5 to S.9, S.12, and S.14 show single-pass SCA; Figures S.10, S.11, S.13, and S.15 show SCA without the schedule. Figure S.10 uses FLUX.2-dev with guidance 4.0 and a FLUX.2-dev reach table. Figure S.14 uses the FLUX.1-Krea-dev transformer with FLUX.1-dev text encoders and VAE and a reach table fitted on FLUX.1-Krea-dev. Its DyPE and SEGA images come from the recorded Krea campaign; seven SEGA outputs were regenerated from an archived runtime because their original source snapshots could not be recovered. The inherited Krea baselines span several runtime revisions. In Figure S.12, UltraImage runs its two-stage defaults (30+30 steps, guidance 3.5 then 7), so that comparison is not equal-cost.

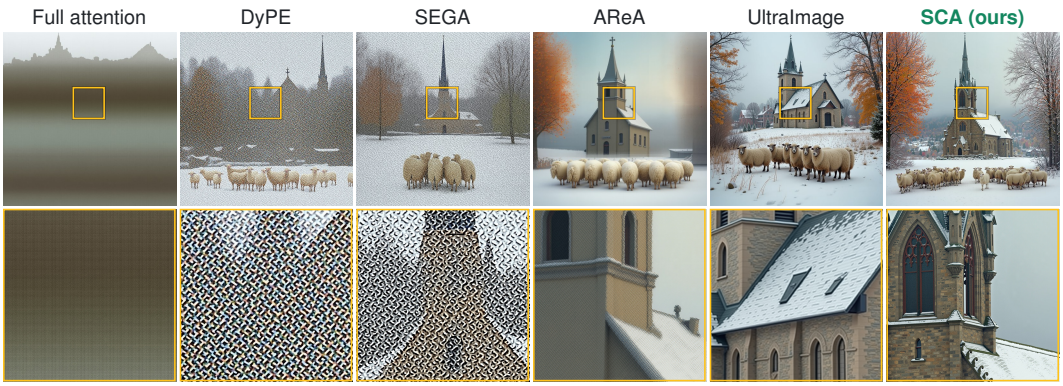

Figure S.1: **Six methods on one 4096^2 prompt.** Single-pass SCA; each column shows the image and a 1:1 crop. Full attention gives a flat field, DyPE and SEGA a lattice, and the other three keep the church and flock. Aesthetic-4K evaluation prompt at seed 0; Figure S.6 shows more prompts.

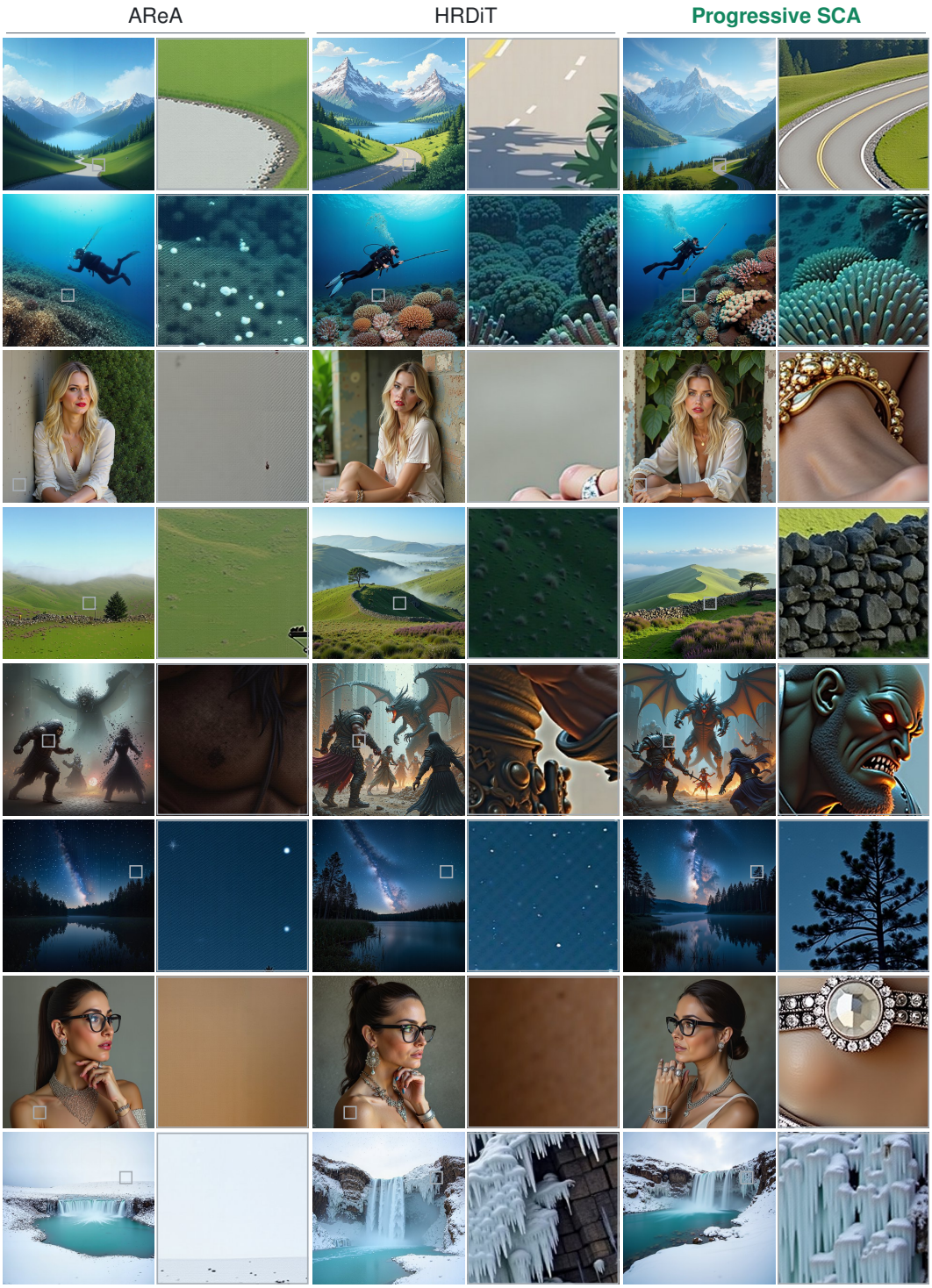

Figure S.2: **Progressive SCA beside AReA and HRDiT at 4096^2.** Sixteen Aesthetic-4K evaluation prompts. FLUX.1-dev, seed 0. AReA: 28 steps at 4096^2, guidance 4.5. HRDiT and progressive SCA: HRDiT's schedule of 30 steps at 1024^2, 17 at 2048^2 and 10 at 4096^2 (guidance 3.5, 4.5, and 6.0); progressive SCA uses SCA attention in the last two stages. Each image is followed by a 1:1 crop of its outlined 320×320 window.

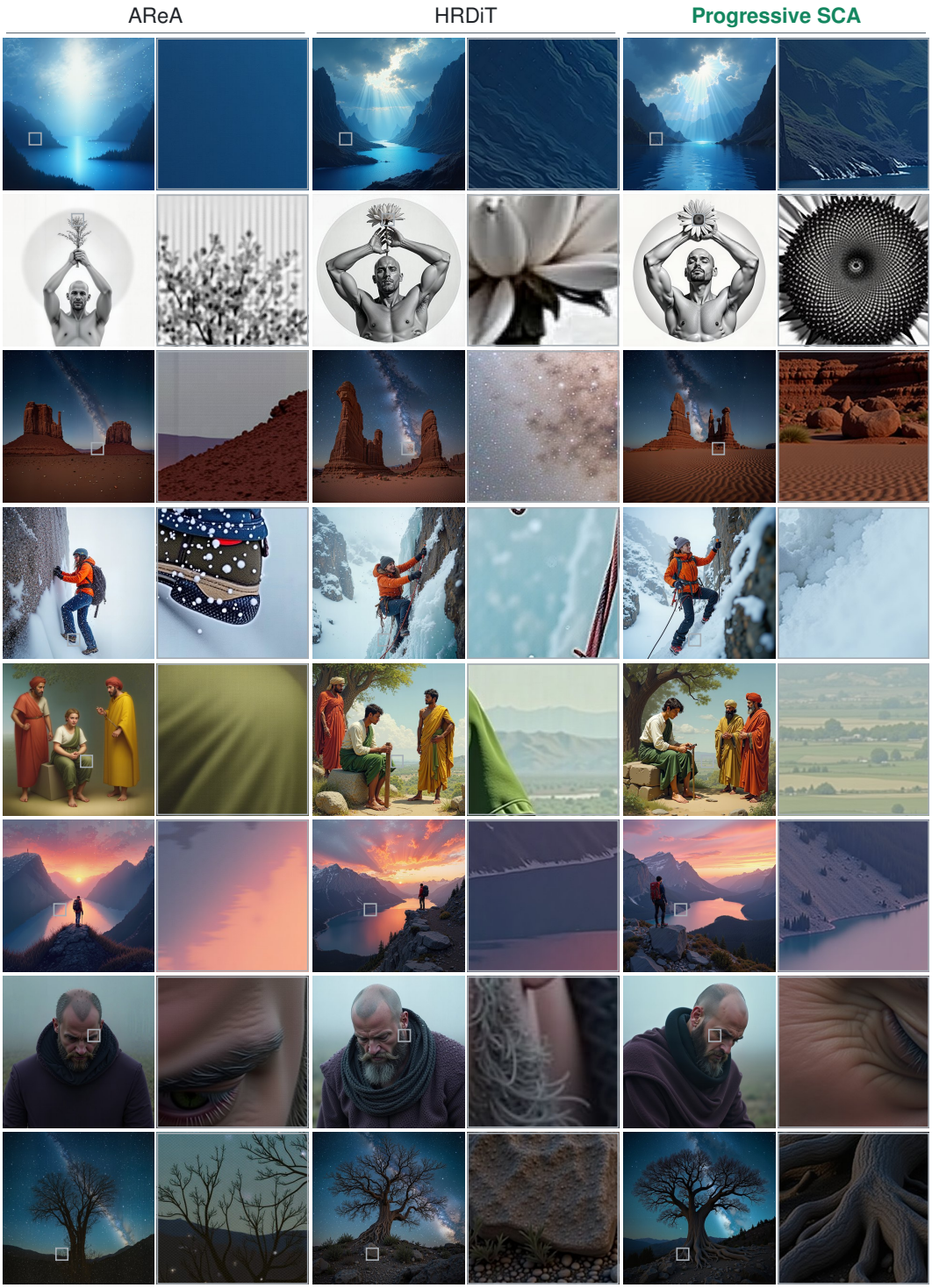

Figure S.3: **Progressive SCA beside AReA and HRDiT at 4096^2, continued.** Prompts nine to sixteen; same protocol as Figure S.2.

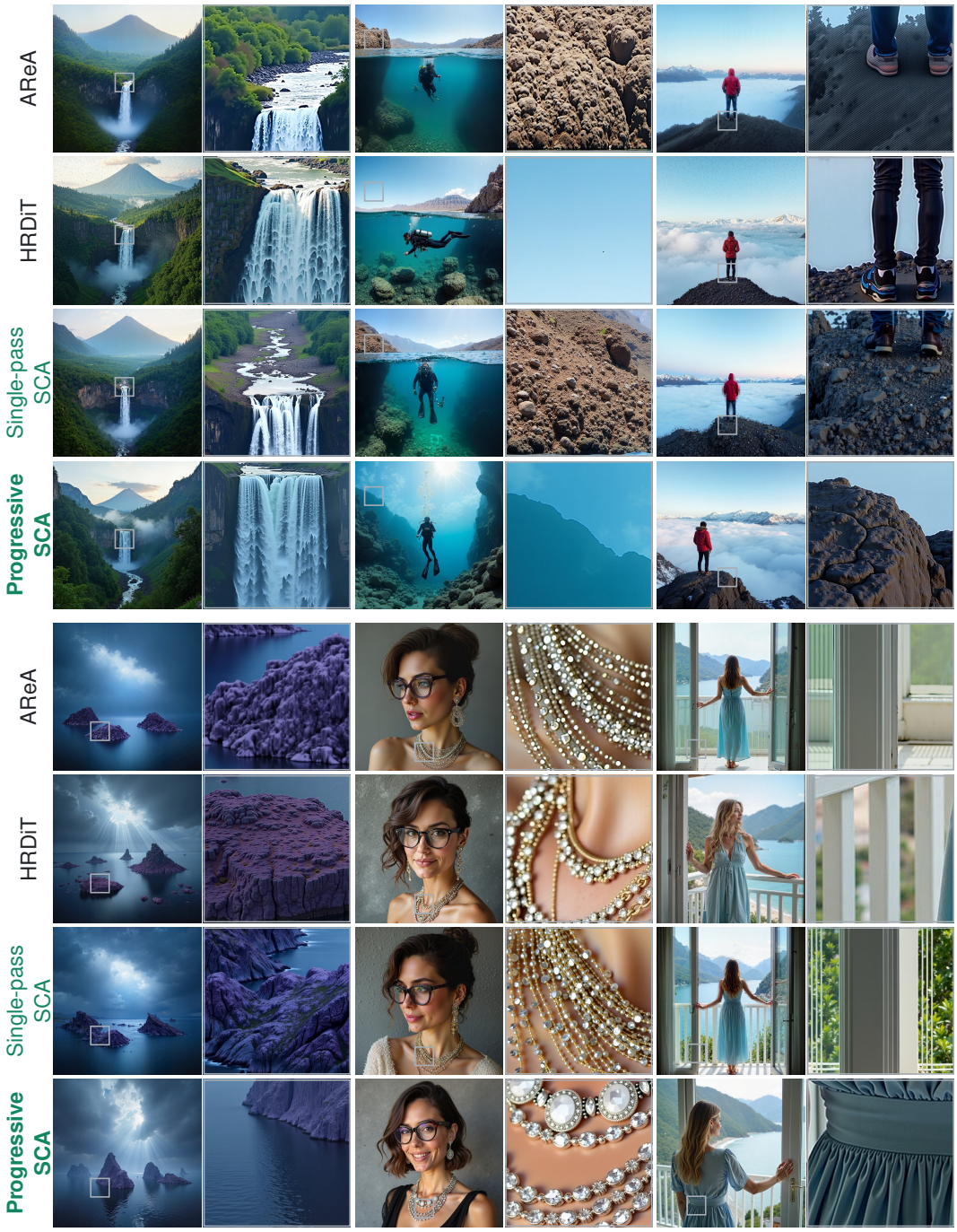

Figure S.4: **Samples at 4096^2.** Six prompts drawn at random (selection seed 0) from the 167 of the 195 evaluation prompts that are neither development prompts nor shown in another figure when the draw was made. FLUX.1-dev, generation seed 0. AReA and single-pass SCA: 28 steps at 4096^2, guidance 4.5. HRDiT and progressive SCA: HRDiT's schedule of 30 steps at 1024^2, 17 at 2048^2 and 10 at 4096^2 (guidance 3.5, 4.5 and 6.0); progressive SCA uses SCA attention in the last two stages. Rows are methods; columns are the prompts in draw order, three per block. Each image is followed by a 1:1 crop of its outlined 512×512 window of highest edge energy averaged over the four methods.

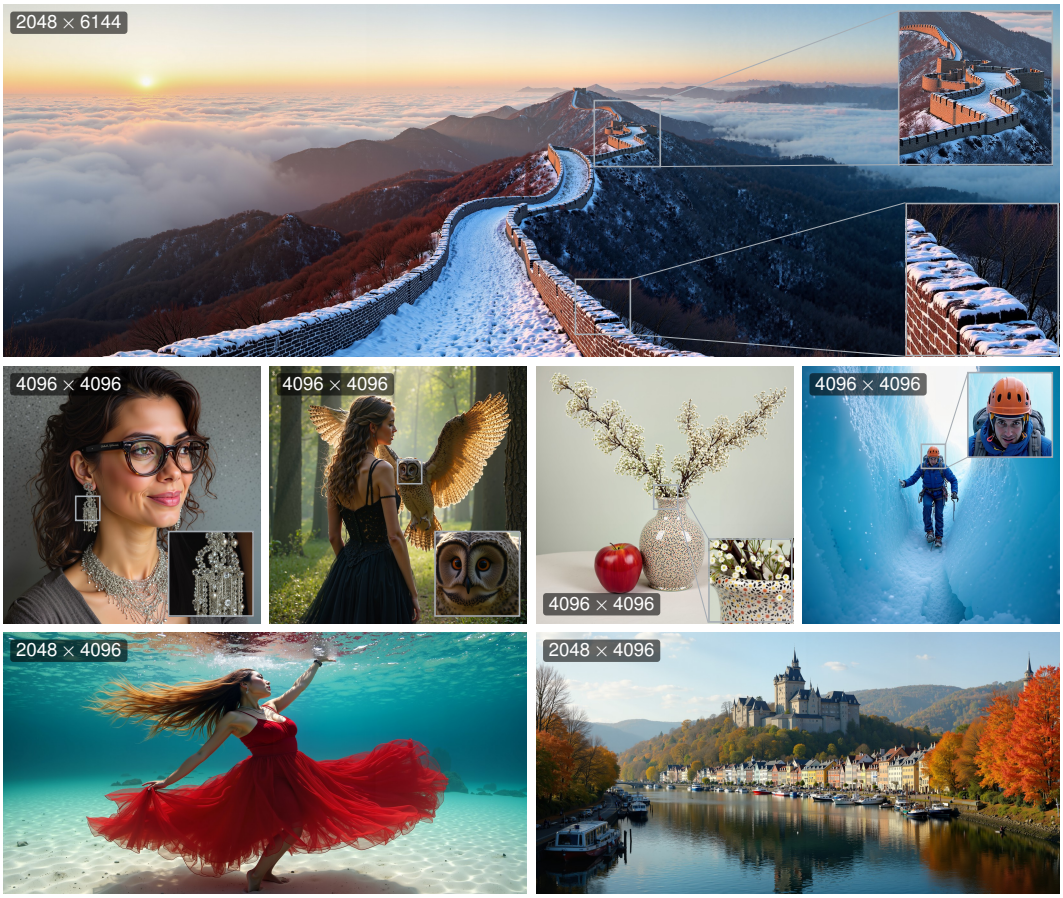

Figure S.5: **SCA keeps layout and fine detail across three aspect ratios without training.** Single-pass SCA outputs at 4096^2, 2048×4096 and 2048×6144 for Aesthetic-4K evaluation prompts at seed 0; insets are 1:1 crops.

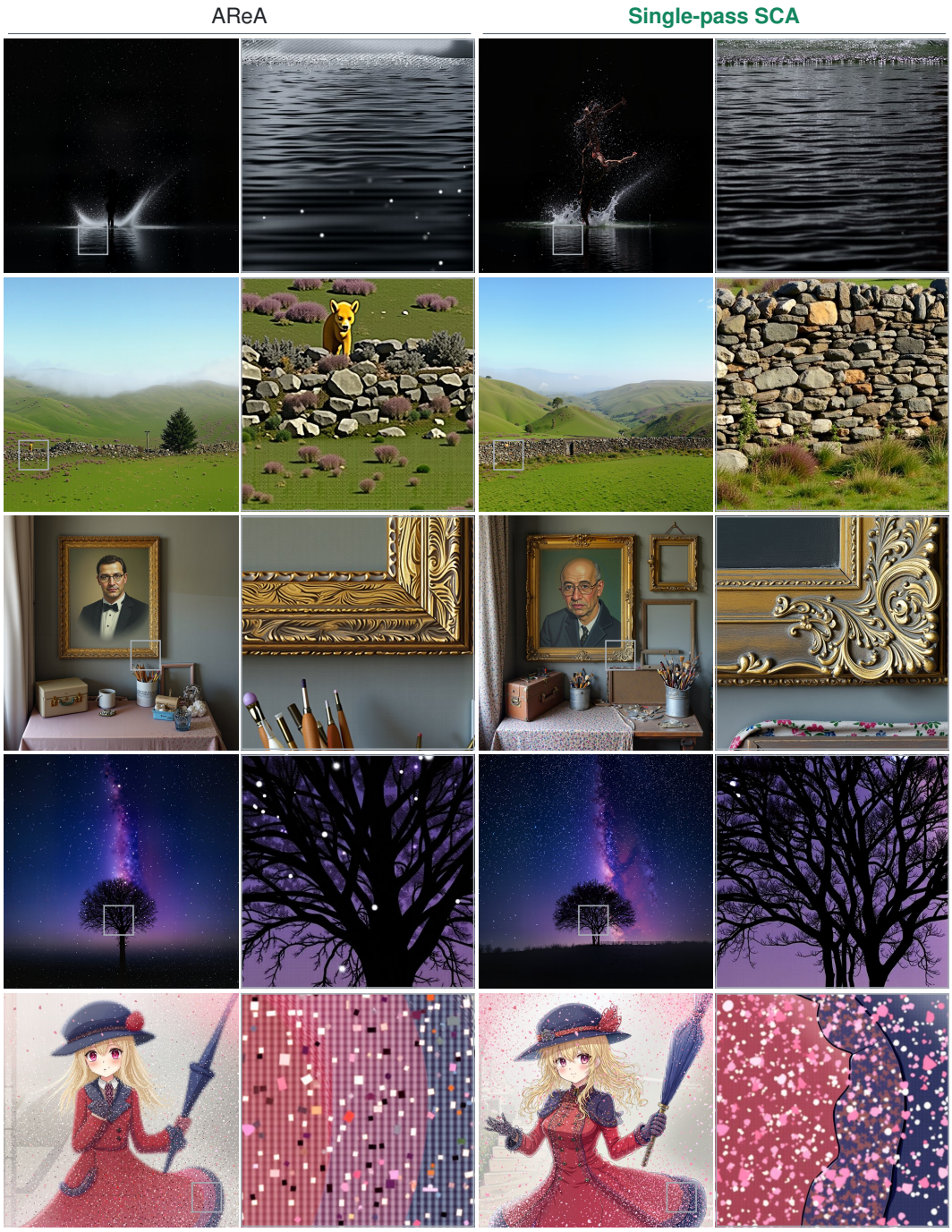

Figure S.6: **At 4096^2 single-pass SCA keeps texture that AReA blurs or grids.** Prompts one to five of 32 drawn at random; 1:1 crops follow.

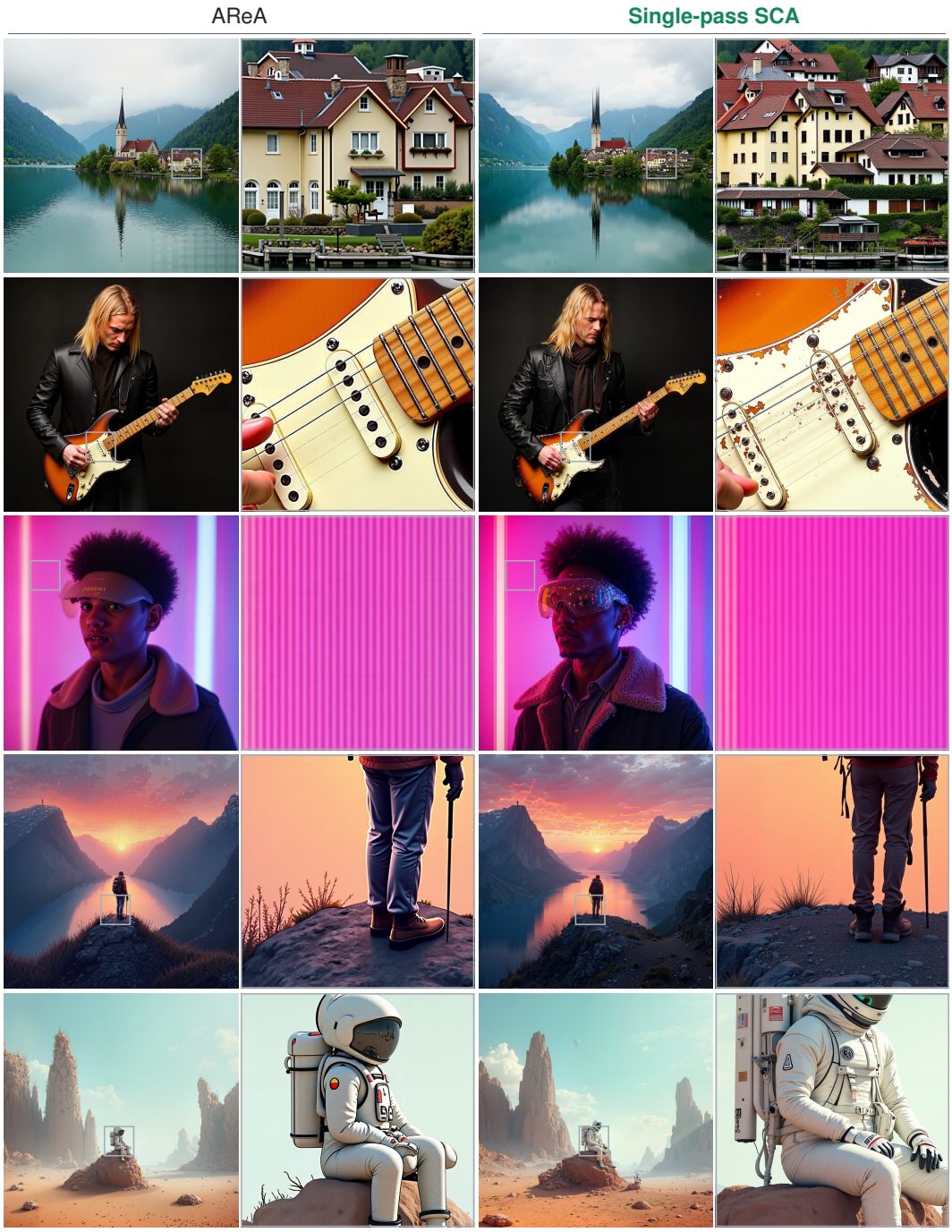

Figure S.7: **At 4096^2, continued: five more prompts, AReA against single-pass SCA.** Prompts six to ten of the random draw in Figure S.6.

Figure S.8: **At 2048 × 4096 both methods keep the layout; SCA avoids AReA's faint weave and keeps sharper rock texture.** First two of six prompts drawn at random; AReA above, single-pass SCA below; 1:1 crops follow.

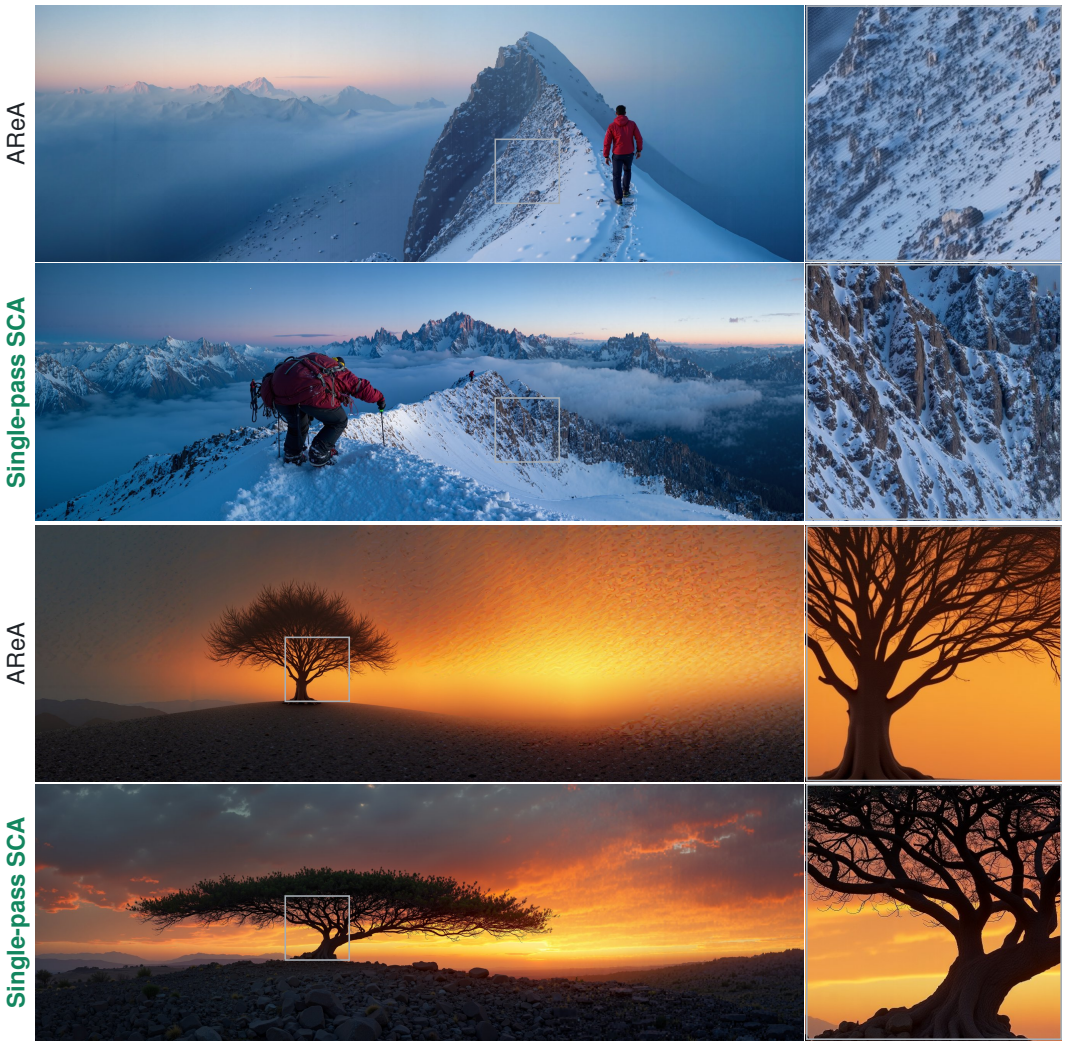

Figure S.9: **At 2048 × 6144 single-pass SCA resolves finer texture than AReA.** First two of six prompts drawn at random; AReA above, single-pass SCA below; 1:1 crops follow.

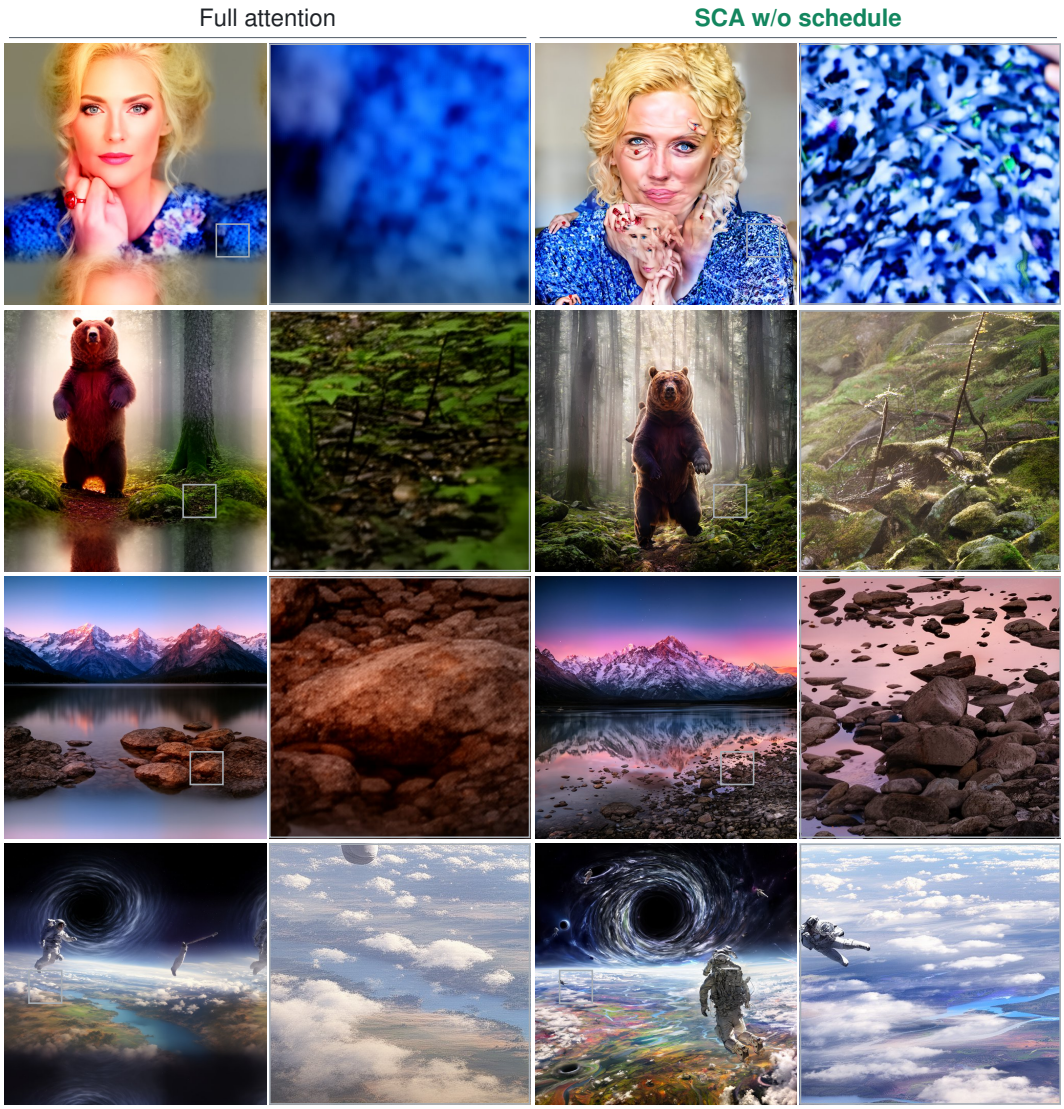

Figure S.10: **FLUX.2-dev, 4096^2: full attention blurs; SCA without the schedule is sharp but distorts the portrait.** Four of 50 prompts drawn at random; 1:1 crops follow each image.

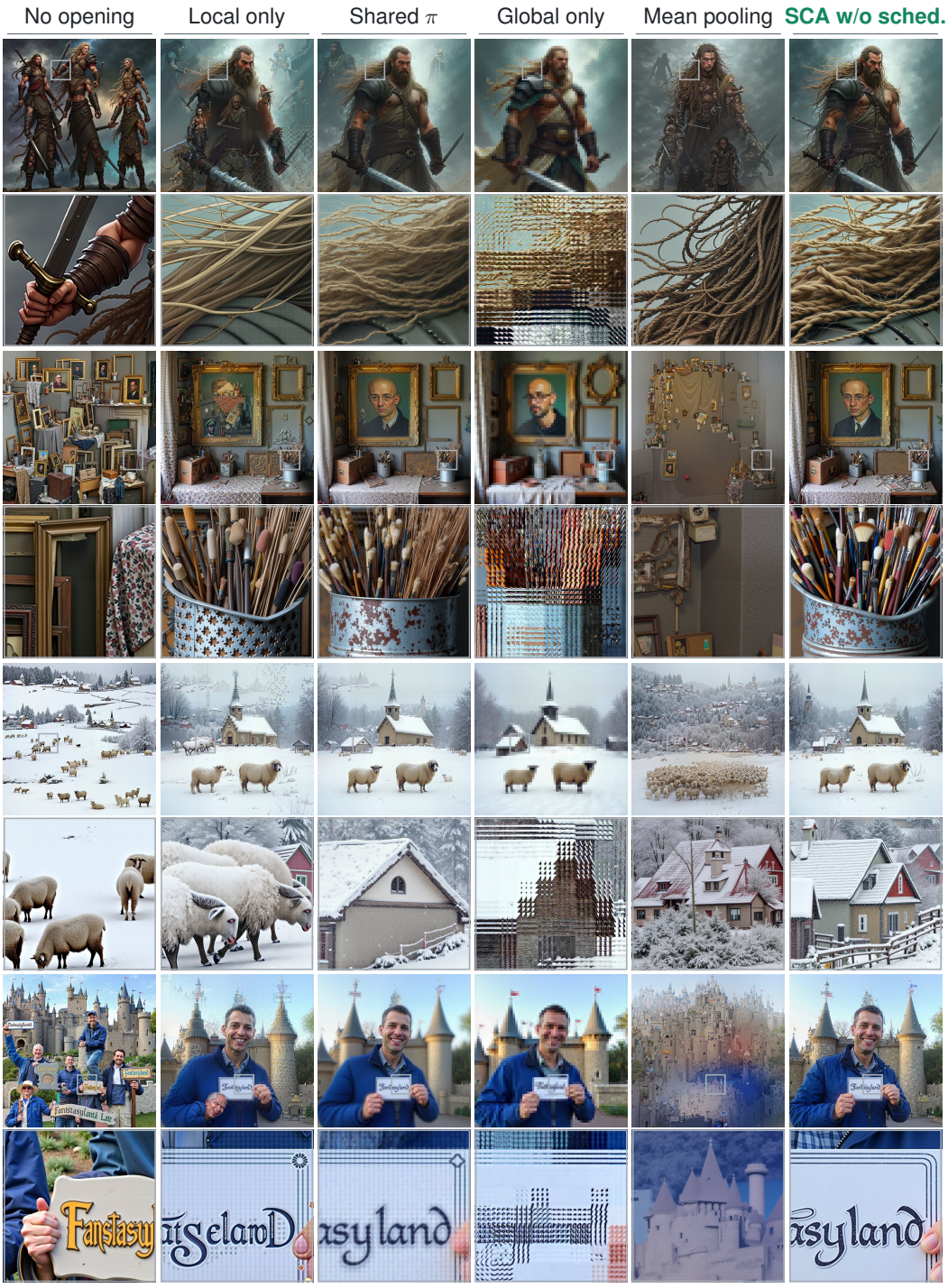

Figure S.11: **Without the opening phase, subjects shrink to a small scale; global only leaves the pooled branch's coarse grid; mean pooling loses the layout.** Four of the 46 ablation prompts at 4096^2 and seed 0, variants of SCA without the schedule (Table 2); 1:1 crops below.

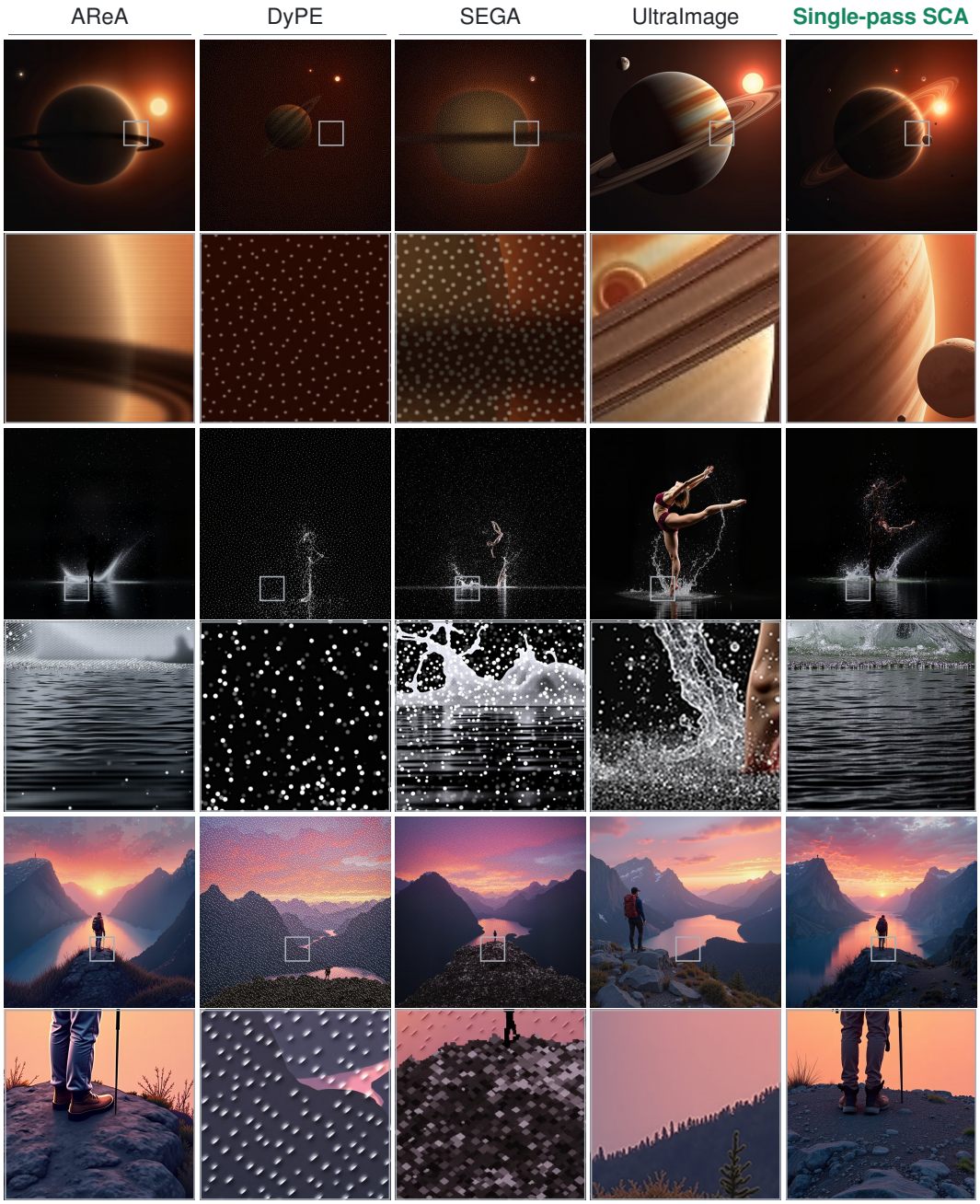

Figure S.12: **At 4096^2 DyPE and SEGA leave speckle, UltraImage softens the planet, and single-pass SCA is sharp.** Three of 191 non-development prompts drawn at random; 1:1 crops below.

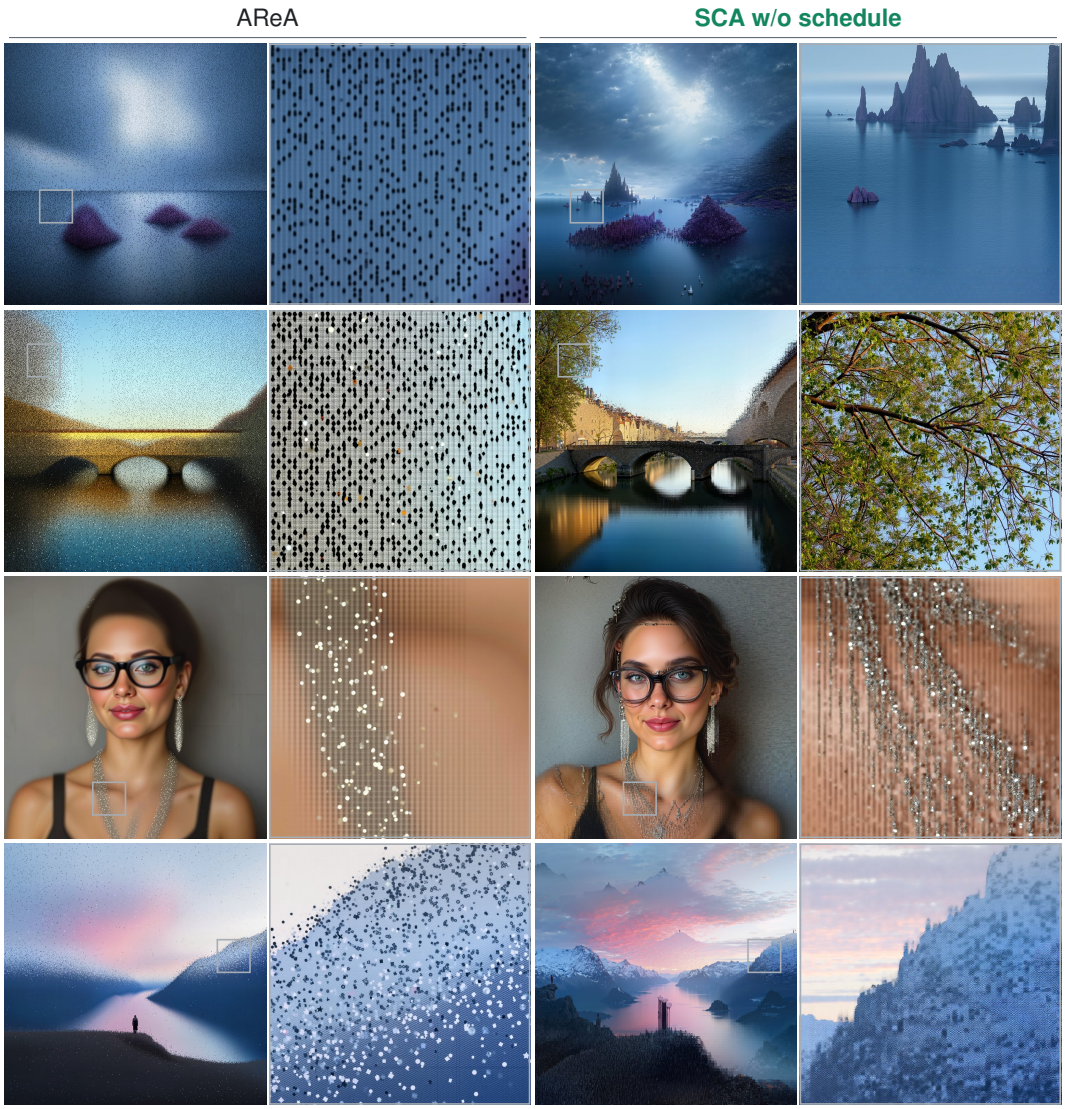

Figure S.13: **At 6144^2 AReA's crops dissolve into speckle or streaks; SCA without the schedule keeps layouts but shows stripes or block speckle.** Four of 100 prompts drawn at random; 1:1 crops follow each image.

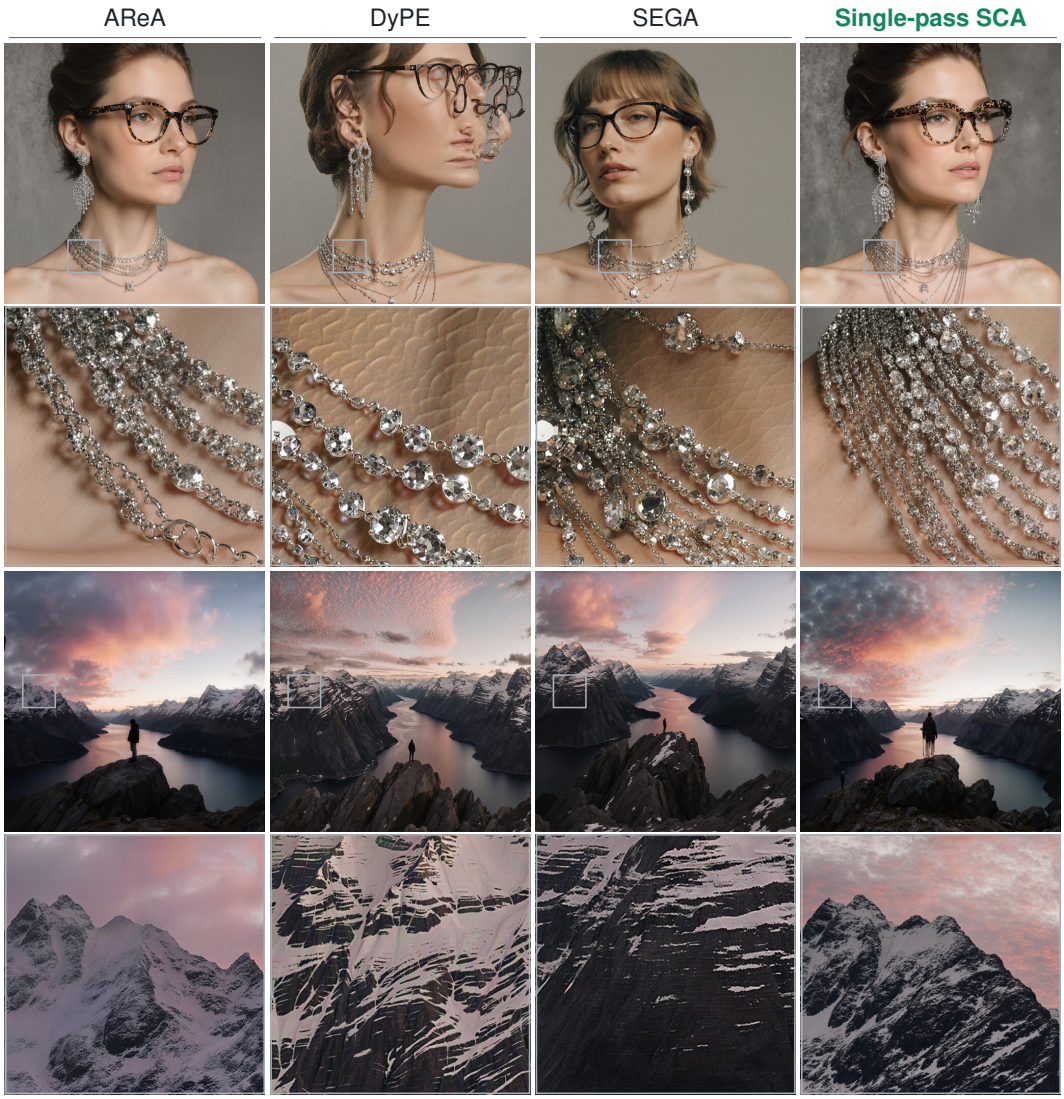

Figure S.14: **On FLUX.1-Krea-dev at 4096^2 DyPE smears the face; single-pass SCA melts the lone figure's legs.** Two of 100 prompts drawn at random; 1:1 crops below.

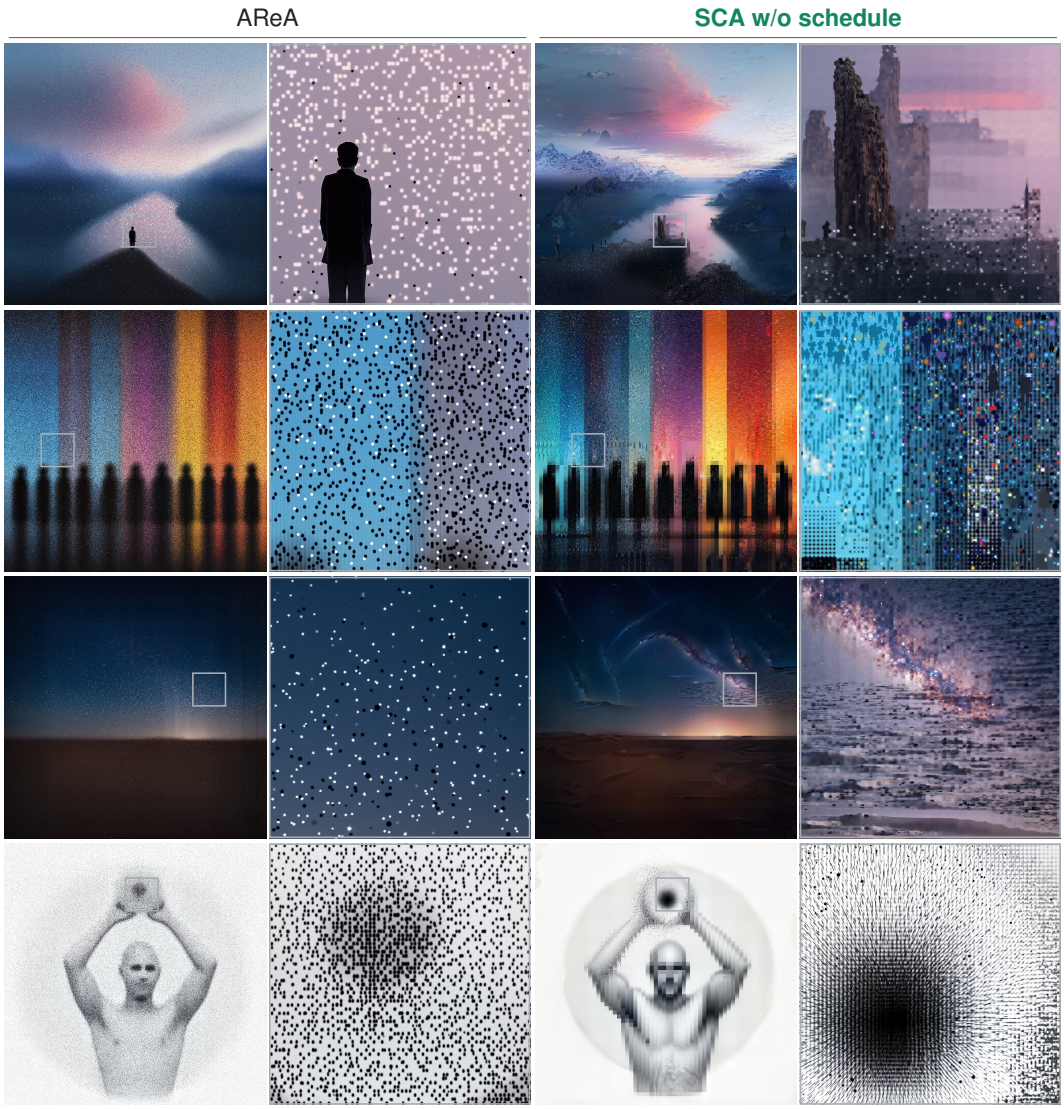

Figure S.15: **At 8192^2 both fail: AReA blurs and speckles, and SCA without the schedule breaks into blocks and stripes.** Exploratory: four of 12 paired prompts drawn at random; 1:1 crops follow each image.

T FAILURE CASES

This section expands the limitations of Section 6. Figure T.1 shows four local errors of SCA without the schedule next to AReA's same windows.

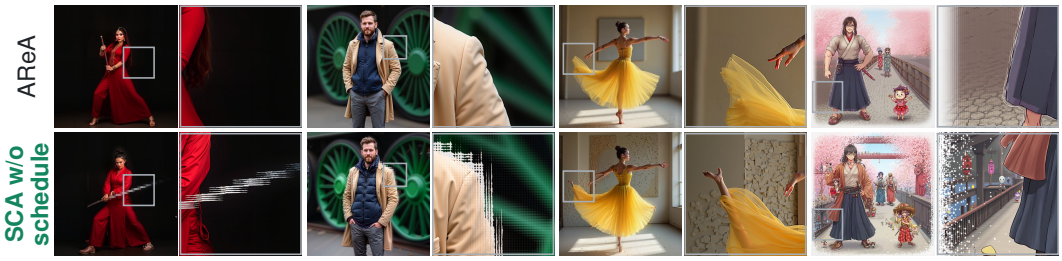

Figure T.1: **Local errors of SCA without the schedule.** Four Aesthetic-4K evaluation prompts at 4096^2 and seed 0, each full image with its window, then the window at 1:1, showing a staircase sword, a blocky patch, a malformed foot, and border speckle. AReA's same windows (top row) show none of them.

Small factors. At 2048×4096 the window covers half the image height, so the local branch alone already sees most of the image along that axis. There the per-head mix gains little over AReA's shrinking windows, and single-pass SCA is significantly worse on aesthetic score, CLIP score, and CLIP-IQA.

Larger scales and another backbone. Both use SCA without the schedule. The gallery sampled from the completed 12-prompt 8192^2 run still shows grid and stripe artifacts in both methods (Figure S.15); this exploratory set does not establish general 8K quality. In FLUX.2-dev, improved aggregate metrics coexist with distorted facial geometry and fine texture in the randomly drawn portrait example (Figure S.10).

Text-image offsets. The local branch keeps out-of-range text-image offsets (Remark 1).

Metrics. No-reference quality metrics reward sharpness even where the layout fails. The operator v0, which mixes both branches with the per-head weights from the first step and has no opening phase, scores higher on CLIP-IQA and ARNIQA (Table K.5). FID against 195 references at 299^2 resolution sees little of the 4K detail. We therefore report every metric and the qualitative galleries.